

Optimal Control Variates for Survey Sampling and Causal Inference

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This paper proposes a family of control variate estimators for variance reduction in design-based survey sampling and causal inference, with and without interference. In these settings, inverse probability weighting (IPW) estimators are widely used, but they can have large variance when sampling or treatment probabilities are heterogeneous. We construct control variates directly from the sampling or treatment indicators, whose expectations are known under the design. We parameterize the proposed control variates by their bases and characterize the optimal bases through a stochastic optimization formulation. The optimal bases are given by leading eigenvectors of matrices that depend on both the randomized sampling or experimental design and the moments of the outcomes. These results provide a unified perspective on control variate estimators and IPW estimators. We benchmark the variance reduction of the proposed control variate estimators against IPW estimators.

1. Introduction

2. Survey Sampling

2.1. Sampling Design

We start with the following survey sampling problem. Let there be n units in a finite population, where each unit is associated with an outcome $Y_1, Y_2, \dots, Y_n$. We collect them into vector form $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)^\top$. We adopt a design-based perspective and condition on the outcomes (Särndal et al. 1978). We are interested in estimating the sample mean of the finite population,

$$\mu = \frac{1}{n} \sum_{i=1}^n Y_i. \quad (1)$$

We follow a randomized sampling design to draw a random subset of the n samples to estimate μ the sample mean of the finite population. We use $W_i = 1$ to stand for a unit being sampled, and $W_i = 0$ otherwise. The randomized sampling design determines the random sampling indicators $\mathbf{W} = (W_1, W_2, \dots, W_n)^\top$ through a known joint probability distribution $\mathcal{W}$. We denote the marginal probabilities as $\pi_i = \Pr(W_i = 1)$ for any $i \in [n] := \{1, 2, \dots, n\}$, and denote the joint probabilities as

$\pi_{ij} = \Pr(W_i = W_j = 1)$ for any $i, j \in [n]$. When $i = j \in [n]$, π_{ii} simply reduces to $\pi_{ii} = \pi_i$. Using the above notations, denote the following covariance matrix

$$\Sigma = \begin{bmatrix} \pi_{11} - \pi_1\pi_1 & \pi_{12} - \pi_1\pi_2 & \dots & \pi_{1n} - \pi_1\pi_n \\ \pi_{12} - \pi_1\pi_2 & \pi_{22} - \pi_2\pi_2 & \dots & \pi_{2n} - \pi_2\pi_n \\ \vdots & \vdots & \ddots & \vdots \\ \pi_{1n} - \pi_1\pi_n & \pi_{2n} - \pi_2\pi_n & \dots & \pi_{nn} - \pi_n\pi_n \end{bmatrix}.$$

The joint probability distribution $\mathbf{W}$ may have some dependency structure. To describe this dependency structure, we define the dependency neighborhood as follows.

Definition 1 (Dependency Neighborhood) *A collection of random variables $\{W_i\}_{i \in [n]}$ has dependency neighborhoods $\mathcal{N}_i \subseteq [n]$ if W_i is independent of $\{W_j\}_{j \notin \mathcal{N}_i}$.*

In this paper, we consider sampling designs $\mathcal{W}$ that satisfy Assumption 1 below.

Assumption 1 *The sampling design $\mathcal{W}$ satisfies two conditions:*

- (i) **(Positivity)** *The marginal probabilities are all bounded away from zero, that is, there exists a constant $\underline{\pi} > 0$ such that for $i \in [n]$,*

$$\pi_i \geq \underline{\pi}.$$

- (ii) **(Dependency Neighborhood)** *The correlation between sampling indicators W_i are restricted to some neighborhoods, that is, there exist constants $\bar{d} > 0$ and $0 \leq \alpha < 1$ such that for $i \in [n]$,*

$$|\mathcal{N}_i| \leq \bar{d}n^\alpha.$$

Assumption 1-(i) is a standard assumption that usually appears in the design based sampling and causal inference literature (Ding 2023, Imbens and Rubin 2015, Wager 2020). Assumption 1-(ii) is a mild assumption to describe weakly dependent random variables (Ross 2011). We will explicitly specify the required upper bound on α when we present our main results. This assumption is satisfied by many common sampling designs. We provide Examples 1 – 3 below that illustrate when this assumption holds and fails.

Example 1 (Bernoulli Sampling) *For each unit $i \in [n]$, we independently sample unit i with probability π_i . The covariance matrix is given by*

$$\Sigma = \begin{bmatrix} \pi_1(1 - \pi_1) & 0 & \dots & 0 \\ 0 & \pi_2(1 - \pi_2) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \pi_n(1 - \pi_n) \end{bmatrix}.$$

This satisfies Assumption 1-(ii) because for $i \in [n]$, $|\mathcal{N}_i| = 1$.

□

Example 2 (Clustered Sampling) We partition n units into equal size clusters where the size of each cluster is s . We independently select each cluster with probability $\frac{1}{2}$, and, if a cluster is selected, we sample all the units within the cluster. Define the $s \times s$ matrix $\Sigma_s = \frac{1}{4} \mathbf{1}_s \mathbf{1}_s^\top$ where $\mathbf{1}_s$ is an s -dimensional vector with each element equal to 1. The covariance matrix is given by

$$\Sigma = \begin{bmatrix} \Sigma_s & 0 & \dots & 0 \\ 0 & \Sigma_s & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \Sigma_s \end{bmatrix}.$$

Note that for $i \in [n]$, $|\mathcal{N}_i| = s$. This satisfies Assumption 1-(ii) if cluster size s is small (e.g., $s < n^\alpha$); this does not satisfy Assumption 1-(ii) if cluster size s is large (e.g., $s \geq n^\alpha$). $\square$

Example 3 (Sampling without Replacement) We randomly sample exactly n_1 of the units from a total of n units. The covariance matrix is given by

$$\Sigma = \begin{bmatrix} \frac{n_1(n-n_1)}{n^2} & -\frac{n_1(n-n_1)}{n^2(n-1)} & \dots & -\frac{n_1(n-n_1)}{n^2(n-1)} \\ -\frac{n_1(n-n_1)}{n^2(n-1)} & \frac{n_1(n-n_1)}{n^2} & \dots & -\frac{n_1(n-n_1)}{n^2(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{n_1(n-n_1)}{n^2(n-1)} & -\frac{n_1(n-n_1)}{n^2(n-1)} & \dots & \frac{n_1(n-n_1)}{n^2} \end{bmatrix}.$$

This does not satisfy Assumption 1-(ii) because for $i \in [n]$, $|\mathcal{N}_i| = n$. $\square$

2.2. Control Variate Estimators for Survey Sampling

Under Assumption 1-(i), one of the most popular ways of estimating μ is to consider the Horvitz-Thompson estimator, which is also referred to as the unnormalized Inverse Probability Weighted (IPW) estimator (Horvitz and Thompson 1952). The Horvitz-Thompson estimator is defined as

$$\hat{\mu}^{\text{HT}} = \frac{1}{n} \sum_{i=1}^n \frac{Y_i \mathbb{1}\{W_i = 1\}}{\pi_i}.$$

It is easy to see that the Horvitz-Thompson estimator is unbiased, that is,

$$\mathbb{E}_{\mathcal{W}}[\hat{\mu}^{\text{HT}}] = \mu.$$

However, the Horvitz-Thompson estimator is sometimes known to have a large variance, especially when the marginal probabilities π_i are heterogeneous and small. To address this challenge, we draw on the control variate techniques common in the simulation literature (Asmussen and Glynn 2007, Glasserman 2004, Owen 2013, Ross 2013). We take the baseline estimator $\hat{\mu}^{\text{HT}}$ and propose a family of control variate estimators to reduce the variance.

We define a control variate $\hat{X}$ to be any random variable that may be correlated with the baseline estimator $\hat{\mu}^{\text{HT}}$. A control variate estimator parameterized by coefficient γ can then be defined as

$$\hat{\mu}^{\text{CV}}(\gamma) = \hat{\mu}^{\text{HT}} - \gamma(\hat{X} - \mathbb{E}[\hat{X}]).$$

When γ is a constant coefficient, the control variate estimator is unbiased, that is,

$$\mathbb{E}_{\mathcal{W}}[\hat{\mu}^{\text{CV}}(\gamma)] = \mathbb{E}_{\mathcal{W}}[\hat{\mu}^{\text{HT}}] - \gamma \mathbb{E}_{\mathcal{W}}[\hat{X} - \mathbb{E}[\hat{X}]] = \mu.$$

If the coefficient γ is selected properly, the control variate estimator has a smaller variance than the baseline Horvitz-Thompson estimator. It is well known that the variance minimizing coefficient γ^* for any generic control variate $\hat{X}$ is given by

$$\gamma^* = \frac{\text{Cov}(\hat{\mu}^{\text{HT}}, \hat{X})}{\text{Var}(\hat{X})}. \quad (2)$$

In this paper, we propose a family of control variates given in the following form

$$\hat{X} = \frac{1}{n} \sum_{i=1}^n a_i \mathbb{1}\{W_i = 1\}, \quad (3)$$

where $a_1, a_2, \dots, a_n$ are any arbitrary constants that we can choose. We collect these constants into vector form $\mathbf{a} = (a_1, a_2, \dots, a_n)^\top$ and refer to them as the “basis” of the control variates. This family of control variates is in contrast to the works that use auxiliary variables such as covariates as the control variates (Fieller and Hartley 1954, Hickernell et al. 2005). Given basis $\mathbf{a}$, the control variates are fully determined by the random sampling indicators $\mathbf{W}$. In other words, we use the sampling indicators as the control variates. The mean value of this family of control variates is given by $\mathbb{E}_{\mathcal{W}}[\hat{X}] = \frac{1}{n} \sum_{i=1}^n a_i \pi_i$ that is usually known in advance.

For this family of control variates, the variance minimizing coefficient γ^* is given by Lemma 1 below. We present Lemma 1 using succinct notations. Recall that $\pi_i = \Pr(W_i = 1)$ for any $i \in [n]$. Denote the following diagonal matrix $\mathbf{\Pi} = \text{diag}(\pi_1, \pi_2, \dots, \pi_n)$.

Lemma 1 (Variance Minimizing Coefficient) *In the survey sampling setting, under Assumption 1-(i), the variance minimizing coefficient γ^* is given by*

$$\gamma^* = \frac{\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a}}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}},$$

and the variance of the control variate estimator under this coefficient is given by

$$\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) = \frac{1}{n^2} \left(\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{\Pi}^{-1} \mathbf{Y} - \frac{(\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a})^2}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \right).$$

The proof of Lemma 1 is given in Section EC.4. Intuitively, we try to use the randomness in $\hat{X}$ to explain as much randomness in $\hat{\mu}^{\text{HT}}$ as possible. The variance $\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))$ is given by the residual from projecting the vector $\mathbf{\Pi}^{-1} \mathbf{Y}$ onto the direction of $\mathbf{a}$. Because the variance minimizing coefficient γ^* depends on unknown outcomes $\mathbf{Y}$, we cannot directly observe γ^* ; instead, we have to estimate γ^* from the data.

2.3. Estimating the Variance Minimizing Coefficient

To estimate γ^* from the data, we can replace the unknown outcomes $\mathbf{Y}$ by their Horvitz-Thompson estimates. Let $\hat{Y}_i = Y_i \frac{\mathbb{1}\{W_i=1\}}{\pi_i}$ and collect $\hat{\mathbf{Y}} = (\hat{Y}_1, \dots, \hat{Y}_n)$. Then the coefficient γ^* can be estimated using

$$\hat{\gamma}^{\text{HT}} = \frac{\hat{\mathbf{Y}}^\top \boldsymbol{\Pi}^{-1} \boldsymbol{\Sigma} \mathbf{a}}{\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}}. \quad (4)$$

It is easy to see that $\hat{Y}_i$ is an unbiased estimator of Y_i , that is, $E_{\mathcal{W}}[\hat{Y}_i] = Y_i$. So we have that $\hat{\gamma}^{\text{HT}}$ is an unbiased estimator of γ^* , that is, $E_{\mathcal{W}}[\hat{\gamma}^{\text{HT}}] = \gamma^*$.

Note that, $\hat{\gamma}^{\text{HT}}$ obtained in this way is not a constant, but a random variable correlated with the control variate $\hat{X}$. Using the estimated coefficient $\hat{\gamma}^{\text{HT}}$ introduces a bias to the control variate estimator, that is,

$$E[\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})] \neq E[\hat{\mu}^{\text{CV}}(\gamma^*)] = \mu.$$

Nonetheless, we can show that the bias is small and that $\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})$ is asymptotically equivalent to $\hat{\mu}^{\text{CV}}(\gamma^*)$, as long as the basis $\mathbf{a}$ satisfies certain regularity conditions. See Assumption 2 below.

Assumption 2 *The basis $\mathbf{a}$ satisfies two conditions:*

(i) (**Boundedness**) *There exists a constant $\bar{a}$ such that for each $i \in [n]$,*

$$|a_i| \leq \bar{a}.$$

(ii) (**Non-degeneracy**) *There exists a constant c_Σ such that*

$$\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} \geq c_\Sigma n.$$

Assumption 2-(i) is a standard boundedness assumption. Assumption 2-(ii) is a standard non-degeneracy assumption on the control variate $\hat{X}$. Note that we have $\text{Var}(\hat{X}) = n^{-2} \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$, and therefore Assumption 2-(ii) is equivalent to requiring that $\text{Var}(\hat{X}) \geq c_\Sigma n^{-1}$. Both assumptions are commonly made in the asymptotic statistics literature (Van Der Vaart and Wellner 1996, Wooldridge 2010). Assumption 2-(i) can be trivially satisfied, for example, by choosing $\mathbf{a} = (1, 1, \dots, 1)^\top$ or choosing $\mathbf{a} = (\frac{1}{\pi_1}, \frac{1}{\pi_2}, \dots, \frac{1}{\pi_n})^\top$. Assumption 2-(ii) can be satisfied if $\mathbf{a}$ is not aligned with a null or near-null direction of $\boldsymbol{\Sigma}$. For example, Assumption 2-(ii) holds under Bernoulli sampling (Example 1) and by choosing $\mathbf{a} = (1, 1, \dots, 1)^\top$ or choosing $\mathbf{a} = (\frac{1}{\pi_1}, \frac{1}{\pi_2}, \dots, \frac{1}{\pi_n})^\top$. Intuitively, putting both assumptions together, Assumption 2 ensures that the basis $\mathbf{a}$ does not contain extreme values that are disproportionately large, or contain too many extreme values that are disproportionately small. Under Assumption 2, we establish Theorem 1 below.

Theorem 1 *Under Assumptions 1 and 2 where in Assumption 1-(ii) we assume $\alpha < \frac{1}{3}$, and assuming the outcomes $|Y_i| \leq \bar{y}$ are all bounded, the control variate estimator using the estimated coefficient $\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})$ is a consistent estimator of μ , that is,*

$$\lim_{n \rightarrow +\infty} \hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) \xrightarrow{p} \mu.$$

Additionally, if in Assumption 1-(ii) we assume $\alpha < \frac{1}{4}$, then the asymptotic variance of the control variate estimator using the estimated coefficient $\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))$ is the same as the asymptotic variance of the control variate estimator using the optimal coefficient $\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^))$, that is,*

$$\lim_{n \rightarrow +\infty} n \left(\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})) - \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) \right) \rightarrow 0.$$

Additionally, if in Assumption 1-(ii) we assume $\alpha < \frac{1}{7}$ and further assuming there exists a constant $\underline{c} > 0$ such that for sufficiently large n , $n \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^)) \geq \underline{c}$, then the control variate estimator using the estimated coefficient $\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})$ is asymptotically normal, that is,*

$$\lim_{n \rightarrow +\infty} \frac{\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) - \mu}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

The proof of Theorem 1 is given in Section EC.4. As we have stronger results, from consistency to asymptotic variance equivalence and to asymptotic normality, the requirement on the dependency neighborhoods becomes correspondingly stronger, with α strengthening from $\alpha < \frac{1}{3}$ to $\alpha < \frac{1}{4}$ and to $\alpha < \frac{1}{7}$. Taken together, Theorem 1 shows that estimating γ^* by $\hat{\gamma}^{\text{HT}}$ does not affect the first-order asymptotic properties of the control variate estimator. In particular, estimating γ^* by $\hat{\gamma}^{\text{HT}}$ leads to the same variance reduction asymptotically, as given by Lemma 1. The magnitude of variance reduction is determined by the alignment between the weighted outcomes $\mathbf{\Pi}^{-1}\mathbf{Y}$ and the basis $\mathbf{a}$. In Section 2.4 below, we study how to choose a basis $\mathbf{a}$ for the optimal variance reduction.

2.4. The Optimal Control Variate

Now we turn our attention to finding the optimal basis $\mathbf{a}$ that minimizes $\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))$ the variance of the control variate estimator for fixed n . If the outcomes $\mathbf{Y}$ were known to take values $Y_i = y_i$ for any $i \in [n]$, the problem

$$\min_{\mathbf{a}} \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) = \min_{\mathbf{a}} \frac{1}{n^2} \left(\mathbf{y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{\Pi}^{-1} \mathbf{y} - \frac{(\mathbf{y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a})^2}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \right)$$

can be easily solved by setting $\mathbf{a} = \mathbf{\Pi}^{-1} \mathbf{y}$. In this case, the variance $\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))$ can be reduced to exactly zero.

However, the above choice of $\mathbf{a}$ is infeasible because the outcomes $\mathbf{Y}$ are unknown. Therefore, we find the optimal basis $\mathbf{a}$ under uncertainty of the outcomes, which leads to a decision making

problem under uncertainty. There are various statistical decision rules for making decisions under uncertainty (Lindley 1972, Manski 2004, Rubin 1978, Savage 1951, Stoye 2009, Wald 1949, Wu 1981, Zhao 2023). We adopt a stochastic optimization perspective (Zhao 2024) and model each unknown outcome Y_i to be independent and identically distributed (i.i.d.) samples from an unknown distribution $\mathcal{Y}$. See Assumption 3 below.

Assumption 3 (i.i.d. Outcomes) *We assume that for each $i \in [n]$, the outcome Y_i is i.i.d. samples from an unknown distribution $\mathcal{Y}$.*

Denote the population mean as $\mu = \mathbb{E}_{Y_i \sim \mathcal{Y}}[Y_i]$ and the population variance as $\sigma^2 = \text{Var}_{Y_i \sim \mathcal{Y}}(Y_i)$. Denote $\mathcal{Y}^n$ to be the joint probability distribution of the outcomes $\mathbf{Y}$. Under Assumption 3, we formulate the following optimization problem

$$\mathbf{a}^* \in \arg \min_{\mathbf{a}} \mathbb{E}_{\mathbf{Y} \sim \mathcal{Y}^n} \left[\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) \right]. \quad (5)$$

The optimal control variate uses the basis that solves (5), which is given by Theorem 2 below. We present Theorem 2 using succinct notations. Recall that Σ is symmetric and positive semidefinite, we can define its square root matrix as $\Sigma^{\frac{1}{2}}$ and its pseudo-inverse square root matrix as $\Sigma^{-\frac{1}{2}}$. See Section EC.1 for their detailed definitions and properties.

Theorem 2 (Optimal Control Variate) *Let $\mathbf{M}$ be a symmetric matrix defined as*

$$\mathbf{M} = \Sigma^{\frac{1}{2}} \Pi^{-1} (\sigma^2 \mathbf{I}_n + \mu^2 \mathbf{1}_n \mathbf{1}_n^\top) \Pi^{-1} \Sigma^{\frac{1}{2}},$$

where $\mathbf{I}_n$ is a $(n \times n)$ identity matrix and $\mathbf{1}_n$ is a n -dimensional vector with each element equal to 1. Let $\lambda_1(\mathbf{M})$ be the largest eigenvalue of $\mathbf{M}$, and let $\mathbf{u}_1(\mathbf{M})$ be the eigenvector that corresponds to $\lambda_1(\mathbf{M})$. Under Assumptions 1-(i) and 3, the optimal basis is given by

$$\mathbf{a}^* = c \Sigma^{-\frac{1}{2}} \mathbf{u}_1(\mathbf{M}),$$

where $c \neq 0$ is any constant. And in expectation, the optimal variance reduction is given by

$$\mathbb{E}_{\mathbf{Y} \sim \mathcal{Y}^n} \left[\text{Var}(\hat{\mu}^{\text{HT}}) - \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) \right] = \frac{\lambda_1(\mathbf{M})}{n^2}.$$

The proof of Theorem 2 is given in Section EC.4. Theorem 2 combines a design-based perspective and a model-based perspective. We assume a model for the unknown outcomes $\mathbf{Y}$ to guide the choice of basis. But we rely on the sampling design as the only source of randomness when we estimate the sample mean. We point it out that Assumption 3 is only required to show optimality of the basis $\mathbf{a}^*$. The basis $\mathbf{a}^*$ is still well defined without Assumption 3.

In Theorem 2, the optimal basis $\mathbf{a}^*$ depends on the unknown quantities μ and σ^2 . But since $\mathbf{u}_1(\mathbf{M})$ is an eigenvector and thus scale-invariant to $\mathbf{M}$, we can always normalize $\mathbf{M}$ by σ^2 if $\sigma^2 > 0$ so the optimal basis $\mathbf{a}^*$ depends on the signal-to-noise ratio $\frac{|\mu|}{\sigma}$. In practice, we may use sample-splitting to estimate the signal-to-noise ratio, and then use it to find the optimal basis $\mathbf{a}^*$. Even an imperfect estimate of the signal-to-noise ratio will yield a basis that can still guarantee a variance reduction over the baseline Horvitz-Thompson estimator.

Theorem 2 provides a refinement of a classical recommendation in the simulations literature. In Section 9.2 of Owen (2013), the author explicitly recommended using the importance weights as the control variate to reduce variance, whenever they are known. In Theorem 2, we further extend this recommendation by studying what is the optimal form of the importance weights to use as the control variates. As we will show below in Corollary 1, when the outcomes have near-zero variance, the optimal control variates are the importance weights themselves.

We next show that the optimal basis $\mathbf{a}^*$ satisfies Assumption 2-(ii).

Proposition 1 *When $c = n^{\frac{1}{2}}$, the optimal basis $\mathbf{a}^*$ in Theorem 2 is given by*

$$\mathbf{a}^* = n^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \mathbf{u}_1(\mathbf{M}),$$

and satisfies the following equality

$$\mathbf{a}^{*\top} \Sigma \mathbf{a}^* = n.$$

The proof of Proposition 1 is given in Section EC.4. Proposition 1 shows that the optimal basis $\mathbf{a}^*$ satisfies Assumption 2-(ii). To conclude this section, we consider a remarkable special case of Theorem 2 when $\sigma^2 = 0$.

Corollary 1 (Noiseless Outcomes) *Under Assumption 1-(i) and assuming that the outcomes $Y_i = y$ take the same unknown constant, the optimal basis is given, for any $i \in [n]$, by*

$$a_i = \frac{c}{\pi_i},$$

where $c \neq 0$ is any constant.

The proof of Corollary 1 is given in Section EC.4. In this special case when all outcomes take the same constant, the only fluctuation of the Horvitz-Thompson estimator comes from the random component $\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=1\}}{\pi_i}$. Corollary 1 suggests to correct it by using the exact same term $\hat{X} = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=1\}}{\pi_i}$ as the control variate. This draws a connection between the control variate estimator and the Hajek estimator, which we will discuss in Section 2.5 below.

2.5. Connections to Hajek Estimator and Normalized Estimators

Hajek Estimator. One popular alternative to the Horvitz-Thompson estimator is the Hajek estimator, which is also referred to as the self-normalized IPW estimator (Basu 1971). The Hajek estimator is defined as

$$\hat{\mu}^{\text{Hajek}} = \frac{\hat{\mu}^{\text{HT}}}{\hat{1}}, \quad \text{where} \quad \hat{1} = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i = 1\}}{\pi_i}.$$

The Hajek estimator normalizes the Horvitz-Thompson estimator by $\hat{1}$, which serves as an estimate of one (Gao and Ding 2025). This self-normalization reduces the impact of observations with extreme inverse probabilities and is therefore generally recognized to have a smaller variance than the Horvitz-Thompson estimator, although such an improvement is not always true.

There is a connection between the Hajek estimator and the control variate estimator. If we view the denominator $\hat{1}$, the estimated one, as a variable, we can make a first order approximation to the Hajek estimator by using Taylor expansion around $\hat{1} \approx 1$, it is given as

$$\hat{\mu}^{\text{Hajek}} \approx \hat{\mu}^{\text{HT}} - \hat{\mu}^{\text{HT}} \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i = 1\}}{\pi_i} - 1 \right).$$

On the other hand, using the basis $a_i = \frac{1}{\pi_i}, \forall i \in [n]$, the control variate estimator is given as

$$\hat{\mu}^{\text{CV}} = \hat{\mu}^{\text{HT}} - \hat{\gamma}^{\text{HT}} \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i = 1\}}{\pi_i} - 1 \right),$$

where $\hat{\gamma}^{\text{HT}}$ is given in (4). Under Bernoulli sampling as defined in Example 1 and when all the sampling probabilities are equal $\pi_i = \pi$, we have

$$\hat{\gamma}^{\text{HT}} = \frac{1}{n} \sum_{i=1}^n \frac{Y_i \mathbb{1}\{W_i = 1\}}{\pi} = \hat{\mu}^{\text{HT}}.$$

Therefore, the first-order approximation to the Hajek estimator is a special case of the control variate estimator using the basis $a_i = \frac{1}{\pi_i}, \forall i \in [n]$. Combining with Corollary 1, it shows that the first-order approximation to the Hajek estimator is asymptotically optimal when the outcomes have near-zero variance, and when data is collected under Bernoulli sampling with all the sampling probabilities being equal. This extends the conventional wisdom in Särndal et al. (1978) that Hajek estimator has smaller variance than Horvitz-Thompson estimator when “ $Y_i - \mu$ is small” (i.e., when the outcomes have near-zero variance) and when “the sample size is not fixed” (e.g., under Bernoulli sampling, Example 1).

Normalized Estimators. Extending the idea of Hajek estimator, Trotter and Tukey (1956) proposed a family of normalized estimators in the form of

$$\hat{\mu}^{\text{Norm}} = \frac{\hat{\mu}^{\text{HT}}}{\lambda \hat{1} + (1 - \lambda)},$$

where $\lambda \in \mathbb{R}$ is a parameter that can take values outside of $[0, 1]$. Two special cases are $\hat{\mu}^{\text{Norm}} = \hat{\mu}^{\text{HT}}$ when $\lambda = 0$ and $\hat{\mu}^{\text{Norm}} = \hat{\mu}^{\text{Hajek}}$ when $\lambda = 1$. For this family of normalized estimators, we can make a first order approximation by using Taylor expansion around $\hat{1} \approx 1$, which gives

$$\hat{\mu}^{\text{Norm}} \approx \hat{\mu}^{\text{HT}} - \lambda \hat{\mu}^{\text{HT}} \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i = 1\}}{\pi_i} - 1 \right).$$

Therefore, the first-order approximations to this family of normalized estimators can also be viewed as a special case of the control variate estimator using the basis $a_i = \frac{1}{\pi_i}, \forall i \in [n]$.

Building on this family of normalized estimators, Khan and Ugander (2023) studied the optimal choice of λ under Bernoulli sampling and from a sampling-based perspective. They further proposed several ways to estimate λ , including an elegant estimator that they termed as the adaptively normalized estimator. The line of work on which Khan and Ugander (2023) builds, including Swaminathan and Joachims (2015), pointed out the connections between the family of normalized estimators and the control variate literature. Yet none of them studied the question of how to choose the optimal control variates.

3. Causal Inference under Network Interference

3.1. Experimental Design under Network Interference

We consider the following causal inference problem under network interference. The classical causal inference problem under the Stable Unit Treatment Value Assumption (SUTVA, Rubin 1980, Holland 1986) can be viewed as a special case of the network interference setup; see Example 4 below for an illustration. In Section EC.2, we present results for causal inference under SUTVA as an important special case.

Let there be n units in a finite population that are connected by an undirected network. Each vertex represents an experimental unit and each edge represents the connection between the two units. Let there be two versions of treatments. We use “treatment” and “control,” or 1 and 0, respectively, to stand for these two versions of treatments. For each $i \in [n]$, let $W_i \in \{0, 1\}$ stand for the treatment assignment that unit i receives. We consider an experimental design setting where we determine the random treatment assignments $\mathbf{W} = (W_1, \dots, W_n)^\top$ through a known joint probability distribution $\mathcal{W}$. Following convention, we use W_i for a random treatment assignment, and w_i for one realization.

We adopt a potential outcomes framework in describing the causal parameters (Neyman 1923, Rubin 1974). In the full generality, for each unit $i \in [n]$, let $Y_i(\mathbf{w})$ be the potential outcome for unit i when the treatment assignment vector is $\mathbf{w}$. The observed outcomes are random variables of the treatment assignments connected through the potential outcomes, that is, if $\mathbf{W} = \mathbf{w}$ then $Y_i = Y_i(\mathbf{w})$. To simplify the connection between observed outcomes and potential outcomes, we borrow the exposure mapping framework (Manski 2013, Aronow and Samii 2017) and assume that the exposure mapping is known. Let Δ_i be the set of all possible exposure conditions for unit i . An exposure mapping for unit i is a dimension reduction mapping $f_i : \{0, 1\}^n \rightarrow \Delta_i$ that maps an assignment vector $\mathbf{w}$ to an exposure condition $f_i(\mathbf{w}) = \delta_i \in \Delta_i$. We assume that the potential outcome of unit i depends on the assignment vector $\mathbf{w}$ only through the exposure condition δ_i .

Assumption 4 (Known Exposure Mapping) *For any two vectors of assignments $\mathbf{w}, \mathbf{w}' \in \{0, 1\}^n$ and for any unit $i \in [n]$, if $f_i(\mathbf{w}) = f_i(\mathbf{w}')$, then*

$$Y_i(\mathbf{w}) = Y_i(\mathbf{w}').$$

Suppose we only focus on two exposure conditions $\delta(1)$ and $\delta(0)$ and suppose they are both attainable, that is, $\delta(1), \delta(0) \in \Delta_i$ for any $i \in [n]$. Under Assumption 4, each observed outcome is related to its respective potential outcomes through

$$Y_i = \begin{cases} Y_i(\delta(1)), & \text{if } f_i(\mathbf{w}) = \delta(1), \\ Y_i(\delta(0)), & \text{if } f_i(\mathbf{w}) = \delta(0), \end{cases}$$

where we reload notation and define $Y_i(\delta(1))$ as the common value of $Y_i(\mathbf{w})$ for any $\mathbf{w}$ such that $f_i(\mathbf{w}) = \delta(1)$, and define $Y_i(\delta(0))$ as the common value of $Y_i(\mathbf{w})$ for any $\mathbf{w}$ such that $f_i(\mathbf{w}) = \delta(0)$. We collect the potential outcomes into vector form $\mathbf{Y}(\delta(1)) = (Y_1(\delta(1)), \dots, Y_n(\delta(1)))^\top$ and $\mathbf{Y}(\delta(0)) = (Y_1(\delta(0)), \dots, Y_n(\delta(0)))^\top$. We reload notation and denote the $2n$ -dimensional potential outcomes vector $\mathbf{Y} = (\mathbf{Y}(\delta(1))^\top, -\mathbf{Y}(\delta(0))^\top)^\top$.

We are interested in estimating the average treatment effect of exposure condition $\delta(1)$ relative to exposure condition $\delta(0)$ in the finite population,

$$\tau(\delta(1), \delta(0)) = \frac{1}{n} \sum_{i=1}^n (Y_i(\delta(1)) - Y_i(\delta(0))).$$

In this paper, we take the design of experiment $\mathcal{W}$ as given. We define the following two events

$$\mathcal{E}_i(1) = \{f_i(\mathbf{W}) = \delta(1)\}, \quad \text{and} \quad \mathcal{E}_i(0) = \{f_i(\mathbf{W}) = \delta(0)\}.$$

Using the definitions of these two events, we reload notation and denote the exposure indicators

$$\mathbf{D} = (\mathbb{1}\{\mathcal{E}_1(1)\}, \dots, \mathbb{1}\{\mathcal{E}_n(1)\}, \mathbb{1}\{\mathcal{E}_1(0)\}, \dots, \mathbb{1}\{\mathcal{E}_n(0)\})^\top.$$

We denote the following two $(n \times n)$ diagonal matrices $\mathbf{\Pi}(1) = \mathbf{diag}(\Pr_{\mathcal{W}}(\mathcal{E}_1(1)), \dots, \Pr_{\mathcal{W}}(\mathcal{E}_n(1)))$ and $\mathbf{\Pi}(0) = \mathbf{diag}(\Pr_{\mathcal{W}}(\mathcal{E}_1(0)), \dots, \Pr_{\mathcal{W}}(\mathcal{E}_n(0)))$. We reload notation and define the following $(2n \times 2n)$ diagonal matrix

$$\mathbf{\Pi} = \begin{bmatrix} \mathbf{\Pi}(1) & \mathbf{0}_{n \times n} \\ \mathbf{0}_{n \times n} & \mathbf{\Pi}(0) \end{bmatrix},$$

where $\mathbf{0}_{n \times n}$ stands for a $(n \times n)$ matrix with each element equal to 0. Matrix $\mathbf{\Pi}$ reflects the marginal probabilities of $\mathbf{D}$. Denote the covariance matrix as $\mathbf{\Omega} = \text{Var}_{\mathcal{W}}(\mathbf{D})$.

We consider designs of experiments $\mathcal{W}$ that satisfy Assumption 1[†] below.

Assumption 1[†] *The design of experiment $\mathcal{W}$ satisfies:*

- (i) (**Positivity**) *The marginal probabilities under both exposure conditions converge to zero at a slow rate, that is, there exist constants $\underline{\pi} > 0$ and $0 \leq \beta < 1$ such that for $i \in [n]$,*

$$\Pr_{\mathcal{W}}(\mathcal{E}_i(1)), \Pr_{\mathcal{W}}(\mathcal{E}_i(0)) \geq \underline{\pi} n^{-\beta} > 0.$$

- (ii) (**Dependency Neighborhood**) *The correlation between the $2n$ exposure indicators $\mathbf{D}$ are restricted to some neighborhoods, that is, there exist constants $\bar{d} > 0$ and $0 \leq \alpha < 1$ such that for $l \in [2n]$, D_l is independent of $\{D_j\}_{j \notin \mathcal{N}_l}$, and*

$$|\mathcal{N}_l| \leq \bar{d} n^{\alpha}.$$

Assumption 1[†]-(i) is weaker than the standard assumptions that usually appear in the design based network interference literature (Aronow and Samii 2017). Assumption 1[†]-(ii) is a mild assumption to describe weakly dependent random variables (Ross 2011, Chin 2018). In the network interference setup, the maximum degree of the network affects both the magnitudes of the marginal probabilities under both exposure conditions, as reflected in Assumption 1[†]-(i), and the size of the dependency neighborhood, as reflected in Assumption 1[†]-(ii). We will explicitly specify the required upper bounds on α and β when we present our main results. We provide Examples 4 and 5 below to illustrate the above notations and assumptions.

Example 4 (SUTVA) *Let the exposure mapping be $f_i(\mathbf{w}) = w_i$. Let the potential outcomes under the two exposure conditions be $Y_i(\delta(1)) = Y_i(1)$ and $Y_i(\delta(0)) = Y_i(0)$. In this case, $\mathcal{E}_i(1) = \{W_i = 1\}$ and $\mathcal{E}_i(0) = \{W_i = 0\}$. The average treatment effect of exposure condition $\delta(1)$ relative to $\delta(0)$, given as $\tau(\delta(1), \delta(0))$, reduces to the average treatment effect $\tau = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))$. The covariance matrix $\mathbf{\Omega}$ is given by*

$$\mathbf{\Omega} = \begin{bmatrix} \mathbf{\Sigma} & -\mathbf{\Sigma} \\ -\mathbf{\Sigma} & \mathbf{\Sigma} \end{bmatrix}.$$

This covariance matrix $\mathbf{\Omega}$ has a special block structure, where each block can be expressed using the covariance matrix $\mathbf{\Sigma} = \text{Var}_{\mathcal{W}}(\mathbf{W})$, where $\mathbf{W}$ is a n -dimensional vector of treatment indicators.

When the experiment is a Bernoulli randomized experiment where, for each unit $i \in [n]$, we independently assign unit i into the treatment or the control group with probability $\frac{1}{2}$, this experimental design under SUTVA satisfies Assumption 1[†]-(i) because $\Pr_{\mathcal{W}}(\mathcal{E}_i(1)) = \frac{1}{2}$ and $\Pr_{\mathcal{W}}(\mathcal{E}_i(0)) = \frac{1}{2}$ for any $i \in [n]$, and satisfies Assumption 1[†]-(ii) because $|\mathcal{N}_i| = 1$ for any $i \in [n]$. See Example 7 in Section EC.2 for more details. $\square$

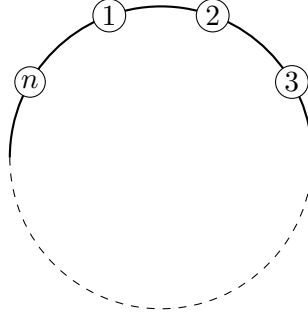


Figure 1 An illustration of a circle network.

Example 5 (Neighborhood Interference) Let there be a circle network as shown in Figure 1. For any $i \in [n]$, let $\mathcal{A}_i$ be the set of units that are adjacent to unit i , including unit i itself. For each unit $i \in [n]$, we independently assign unit i into the treatment or the control group with probability $\frac{1}{2}$. Let the exposure mapping be $f_i(\mathbf{w}) = \mathbf{w}_{\mathcal{A}_i}$, where $\mathbf{w}_{\mathcal{A}_i}$ stands for the sub-vector of treatment assignments for unit i and its adjacent units. Let the potential outcomes under the two exposure conditions be $Y_i(\delta(1)) = Y_i(\mathbf{1}_{\mathcal{A}_i})$ and $Y_i(\delta(0)) = Y_i(\mathbf{0}_{\mathcal{A}_i})$. In this case, $\mathcal{E}_i(1) = \{\mathbf{W}_{\mathcal{A}_i} = \mathbf{1}_{\mathcal{A}_i}\}$ and $\mathcal{E}_i(0) = \{\mathbf{W}_{\mathcal{A}_i} = \mathbf{0}_{\mathcal{A}_i}\}$. The average treatment effect of exposure condition $\delta(1)$ relative to $\delta(0)$, given as $\tau(\delta(1), \delta(0))$, reduces to the global average treatment effect $\tau^{\text{GATE}} = \frac{1}{n} \sum_{i=1}^n (Y_i(\mathbf{1}_{\mathcal{A}_i}) - Y_i(\mathbf{0}_{\mathcal{A}_i}))$. The covariance matrix $\mathbf{\Omega}$ is given by

$$\mathbf{\Omega} = \begin{bmatrix} \mathbf{\Omega}^{(1)} & \mathbf{\Omega}^{(2)} \\ \mathbf{\Omega}^{(2)} & \mathbf{\Omega}^{(1)} \end{bmatrix},$$

where

$$\mathbf{\Omega}^{(1)} = \begin{bmatrix} \frac{7}{64} & \frac{3}{64} & \frac{1}{64} & 0 & \cdots & \frac{1}{64} & \frac{3}{64} \\ \frac{3}{64} & \frac{7}{64} & \frac{3}{64} & \frac{1}{64} & \cdots & 0 & \frac{1}{64} \\ \frac{1}{64} & \frac{3}{64} & \frac{7}{64} & \frac{3}{64} & \cdots & 0 & 0 \\ 0 & \frac{1}{64} & \frac{3}{64} & \frac{7}{64} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{1}{64} & 0 & 0 & 0 & \cdots & \frac{7}{64} & \frac{3}{64} \\ \frac{3}{64} & \frac{1}{64} & 0 & 0 & \cdots & \frac{3}{64} & \frac{7}{64} \end{bmatrix}, \quad \mathbf{\Omega}^{(2)} = \begin{bmatrix} -\frac{1}{64} & -\frac{1}{64} & -\frac{1}{64} & 0 & \cdots & -\frac{1}{64} & -\frac{1}{64} \\ -\frac{1}{64} & -\frac{1}{64} & -\frac{1}{64} & -\frac{1}{64} & \cdots & 0 & -\frac{1}{64} \\ -\frac{1}{64} & -\frac{1}{64} & -\frac{1}{64} & -\frac{1}{64} & \cdots & 0 & 0 \\ 0 & -\frac{1}{64} & -\frac{1}{64} & -\frac{1}{64} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -\frac{1}{64} & 0 & 0 & 0 & \cdots & -\frac{1}{64} & -\frac{1}{64} \\ -\frac{1}{64} & -\frac{1}{64} & 0 & 0 & \cdots & -\frac{1}{64} & -\frac{1}{64} \end{bmatrix}.$$

Here, $\mathbf{\Omega}^{(1)}$ captures the covariance between exposure indicators under the same exposure condition, and $\mathbf{\Omega}^{(2)}$ captures the covariance between exposure indicators under opposite exposure conditions. This experimental design under neighborhood interference satisfies Assumption 1[†]-(i) because $\Pr_{\mathcal{W}}(\mathcal{E}_i(1)) = \frac{1}{8}$ and $\Pr_{\mathcal{W}}(\mathcal{E}_i(0)) = \frac{1}{8}$ for any $i \in [n]$, and satisfies Assumption 1[†]-(ii) because $|\mathcal{N}_i| = 10$ for any $i \in [n]$. $\square$

3.2. Control Variate Estimators for Causal Inference under Network Interference

Under Assumption 1[†]-(i), one of the most popular way of estimating $\tau(\delta(1), \delta(0))$ is to consider the Horvitz-Thompson estimator

$$\hat{\tau}^{\text{HT}} = \hat{\mu}^{\text{HT}}(\delta(1)) - \hat{\mu}^{\text{HT}}(\delta(0)) = \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i \mathbb{1}\{\mathcal{E}_i(1)\}}{\Pr(\mathcal{E}_i(1))} - \frac{Y_i \mathbb{1}\{\mathcal{E}_i(0)\}}{\Pr(\mathcal{E}_i(0))} \right).$$

It is easy to see that the Horvitz-Thompson estimator is unbiased, that is,

$$\mathbb{E}_{\mathcal{W}}[\hat{\tau}^{\text{HT}}] = \tau(\delta(1), \delta(0)).$$

The Horvitz-Thompson estimator under network interference is usually known to have a large variance, because the marginal probabilities $\Pr(\mathcal{E}_i(1))$ and $\Pr(\mathcal{E}_i(0))$ depend on the heterogeneous degrees of the network, and therefore they are usually heterogeneous and small. Again, we draw on the control variate techniques, take the baseline estimator $\hat{\tau}^{\text{HT}}$, and propose a family of control variate estimators to reduce the variance.

We consider the following control variate estimator using two control variates $\hat{X}(1)$ and $\hat{X}(0)$ and two coefficients $\gamma(1)$ and $\gamma(0)$,

$$\hat{\tau}^{\text{CV}}(\gamma(1), \gamma(0)) = \hat{\tau}^{\text{HT}} - \gamma(1) \left(\hat{X}(1) - \mathbb{E}[\hat{X}(1)] \right) - \gamma(0) \left(\hat{X}(0) - \mathbb{E}[\hat{X}(0)] \right).$$

When $\gamma(1)$ and $\gamma(0)$ are two constants, the control variate estimator is unbiased, that is,

$$\mathbb{E}_{\mathcal{W}}[\hat{\tau}^{\text{CV}}(\gamma(1), \gamma(0))] = \mathbb{E}_{\mathcal{W}}[\hat{\tau}^{\text{HT}}] - \gamma(1) \mathbb{E}_{\mathcal{W}}[\hat{X}(1) - \mathbb{E}[\hat{X}(1)]] - \gamma(0) \mathbb{E}_{\mathcal{W}}[\hat{X}(0) - \mathbb{E}[\hat{X}(0)]] = \tau.$$

If the coefficients $\gamma(1)$ and $\gamma(0)$ are selected properly, the control variate estimator has a smaller variance than the baseline Horvitz-Thompson estimator. It is easy to show that the variance minimizing coefficients $\gamma^*(1)$ and $\gamma^*(0)$ for any generic control variates $\hat{X}(1)$ and $\hat{X}(0)$ are given by

$$\begin{bmatrix} \gamma^*(1) \\ \gamma^*(0) \end{bmatrix} = \begin{bmatrix} \text{Var}(\hat{X}(1)) & \text{Cov}(\hat{X}(1), \hat{X}(0)) \\ \text{Cov}(\hat{X}(1), \hat{X}(0)) & \text{Var}(\hat{X}(0)) \end{bmatrix}^{-1} \cdot \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix}.$$

In the network interference setting, we propose a family of control variates given in the following form

$$\hat{X}(1) = \frac{1}{n} \sum_{i=1}^n a_i(1) \mathbb{1}\{\mathcal{E}_i(1)\}, \quad \hat{X}(0) = \frac{1}{n} \sum_{i=1}^n a_i(0) \mathbb{1}\{\mathcal{E}_i(0)\},$$

where $a_1(1), a_2(1), \dots, a_n(1)$ and $a_1(0), a_2(0), \dots, a_n(0)$ are any arbitrary constants that we can choose. We collect these constants into vector form $\mathbf{a}(1) = (a_1(1), a_2(1), \dots, a_n(1))^\top$ and $\mathbf{a}(0) = (a_1(0), a_2(0), \dots, a_n(0))^\top$ and refer to them as the bases of the control variates. The mean values of this family of control variates are given by $E_{\mathcal{W}}[\widehat{X}(1)] = \frac{1}{n} \sum_{i=1}^n a_i(1) \Pr_{\mathcal{W}}(\mathcal{E}_i(1))$ and $E_{\mathcal{W}}[\widehat{X}(0)] = \frac{1}{n} \sum_{i=1}^n a_i(0) \Pr_{\mathcal{W}}(\mathcal{E}_i(0))$, which are known because the experimental design $\mathcal{W}$ is known.

For this family of control variates, we can derive the variance minimizing coefficients $\gamma^*(1)$ and $\gamma^*(0)$ in Lemma 2. We present Lemma 2 using succinct notations. Recall the $2n$ -dimensional vector $\mathbf{Y}$, diagonal matrix $\mathbf{\Pi}$, and the covariance matrix $\mathbf{\Omega}$. We denote the $(2n \times 2)$ block diagonal bases matrix

$$\mathbf{B} = \begin{bmatrix} \mathbf{a}(1) & \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{a}(0) \end{bmatrix},$$

where $\mathbf{0}_n$ is a n -dimensional vector with each element equal to 0. Let $\mathcal{B}$ denote the set of all such block diagonal matrices. Using these notations, we introduce Lemma 2 as follows.

Lemma 2 (Variance Minimizing Coefficients) *In the network interference setting, under Assumptions 1[†]-(i) and 4, the variance minimizing coefficients $\gamma^*(1)$ and $\gamma^*(0)$ are given by*

$$(\gamma^*(1), \gamma^*(0))^\top = (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y}.$$

and the variance of the control variate estimator under these coefficients is given by

$$\text{Var}(\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) = \frac{1}{n^2} \left(\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y} - \mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y} \right).$$

The proof of Lemma 2 is given in Section EC.4. Intuitively, we try to use the randomness in two control variates $\widehat{X}(1)$ and $\widehat{X}(0)$ to explain as much randomness in $\widehat{\tau}^{\text{HT}}$ as possible. The variance $\text{Var}(\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$ is given by the residual from projecting the potential outcomes $\mathbf{\Pi}^{-1} \mathbf{Y}$ onto the column space of $\mathbf{B}$. Because the variance minimizing coefficients $\gamma^*(1)$ and $\gamma^*(0)$ depend on unknown outcomes $\mathbf{Y}$, we cannot directly observe $\gamma^*(1)$ and $\gamma^*(0)$; instead, we have to estimate $\gamma^*(1)$ and $\gamma^*(0)$ from the data.

3.3. Estimating the Variance Minimizing Coefficients under Network Interference

To estimate $\gamma^*(1)$ and $\gamma^*(0)$ from the data, we can replace the unknown potential outcomes by their Horvitz-Thompson estimates. Let $\widehat{Y}_i(\delta(1)) = Y_i \frac{1\{\mathcal{E}_i(1)\}}{\Pr(\mathcal{E}_i(1))}$ and $\widehat{Y}_i(\delta(0)) = Y_i \frac{1\{\mathcal{E}_i(0)\}}{\Pr(\mathcal{E}_i(0))}$ for $i \in [n]$. We collect $\widehat{\mathbf{Y}} = (\widehat{Y}_1(\delta(1)), \dots, \widehat{Y}_n(\delta(1)), -\widehat{Y}_1(\delta(0)), \dots, -\widehat{Y}_n(\delta(0)))$. Then the coefficients $(\gamma^*(1), \gamma^*(0))$ can be estimated using

$$(\widehat{\gamma}^{\text{HT}}(1), \widehat{\gamma}^{\text{HT}}(0))^\top = (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \widehat{\mathbf{Y}}.$$

Since $\hat{Y}_i(\delta(1))$ is an unbiased estimator of $Y_i(\delta(1))$ and $\hat{Y}_i(\delta(0))$ is an unbiased estimator of $Y_i(\delta(0))$, $(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ as defined above are also unbiased estimators of $(\gamma^*(1), \gamma^*(0))$.

Note that, $(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ obtained in this way are not constants, but random variables correlated with the control variates $\hat{X}(1)$ and $\hat{X}(0)$. Using the estimated coefficients $(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ introduces a bias to the control variate estimator. Nonetheless, we can show that the bias is small and that $\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ is asymptotically equivalent to $\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))$, as long as the bases matrix $\mathbf{B}$ satisfies certain regularity conditions. See Assumption 2[†] below.

Assumption 2[†] *The bases matrix $\mathbf{B}$ satisfies two conditions:*

(i) (**Boundedness**) *There exists a constant $\bar{a}$ such that for each $i \in [n]$,*

$$|a_i(1)| \leq \bar{a}, \quad |a_i(0)| \leq \bar{a}.$$

(ii) (**Non-degeneracy**) *Let $\lambda_{\min}(\cdot)$ be the smallest eigenvalue of a matrix. There exists a constant $\underline{\lambda}_{\Omega}$ such that*

$$\lambda_{\min}(\mathbf{B}^{\top} \mathbf{\Omega} \mathbf{B}) \geq \underline{\lambda}_{\Omega} n.$$

Assumption 2[†]-(i) is a standard boundedness assumption. Assumption 2[†]-(ii) is a standard non-degeneracy assumption on the control variates $\hat{X}(1)$ and $\hat{X}(0)$. Under Assumption 2[†], we establish Theorem 3 below.

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Theorem 3 *Under Assumptions 1[†] and 2[†] where in Assumption 1[†]-(ii) we assume $\alpha < \frac{1}{3}$, and assuming the potential outcomes $|Y_i(\delta(1))| \leq \bar{y}$ and $|Y_i(\delta(0))| \leq \bar{y}$ are all bounded, the control variate estimator using the estimated coefficients $\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ is a consistent estimator of $\tau(\delta(1), \delta(0))$, that is,*

$$\lim_{n \rightarrow +\infty} \hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) \xrightarrow{p} \tau(\delta(1), \delta(0)).$$

Additionally, if in Assumption 1[†]-(ii) we assume $\alpha < \frac{1}{4}$, then the asymptotic variance of the control variate estimator using the estimated coefficients $\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))$ is the same as the asymptotic variance of the control variate estimator using the optimal coefficients $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^(1), \gamma^*(0)))$, that is,*

$$\lim_{n \rightarrow +\infty} n \left(\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right) \rightarrow 0.$$

Additionally, if in Assumption 1[†]-(ii) we assume $\alpha < \frac{1}{7}$ and further assuming there exists a constant $\underline{c} > 0$ such that for sufficiently large n , $n \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^(1), \gamma^*(0))) \geq \underline{c}$, then the control variate estimator using the estimated coefficient $\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ is asymptotically normal, that is,*

$$\lim_{n \rightarrow +\infty} \frac{\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \tau(\delta(1), \delta(0))}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

The proof of Theorem 3 is given in Section EC.4. As we have stronger results, from consistency to asymptotic variance equivalence and to asymptotic normality, the requirement on the dependency neighborhoods becomes correspondingly stronger, with α strengthening from $\alpha < \frac{1}{3}$ to $\alpha < \frac{1}{4}$ and to $\alpha < \frac{1}{7}$. Taken together, Theorem 1 shows that estimating $\gamma^*(1)$ and $\gamma^*(0)$ by $\hat{\gamma}^{\text{HT}}(1)$ and $\hat{\gamma}^{\text{HT}}(0)$ does not affect the first-order asymptotic properties of the control variate estimator. The magnitude of variance reduction is determined by the choice of the bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$. In Section 3.4 below, we study how to choose the bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$ for the optimal variance reduction.

3.4. The Optimal Control Variates

Now we focus on finding the optimal bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$ that minimize $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$ the variance of the control variate estimator for fixed n . If the vector of potential outcomes $\mathbf{Y}(\delta(1))$ and $\mathbf{Y}(\delta(0))$ were known to take values $Y_i(\delta(1)) = y_i(\delta(1))$ and $Y_i(\delta(0)) = y_i(\delta(0))$ for any $i \in [n]$, then we denote $\mathbf{y} = (\mathbf{y}(\delta(1))^\top, -\mathbf{y}(\delta(0))^\top)^\top$. We see that the problem

$$\min_{\mathbf{B} \in \mathcal{B}} \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) = \min_{\mathbf{B} \in \mathcal{B}} \frac{1}{n^2} \left(\mathbf{y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{y} - \mathbf{y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{y} \right).$$

can be easily solved by choosing $\mathbf{B}$ such that $\mathbf{\Pi}^{-1} \mathbf{y}$ is in the column space of $\mathbf{B}$, such as $\mathbf{a}(1) = \mathbf{\Pi}(1)^{-1} \mathbf{y}(\delta(1))$ and $\mathbf{a}(0) = -\mathbf{\Pi}(0)^{-1} \mathbf{y}(\delta(0))$. In this case, the variance $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$ can be reduced to exactly zero.

However, the above choice of $\mathbf{B}$ is infeasible because the potential outcomes $\mathbf{Y}(\delta(1))$ and $\mathbf{Y}(\delta(0))$ are unknown. Therefore, we find the optimal bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$ under uncertainty of the potential outcomes, which leads to a decision making problem under uncertainty. We adopt a stochastic optimization perspective (Zhao 2024) and model the unknown outcomes $(Y_i(\delta(1)), Y_i(\delta(0)))$ to be independent and identically distributed (i.i.d.) samples from an unknown joint distribution $\mathcal{Y}_\delta$. See Assumption 3[†] below.

Assumption 3[†] *We assume that for each $i \in [n]$, the pair of potential outcomes $(Y_i(\delta(1)), Y_i(\delta(0)))$ are i.i.d. samples from an unknown joint distribution $\mathcal{Y}_\delta$, whose marginal distributions are denoted as $\mathcal{Y}_\delta(1)$ and $\mathcal{Y}_\delta(0)$.*

Denote the population means as $\mu(1) = \mathbb{E}_{Y_i(\delta(1)) \sim \mathcal{Y}_\delta(1)}[Y_i(\delta(1))]$ and $\mu(0) = \mathbb{E}_{Y_i(\delta(0)) \sim \mathcal{Y}_\delta(0)}[Y_i(\delta(0))]$, the population variances as $\sigma^2(1) = \text{Var}_{Y_i(\delta(1)) \sim \mathcal{Y}_\delta(1)}(Y_i(\delta(1)))$ and $\sigma^2(0) = \text{Var}_{Y_i(\delta(0)) \sim \mathcal{Y}_\delta(0)}(Y_i(\delta(0)))$, and the population covariance as $\sigma(1, 0) = \text{Cov}_{Y_i(\delta(1)), Y_i(\delta(0)) \sim \mathcal{Y}_\delta}(Y_i(\delta(1)), Y_i(\delta(0)))$. Denote $\mathcal{Y}_\delta^n$ to be the joint probability distribution of the n pairs of potential outcomes $\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0))$. Under Assumption 3[†], we formulate the following constrained stochastic optimization problem

$$\mathbf{B}^* \in \arg \min_{\mathbf{B} \in \mathcal{B}} \mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right]. \quad (6)$$

The optimal control variates use the bases that solve (6), which is given by Theorem 4 below. Recall that $\mathbf{\Omega}$ is symmetric and positive semidefinite, we can define the square root matrix $\mathbf{\Omega}^{\frac{1}{2}}$ and the pseudo-inverse square root matrix $\mathbf{\Omega}^{-\frac{1}{2}}$.

Theorem 4 (Optimal Control Variates) *For notational simplicity, we reload the notation $\mathbf{M}$ and define $\mathbf{M}$ to be a $2n \times 2n$ symmetric matrix*

$$\mathbf{M} = \mathbf{\Omega}^{\frac{1}{2}} \mathbf{\Pi}^{-1} \begin{bmatrix} \mu^2(1)\mathbf{1}\mathbf{1}^\top + \sigma^2(1)\mathbf{I}_n & -\mu(1)\mu(0)\mathbf{1}\mathbf{1}^\top - \sigma(1,0)\mathbf{I}_n \\ -\mu(1)\mu(0)\mathbf{1}\mathbf{1}^\top - \sigma(1,0)\mathbf{I}_n & \mu^2(0)\mathbf{1}\mathbf{1}^\top + \sigma^2(0)\mathbf{I}_n \end{bmatrix} \mathbf{\Pi}^{-1} \mathbf{\Omega}^{\frac{1}{2}},$$

where $\mathbf{I}_n$ is a $(n \times n)$ identity matrix and $\mathbf{1}$ is a n -dimensional vector with each element equal to 1. Let $\lambda_l(\mathbf{M})$ be the l -th largest eigenvalue of $\mathbf{M}$, and let $\mathbf{u}_l(\mathbf{M})$ be the principal eigenvector that corresponds to $\lambda_l(\mathbf{M})$. Under Assumptions 1[†]-(i), 3[†], and 4, the optimal bases are given by

$$\mathbf{B}^* \in \arg \max_{\mathbf{B} \in \mathcal{B}} \text{Tr} \left((\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega}^{\frac{1}{2}} \mathbf{M} \mathbf{\Omega}^{\frac{1}{2}} \mathbf{B} \right). \quad (7)$$

And in expectation, the optimal variance reduction can be upper bounded by

$$\mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\text{Var}(\hat{\tau}^{\text{HT}}) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right] \leq \frac{\lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M})}{n^2}.$$

The proof of Theorem 4 is given in Section EC.4. Theorem 4 involves solving a non-convex maximization problem (7), for which a closed-form solution is generally unavailable. In Section 3.6, we discuss how to compute the near-optimal control variates. We present a $\frac{1}{2}$ -approximate solution for (7), and then use its solution to initialize an alternating local search heuristic, which monotonically improves the objective and converges to a local maximum.

Similar to Theorems 2 and EC.2, Theorem 4 also combines a design-based perspective and a model-based perspective, where we assume a model for the unknown potential outcomes $\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0))$ to guide the choice of bases. But we rely on the experimental design as the only source of randomness when we estimate the average treatment effect. We point it out that Assumption 3[†] is only required to show optimality of the bases $\mathbf{B}^*$. The basis $\mathbf{B}^*$ is still well defined without Assumption 3[†].

In Theorem 4, the optimal bases $\mathbf{B}^*$ depend on the unknown moments of the joint distribution $\mathcal{Y}_\delta$. In practice, we may use domain knowledge to estimate the moments of the joint distribution $\mathcal{Y}_\delta$, and then use them to find the optimal bases $\mathbf{B}^*$. Even an imperfect guess of the moments will yield a set of bases that can still reduce the variance over the baseline Horvitz-Thompson estimator.

Theorem 4 in the network interference setting nests Theorem EC.2 in the SUTVA setting as a special case. See Example 6 below.

Example 6 (SUTVA, Example 4 Continued) *In Example 4 we show that the network interference setting nests the SUTVA setting, and Ω has a special block structure under SUTVA,*

$$\Omega = \begin{bmatrix} \Sigma & -\Sigma \\ -\Sigma & \Sigma \end{bmatrix}. \quad (8)$$

First, we compare Lemmas 2 and EC.1. Note that there is a one-to-one correspondence between a set of bases in the network interference setting $\mathbf{B} = \begin{bmatrix} \mathbf{a}(1) & \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{a}(0) \end{bmatrix}$ and a set of bases in the SUTVA setting $\mathbf{A} = [\mathbf{a}(1), -\mathbf{a}(0)]$. So we have $\mathbf{B}^\top \Omega \mathbf{B} = \mathbf{A}^\top \Sigma \mathbf{A}$. Next, recall that $\mathbf{G} = \Pi(1)^{-1} \mathbf{Y}(1) + \Pi(0)^{-1} \mathbf{Y}(0)$. So we have $\mathbf{B}^\top \Omega \Pi^{-1} \mathbf{Y} = \mathbf{A}^\top \Sigma \mathbf{G}$. Therefore, Lemma 2 in the network interference setting nests Lemma EC.1 in the SUTVA setting as a special case.

Second, we compare Theorems 4 and EC.2. In the general network interference setting, the block diagonal constraint $\mathbf{B} \in \mathcal{B}$ can make the optimization problem harder than the unconstrained problem. Under SUTVA, however, the special block structure in (8) makes this constraint nonbinding: for any $2n \times 2$ bases matrix, we can explicitly construct a bases matrix satisfying the block diagonal constraint $\mathbf{B} \in \mathcal{B}$ that attains exactly the same objective value. See Section EC.4 for details. This shows that Theorem 4 in the network interference setting nests Theorem EC.2 in the SUTVA setting as a special case. $\square$

To conclude this section, we consider a remarkable special case of Theorem 4 when $\sigma^2(1) = \sigma^2(0) = \sigma(1, 0) = 0$.

Corollary 2 (Noiseless Potential Outcomes under Network Interference) *Under*

Assumptions 1[†]-(i) and 4, and assuming that the potential outcomes $Y_i(\delta(1)) = y(\delta(1))$ and $Y_i(\delta(0)) = y(\delta(0))$ take the same unknown constants, then one set of optimal bases is given, for any $i \in [n]$, by

$$a_i(1) = \frac{1}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))}, \quad \text{and} \quad a_i(0) = -\frac{1}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))}.$$

The proof of Corollary 2 is given in Section EC.4. In this special case when all potential outcomes take the same two constants, the only fluctuation of the Horvitz-Thompson estimator comes from the random components $\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(1)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))}$ and $\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(0)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))}$. Corollary 2 suggests to correct it by using the exact same terms $\hat{X}(1) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(1)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))}$ and $\hat{X}(0) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(0)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))}$ as the control variates. This draws a connection between the control variate estimator and the Hajek estimator, which we will discuss in Section 3.5 below.

3.5. Connections to Hajek Estimator

Hajek Estimator. In the network interference setup, the Hajek estimator is defined as

$$\hat{\tau}^{\text{Hajek}}(\delta(1), \delta(0)) = \frac{\hat{\mu}^{\text{HT}}(\delta(1))}{\hat{1}(\delta(1))} - \frac{\hat{\mu}^{\text{HT}}(\delta(0))}{\hat{1}(\delta(0))},$$

where

$$\hat{1}(\delta(1)) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(1)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))}, \quad \hat{1}(\delta(0)) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(0)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))}.$$

The Hajek estimator in the network interference setup simply combines two Hajek estimators. Following similar discussions as in Section 2.5, we make a first order approximation by using Taylor expansion around $\hat{1}(\delta(1)) \approx 1$ and $\hat{1}(\delta(0)) \approx 1$, which gives

$$\begin{aligned} \hat{\tau}^{\text{Hajek}}(\delta(1), \delta(0)) &\approx \hat{\tau}^{\text{HT}}(\delta(1), \delta(0)) \\ &\quad - \hat{\mu}^{\text{HT}}(\delta(1)) \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(1)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))} - 1 \right) + \hat{\mu}^{\text{HT}}(\delta(0)) \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(0)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))} - 1 \right). \end{aligned}$$

The first order approximation of the Hajek estimator can be viewed as a control variate estimator using the bases $a_i(1) = \frac{1}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))}$ and $a_i(0) = -\frac{1}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))}$, $\forall i \in [n]$.

The variance difference between the control variate estimator and the Hajek estimator is especially significant under network interference. This is because the two control variates $\hat{1}(\delta(1)) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(1)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))}$ and $\hat{1}(\delta(0)) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{\mathcal{E}_i(0)\}}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))}$ can be correlated because the exposure events $\mathcal{E}_i(1)$ and $\mathcal{E}_i(0)$ are induced by both the same treatment assignment vector $\mathbf{W}$ and the same underlying network structure. The Hajek estimator implicitly chooses its coefficients by viewing the two control variates $\hat{1}(\delta(1))$ and $\hat{1}(\delta(0))$ separately, whereas the optimal control variate coefficients account for the correlation between them. Consequently, even when restricted to the same control variates $\hat{1}(\delta(1))$ and $\hat{1}(\delta(0))$ as used by the Hajek estimator, the the optimal control variate estimator can achieve additional variance reduction by accounting for the cross exposure correlation.

3.6. Computation of Optimal Control Variates

We now discuss the computation of the optimal control variates, which solves the constrained maximization problem (7). For convenience, we restate the problem below,

$$\mathbf{B}^* \in \arg \max_{\mathbf{B} \in \mathcal{B}} \text{Tr} \left((\mathbf{B}^\top \boldsymbol{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \boldsymbol{\Omega}^{\frac{1}{2}} \mathbf{M} \boldsymbol{\Omega}^{\frac{1}{2}} \mathbf{B} \right). \quad (7)$$

First, we propose a $\frac{1}{2}$ -approximate solution for (7). Second, we propose an alternating local search heuristic. We use the $\frac{1}{2}$ -approximate solution to initialize the alternating local search heuristic, which monotonically improves the objective and converges to a local maximum.

Algorithm 1 An Alternating Local Search Heuristic

Input: Numerical tolerance ϵ .

```

1: Initialize:  $k \leftarrow 1$ ;  $\mathbf{a}^0(1) \leftarrow \tilde{\mathbf{b}}_1$ ;  $\mathbf{a}^0(0) \leftarrow \tilde{\mathbf{b}}_2$ ;  $\triangleright \tilde{\mathbf{b}}_1$  and  $\tilde{\mathbf{b}}_2$  are obtained in (9)
2: for  $k = 1, 2, \dots$  do
3:    $\mathbf{a}^k(1) \leftarrow \arg \max_{\mathbf{a}(1)} \psi(\mathbf{a}(1), \mathbf{a}^{k-1}(0))$ ;
4:    $\mathbf{a}^k(0) \leftarrow \arg \max_{\mathbf{a}(0)} \psi(\mathbf{a}^k(1), \mathbf{a}(0))$ ;
5:   if  $\psi(\mathbf{a}^k(1), \mathbf{a}^k(0)) - \psi(\mathbf{a}^{k-1}(1), \mathbf{a}^{k-1}(0)) \leq \epsilon$  then
6:     break
7: return  $(\mathbf{a}^k(1), \mathbf{a}^k(0))$ 

```

A $\frac{1}{2}$ -Approximate Solution. We present a $\frac{1}{2}$ -approximate solution for problem (7). We first consider a rank-1 problem and use its solution to construct a feasible solution for (7). The rank-1 problem is as follows,

$$\max_{\mathbf{b} \in \mathcal{B}} \frac{\mathbf{b}^\top \Omega^{\frac{1}{2}} M \Omega^{\frac{1}{2}} \mathbf{b}}{\mathbf{b}^\top \Omega \mathbf{b}}.$$

Suppose $\tilde{\mathbf{b}} = (\tilde{\mathbf{b}}_1^\top, \tilde{\mathbf{b}}_2^\top)^\top$ is one optimal solution to the above rank-1 problem. Then we pick the bases to be $\mathbf{a}(1) = \tilde{\mathbf{b}}_1$ and $\mathbf{a}(0) = \tilde{\mathbf{b}}_2$, which gives a feasible solution

$$\tilde{\mathbf{B}} = \begin{bmatrix} \tilde{\mathbf{b}}_1 & \mathbf{0}_n \\ \mathbf{0}_n & \tilde{\mathbf{b}}_2 \end{bmatrix}. \quad (9)$$

We show that $\tilde{\mathbf{B}}$ as constructed in (9) above is a $\frac{1}{2}$ -approximate solution.

Theorem 5 ($\frac{1}{2}$ -Approximation) $\tilde{\mathbf{B}}$ is a $\frac{1}{2}$ -approximate solution to (7), that is,

$$\text{Tr} \left((\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-1} \tilde{\mathbf{B}}^\top \Omega^{\frac{1}{2}} M \Omega^{\frac{1}{2}} \tilde{\mathbf{B}} \right) \geq \frac{1}{2} \max_{\mathbf{B} \in \mathcal{B}} \text{Tr} \left((\mathbf{B}^\top \Omega \mathbf{B})^{-1} \mathbf{B}^\top \Omega^{\frac{1}{2}} M \Omega^{\frac{1}{2}} \mathbf{B} \right).$$

An Alternating Local Search Heuristic. We present an alternating local search heuristic that takes the $\frac{1}{2}$ -approximate solution $\tilde{\mathbf{B}}$ to initialize. For notational convenience, denote

$$\psi(\mathbf{b}_1, \mathbf{b}_2) = \text{Tr} \left((\mathbf{B}^\top \Omega \mathbf{B})^{-1} \mathbf{B}^\top \Omega^{\frac{1}{2}} M \Omega^{\frac{1}{2}} \mathbf{B} \right), \quad \text{where} \quad \mathbf{B} = \begin{bmatrix} \mathbf{b}_1 & \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{b}_2 \end{bmatrix},$$

to emphasize the dependence on the two directions. Using this notation, we present the heuristic in Algorithm 1.

The alternating local search heuristic alternates between the two directions $\mathbf{a}(1)$ and $\mathbf{a}(0)$ to monotonically improve the objective. We will show in Theorem 6 below that the objective value converges. On the other hand, the solutions may not converge. Note that any scaling does not change the objective of (7), that is, for any $c_1, c_0 \neq 0$, $\psi(\mathbf{b}_1, \mathbf{b}_2) = \psi(c_1 \mathbf{b}_1, c_0 \mathbf{b}_2)$. Therefore, convergence of the optimal solutions requires an additional normalization as well as a spectral gap assumption to avoid multiple local optimal solutions.

Theorem 6 (Monotone Convergence) Let $\{(\mathbf{a}^k(1), \mathbf{a}^k(0))\}_{k \geq 0}$ be the sequence generated by Algorithm 1 without the stopping rule. Then the objective values $\psi^k := \psi(\mathbf{a}^k(1), \mathbf{a}^k(0))$ are non-decreasing and converge, that is, there exists $\psi^\infty \in \mathbb{R}$ such that

$$\lim_{k \rightarrow +\infty} \psi^k = \psi^\infty.$$

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Online Appendix

EC.1. Additional Matrix Definitions

For any symmetric and positive semidefinite matrix $\mathbf{E}$, it has an eigen-decomposition in the form of

$$\mathbf{E} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1},$$

where $\mathbf{\Lambda} = \mathbf{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ is a diagonal matrix consisting of all the n eigenvalues and each eigenvalue $\lambda_i \geq 0$ is non-negative, and $\mathbf{U}$ is an orthonormal matrix. Using the above eigen-decomposition, define the square root matrix as

$$\mathbf{E}^{\frac{1}{2}} = \mathbf{U}\mathbf{\Lambda}^{\frac{1}{2}}\mathbf{U}^{-1}$$

where $\mathbf{\Lambda}^{\frac{1}{2}} = \mathbf{diag}(\lambda_1^{\frac{1}{2}}, \lambda_2^{\frac{1}{2}}, \dots, \lambda_n^{\frac{1}{2}})$. We have $\mathbf{E}^{\frac{1}{2}}\mathbf{E}^{\frac{1}{2}} = \mathbf{E}$.

For any symmetric and positive semidefinite matrix $\mathbf{E}$, define the pseudo-inverse matrix as

$$\mathbf{E}^{-1} = \mathbf{U}\mathbf{\Lambda}^{-1}\mathbf{U}^{-1}$$

where $\mathbf{\Lambda}^{-1} = \mathbf{diag}(\eta_1, \eta_2, \dots, \eta_n)$ is a diagonal matrix where each element η_i is defined as

$$\eta_i = \begin{cases} \lambda_i^{-1}, & \text{if } \lambda_i > 0, \\ 0, & \text{if } \lambda_i = 0. \end{cases}$$

We have $\mathbf{E}^{-1}\mathbf{E} = \mathbf{U}\mathbb{1}^+\mathbf{U}^{-1}$, where $\mathbb{1}^+ = \mathbf{diag}(\mathbb{1}\{\lambda_1 > 0\}, \mathbb{1}\{\lambda_2 > 0\}, \dots, \mathbb{1}\{\lambda_n > 0\})$ is a diagonal matrix and λ_i stands for the i -th largest eigenvalue of matrix $\mathbf{E}$.

For any symmetric and positive semidefinite matrix $\mathbf{E}$, define the pseudo-inverse square root matrix as

$$\mathbf{E}^{-\frac{1}{2}} = \mathbf{U}\mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{U}^{-1}$$

where $\mathbf{\Lambda}^{-\frac{1}{2}} = \mathbf{diag}(\eta_1, \eta_2, \dots, \eta_n)$ is a diagonal matrix where each element η_i is defined as

$$\eta_i = \begin{cases} \lambda_i^{-\frac{1}{2}}, & \text{if } \lambda_i > 0, \\ 0, & \text{if } \lambda_i = 0. \end{cases}$$

We have $\mathbf{E}^{-\frac{1}{2}}\mathbf{E}^{\frac{1}{2}} = \mathbf{U}\mathbb{1}^+\mathbf{U}^{-1}$, where $\mathbb{1}^+ = \mathbf{diag}(\mathbb{1}\{\lambda_1 > 0\}, \mathbb{1}\{\lambda_2 > 0\}, \dots, \mathbb{1}\{\lambda_n > 0\})$ is a diagonal matrix and λ_i stands for the i -th largest eigenvalue of matrix $\mathbf{E}$.

EC.2. Causal Inference under SUTVA

EC.2.1. Experimental Design

We consider the following causal inference problem. Let there be n units in a finite population. Let there be two versions of treatments. We use “treatment” and “control,” or 1 and 0, respectively, to stand for these two versions of treatments. For each $i \in [n]$, let $W_i \in \{0, 1\}$ stand for the treatment assignment that unit i receives. We consider an experimental design setting where we determine the random treatment assignments $\mathbf{W} = (W_1, \dots, W_n)^\top$ through a known joint probability distribution $\mathcal{W}$. Following convention, we use W_i for a random treatment assignment, and w_i for one realization.

Following the potential outcomes framework (Neyman 1923, Rubin 1974) and under the Stable Unit Treatment Value Assumption (SUTVA, Rubin 1980, Holland 1986), each unit i has a pair of potential outcomes $(Y_i(1), Y_i(0))$. We collect them into vector form $\mathbf{Y}(1) = (Y_1(1), Y_2(1), \dots, Y_n(1))^\top$ and $\mathbf{Y}(0) = (Y_1(0), Y_2(0), \dots, Y_n(0))^\top$. Each observed outcome is related to its respective potential outcomes through

$$Y_i = \begin{cases} Y_i(1), & \text{if } W_i = 1, \\ Y_i(0), & \text{if } W_i = 0. \end{cases}$$

We are interested in estimating the average treatment effect of the finite population,

$$\tau = \mu(1) - \mu(0) = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)). \quad (\text{EC.1})$$

In this paper, we take the design of experiment $\mathcal{W}$ as given. Denote the vector of treatment indicators as $\mathbf{D} = (\mathbb{1}\{W_1 = 1\}, \dots, \mathbb{1}\{W_n = 1\})^\top$. Denote the marginal treatment probabilities as $\pi_i = \Pr_{\mathcal{W}}(W_i = 1)$ for any $i \in [n]$. Denote the following two diagonal matrices as $\mathbf{\Pi}(1) = \mathbf{diag}(\pi_1, \pi_2, \dots, \pi_n)$ and $\mathbf{\Pi}(0) = \mathbf{diag}(1 - \pi_1, 1 - \pi_2, \dots, 1 - \pi_n)$, which reflect the marginal probabilities of $\mathbf{D}$. Denote the covariance matrix as $\mathbf{\Sigma} = \text{Var}_{\mathcal{W}}(\mathbf{D})$. We consider designs of experiments $\mathcal{W}$ that satisfy Assumption 1* below.

Assumption 1* *The design of experiment $\mathcal{W}$ satisfies:*

- (i) **(Positivity)** *The marginal treatment probabilities are all bounded away from zero and one, that is, there exists a constant $\underline{\pi} > 0$ such that for $i \in [n]$,*

$$\underline{\pi} \leq \pi_i \leq 1 - \underline{\pi}.$$

- (ii) **(Dependency Neighborhood)** *The correlation between treatment indicators are restricted to some neighborhoods, that is, there exist constants $\bar{d} > 0$ and $0 \leq \alpha < 1$ such that for $i \in [n]$,*

$$|\mathcal{N}_i| \leq \bar{d}n^\alpha.$$

Assumption 1^{*}-(i) is a standard assumption in the design based causal inference literature (Ding 2023, Imbens and Rubin 2015, Wager 2020). Assumption 1-(ii) is a mild assumption to describe weakly dependent random variables (Ross 2011). We will explicitly specify the required upper bound on α when we describe our main results. This assumption is satisfied by many common experimental designs. We provide Example 7 below to illustrate the above notations and assumptions.

Example 7 (Bernoulli Randomized Experiments) *For each unit $i \in [n]$, we independently assign unit i into treatment or control with probability π_i or $1 - \pi_i$, respectively. The covariance matrix Σ is given by*

$$\Sigma = \begin{bmatrix} \pi_1(1 - \pi_1) & 0 & \dots & 0 \\ 0 & \pi_2(1 - \pi_2) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \pi_n(1 - \pi_n) \end{bmatrix}.$$

This is exactly the same covariance matrix that we have seen in Example 1. This experimental design satisfies Assumption 1^{}-(ii) because for $i \in [n]$, $|\mathcal{N}_i| = 1$. $\square$*

EC.2.2. Control Variate Estimators for Causal Inference under SUTVA

Under Assumption 1^{*}-(i), one of the most popular way of estimating τ is to consider the Horvitz-Thompson estimator

$$\hat{\tau}^{\text{HT}} = \hat{\mu}^{\text{HT}}(1) - \hat{\mu}^{\text{HT}}(0) = \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i \mathbb{1}\{W_i = 1\}}{\pi_i} - \frac{Y_i \mathbb{1}\{W_i = 0\}}{1 - \pi_i} \right).$$

It is easy to see that the Horvitz-Thompson estimator is unbiased, that is,

$$\mathbb{E}_{\mathcal{W}}[\hat{\tau}^{\text{HT}}] = \tau.$$

Similar to survey sampling, the Horvitz-Thompson estimator for causal inference is sometimes known to have a large variance, especially when the marginal probabilities π_i are heterogeneous and when some of them are close to 0 or 1. Again, we draw on the control variate techniques, take the baseline estimator $\hat{\tau}^{\text{HT}}$, and propose a family of control variate estimators to reduce the variance.

We consider the following control variate estimator using two control variates $\hat{X}(1)$ and $\hat{X}(0)$ and two coefficients $\gamma(1)$ and $\gamma(0)$,

$$\hat{\tau}^{\text{CV}}(\gamma(1), \gamma(0)) = \hat{\tau}^{\text{HT}} - \gamma(1) \left(\hat{X}(1) - \mathbb{E}[\hat{X}(1)] \right) - \gamma(0) \left(\hat{X}(0) - \mathbb{E}[\hat{X}(0)] \right).$$

When $\gamma(1)$ and $\gamma(0)$ are two constant coefficients, the control variate estimator is unbiased, that is,

$$\mathbb{E}_{\mathcal{W}}[\hat{\tau}^{\text{CV}}(\gamma(1), \gamma(0))] = \mathbb{E}_{\mathcal{W}}[\hat{\tau}^{\text{HT}}] - \gamma(1)\mathbb{E}_{\mathcal{W}}[\hat{X}(1) - \mathbb{E}[\hat{X}(1)]] - \gamma(0)\mathbb{E}_{\mathcal{W}}[\hat{X}(0) - \mathbb{E}[\hat{X}(0)]] = \tau.$$

If the coefficients $\gamma(1)$ and $\gamma(0)$ are selected properly, the control variate estimator has a smaller variance than the baseline Horvitz-Thompson estimator. It is easy to show that the variance minimizing coefficients $\gamma^*(1)$ and $\gamma^*(0)$ for any generic control variates $\hat{X}(1)$ and $\hat{X}(0)$ are given by

$$\begin{bmatrix} \gamma^*(1) \\ \gamma^*(0) \end{bmatrix} = \begin{bmatrix} \text{Var}(\hat{X}(1)) & \text{Cov}(\hat{X}(1), \hat{X}(0)) \\ \text{Cov}(\hat{X}(1), \hat{X}(0)) & \text{Var}(\hat{X}(0)) \end{bmatrix}^{-1} \cdot \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix}. \quad (\text{EC.2})$$

Similar to survey sampling, we propose a family of control variates given in the following form

$$\hat{X}(1) = \frac{1}{n} \sum_{i=1}^n a_i(1) \mathbb{1}\{W_i = 1\}, \quad \hat{X}(0) = \frac{1}{n} \sum_{i=1}^n a_i(0) \mathbb{1}\{W_i = 0\}, \quad (\text{EC.3})$$

where $a_1(1), a_2(1), \dots, a_n(1)$ and $a_1(0), a_2(0), \dots, a_n(0)$ are any arbitrary constants that we can choose. We collect these constants into vector form $\mathbf{a}(1) = (a_1(1), a_2(1), \dots, a_n(1))^\top$ and $\mathbf{a}(0) = (a_1(0), a_2(0), \dots, a_n(0))^\top$ and refer to them as the bases of the control variates. The mean values of this family of control variates are given by $\mathbb{E}_{\mathcal{W}}[\hat{X}(1)] = \frac{1}{n} \sum_{i=1}^n a_i(1) \pi_i$ and $\mathbb{E}_{\mathcal{W}}[\hat{X}(0)] = \frac{1}{n} \sum_{i=1}^n a_i(0) (1 - \pi_i)$ that are usually known in advance.

Similar to Lemma 1, for this family of control variates in (EC.3), we can derive the variance minimizing coefficients $\gamma^*(1)$ and $\gamma^*(0)$ in Lemma EC.1. We present Lemma EC.1 using succinct notations. Recall the diagonal matrices $\mathbf{\Pi}(1), \mathbf{\Pi}(0)$ and the covariance matrix $\mathbf{\Sigma}$. Denote the baseline outcomes $\mathbf{G} = \mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0)$ as similarly defined in Bai (2022). Denote $\mathbf{A} = [\mathbf{a}(1), -\mathbf{a}(0)]$ to be a $(n \times 2)$ bases matrix. Because $\mathbf{\Sigma}$ is a symmetric and positive semidefinite matrix, $\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A}$ is also a symmetric and positive semidefinite matrix so its pseudo inverse matrix $(\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1}$ exists. Using these notations, we introduce Lemma EC.1 as follows.

Lemma EC.1 (Variance Minimizing Coefficients) *In the causal inference setting under SUTVA, under Assumption 1*-(i), the variance minimizing coefficients $\gamma^*(1)$ and $\gamma^*(0)$ are given by*

$$(\gamma^*(1), \gamma^*(0))^\top = (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{\Sigma} \mathbf{G}.$$

and the variance of the control variate estimator under these coefficients is given by

$$\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) = \frac{1}{n^2} \left(\mathbf{G}^\top \mathbf{\Sigma} \mathbf{G} - \mathbf{G}^\top \mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{\Sigma} \mathbf{G} \right).$$

The proof of Lemma EC.1 is given in Section EC.4. Intuitively, we try to use the randomness in two control variates $\widehat{X}(1)$ and $\widehat{X}(0)$ to explain as much randomness in $\widehat{\tau}^{\text{HT}}$ as possible. The variance $\text{Var}(\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$ is given by the residual from projecting the baseline outcomes $\mathbf{G}$ onto the column space of $\mathbf{A} = [\mathbf{a}(1), -\mathbf{a}(0)]$. So the magnitude of variance reduction is determined by the choice of the bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$. In Section EC.2.4 below, we study how to choose the bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$ for the optimal variance reduction.

EC.2.3. Estimating the Variance Minimizing Coefficients under SUTVA

To estimate $(\gamma^*(1), \gamma^*(0))$ from the data, we can replace the unknown potential outcomes by their Horvitz-Thompson estimates. Let $\widehat{G}_i = Y_i \left(\frac{\mathbb{1}\{W_i=1\}}{\pi_i^2} + \frac{\mathbb{1}\{W_i=0\}}{(1-\pi_i)^2} \right)$ and collect $\widehat{\mathbf{G}} = (\widehat{G}_1, \dots, \widehat{G}_n)$. Then the coefficients $(\gamma^*(1), \gamma^*(0))$ can be estimated using

$$(\widehat{\gamma}^{\text{HT}}(1), \widehat{\gamma}^{\text{HT}}(0))^\top = (\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A})^{-1} \mathbf{A}^\top \boldsymbol{\Sigma} \widehat{\mathbf{G}}.$$

Since $\widehat{G}_i$ is an unbiased estimator of G_i , $(\widehat{\gamma}^{\text{HT}}(1), \widehat{\gamma}^{\text{HT}}(0))$ as defined above are also unbiased estimators of $(\gamma^*(1), \gamma^*(0))$.

Note that, $(\widehat{\gamma}^{\text{HT}}(1), \widehat{\gamma}^{\text{HT}}(0))$ obtained in this way are not constants, but random variables correlated with the control variates $\widehat{X}(1)$ and $\widehat{X}(0)$. Using the estimated coefficients $(\widehat{\gamma}^{\text{HT}}(1), \widehat{\gamma}^{\text{HT}}(0))$ introduces a bias to the control variate estimator. Nonetheless, we can show that the bias is small and that $\widehat{\tau}^{\text{CV}}(\widehat{\gamma}^{\text{HT}}(1), \widehat{\gamma}^{\text{HT}}(0))$ is asymptotically equivalent to $\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))$, as long as the bases matrix $\mathbf{A} = [\mathbf{a}(1), -\mathbf{a}(0)]$ satisfies certain regularity conditions. See Assumption 2* below.

Assumption 2* *The bases matrix $\mathbf{A} = [\mathbf{a}(1), -\mathbf{a}(0)]$ satisfies two conditions:*

(i) **(Boundedness)** *There exists a constant $\bar{a}$ such that for each $i \in [n]$,*

$$|a_i(1)| \leq \bar{a}, \quad |a_i(0)| \leq \bar{a}.$$

(ii) **(Non-degeneracy)** *Let $\lambda_{\min}(\cdot)$ be the smallest eigenvalue of a matrix. There exists a constant $\underline{\lambda}_\Sigma$ such that*

$$\lambda_{\min}(\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A}) \geq \underline{\lambda}_\Sigma n.$$

Assumption 2*-(i) is a standard boundedness assumption. Assumption 2*-(ii) is a standard non-degeneracy assumption on the control variates $\widehat{X}(1)$ and $\widehat{X}(0)$. Denote $\widehat{\mathbf{X}} = (\widehat{X}(1), \widehat{X}(0))^\top$. Note that we have $\text{Var}_{\mathcal{W}}(\widehat{\mathbf{X}}) = n^{-2} \mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A}$, and therefore Assumption 2*-(ii) is equivalent to requiring that $\text{Var}(\widehat{\mathbf{X}}) \succeq \underline{\lambda}_\Sigma n^{-1} \mathbf{I}_2$. Both assumptions are commonly made in the asymptotic statistics literature (Van Der Vaart and Wellner 1996, Wooldridge 2010). Under Assumption 2*, we establish Theorem EC.1 below.

Theorem EC.1 Under Assumptions 1* and 2* where in Assumption 1*-(ii) we assume $\alpha < \frac{1}{3}$, and assuming the potential outcomes $|Y_i(1)| \leq \bar{y}$ and $|Y_i(0)| \leq \bar{y}$ are all bounded, the control variate estimator using the estimated coefficients $\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ is a consistent estimator of τ , that is,

$$\lim_{n \rightarrow +\infty} \hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) \xrightarrow{p} \tau.$$

Additionally, if in Assumption 1*-(ii) we assume $\alpha < \frac{1}{4}$, then the asymptotic variance of the control variate estimator using the estimated coefficients $\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))$ is the same as the asymptotic variance of the control variate estimator using the optimal coefficients $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$, that is,

$$\lim_{n \rightarrow +\infty} n \left(\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right) \rightarrow 0.$$

Additionally, if in Assumption 1*-(ii) we assume $\alpha < \frac{1}{7}$ and further assuming there exists a constant $\underline{c} > 0$ such that for sufficiently large n , $n \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c}$, then the control variate estimator using the estimated coefficient $\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ is asymptotically normal, that is,

$$\lim_{n \rightarrow +\infty} \frac{\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \tau}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

The proof of Theorem EC.1 is given in Section EC.4. As we have stronger results, from consistency to asymptotic variance equivalence and to asymptotic normality, the requirement on the dependency neighborhoods becomes correspondingly stronger, with α strengthening from $\alpha < \frac{1}{3}$ to $\alpha < \frac{1}{4}$ and to $\alpha < \frac{1}{7}$. Taken together, Theorem EC.1 shows that estimating the coefficients $\gamma^*(1)$ and $\gamma^*(0)$ by $\hat{\gamma}^{\text{HT}}(1)$ and $\hat{\gamma}^{\text{HT}}(0)$ does not affect the first-order asymptotic properties of the control variate estimator. In particular, estimating $\gamma^*(1)$ and $\gamma^*(0)$ by $\hat{\gamma}^{\text{HT}}(1)$ and $\hat{\gamma}^{\text{HT}}(0)$ leads to the same variance reduction asymptotically, as given by Lemma EC.1. The magnitude of variance reduction is determined by the alignment between the baseline outcomes $\mathbf{G}$ and the basis $\mathbf{A}$. In Section EC.2.4 below, we study how to choose a basis matrix $\mathbf{A}$ for the optimal variance reduction.

EC.2.4. The Optimal Control Variates

Now we focus on finding the optimal bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$ that minimize $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$ the variance of the control variate estimator for fixed n . If the potential outcomes $\mathbf{Y}(1)$ and $\mathbf{Y}(0)$ were known to take values $Y_i(1) = y_i(1)$ and $Y_i(0) = y_i(0)$ for any $i \in [n]$, then denote $\mathbf{g} = \mathbf{\Pi}(1)^{-1}\mathbf{y}(1) + \mathbf{\Pi}(0)^{-1}\mathbf{y}(0)$, and we see that the problem

$$\min_{\mathbf{A}} \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) = \min_{\mathbf{A}} \frac{1}{n^2} \left(\mathbf{g}^\top \mathbf{\Sigma} \mathbf{g} - \mathbf{g}^\top \mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{\Sigma} \mathbf{g} \right)$$

can be easily solved by choosing $\mathbf{A}$ such that $\mathbf{g}$ is in the column space of $\mathbf{A}$, such as $\mathbf{a}(1) = \mathbf{\Pi}(1)^{-1}\mathbf{y}(1)$ and $\mathbf{a}(0) = -\mathbf{\Pi}(0)^{-1}\mathbf{y}(0)$. In this case, the variance $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$ can be reduced to exactly zero.

However, the above choice of $\mathbf{A}$ is infeasible because the potential outcomes $\mathbf{Y}(1)$ and $\mathbf{Y}(0)$ are unknown. Therefore, we find the optimal bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$ under uncertainty of the potential outcomes, which leads to a decision making problem under uncertainty. We adopt a stochastic optimization perspective (Zhao 2024) and model each pair of unknown outcomes $Y_i(1), Y_i(0)$ to be independent and identically distributed (i.i.d.) samples from an unknown joint distribution $\mathcal{Y}_{\text{PO}}$. See Assumption 3* below.

Assumption 3* *We assume that for each $i \in [n]$, the potential outcomes $(Y_i(1), Y_i(0))$ are i.i.d. samples from an unknown joint distribution $\mathcal{Y}_{\text{PO}}$, whose marginal distributions are denoted as $\mathcal{Y}(1)$ and $\mathcal{Y}(0)$.*

Denote the population means as $\mu(1) = \mathbb{E}_{Y_i(1) \sim \mathcal{Y}(1)}[Y_i(1)]$ and $\mu(0) = \mathbb{E}_{Y_i(0) \sim \mathcal{Y}(0)}[Y_i(0)]$, the population variances as $\sigma^2(1) = \text{Var}_{Y_i(1) \sim \mathcal{Y}(1)}(Y_i(1))$ and $\sigma^2(0) = \text{Var}_{Y_i(0) \sim \mathcal{Y}(0)}(Y_i(0))$, and the population covariance as $\sigma(1, 0) = \text{Cov}_{Y_i(1), Y_i(0) \sim \mathcal{Y}_{\text{PO}}}(Y_i(1), Y_i(0))$. **But $\sigma(1, 0)$ is not directly estimable.** Denote $\mathcal{Y}_{\text{PO}}^n$ to be the joint probability distribution of the outcomes $\mathbf{Y}(1), \mathbf{Y}(0)$. Under Assumption 3*, we formulate the following optimization problem

$$\mathbf{A}^* \in \arg \min_{\mathbf{A}} \mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right]. \quad (\text{EC.4})$$

The optimal control variates use the bases that solve (EC.4), which is given by Theorem EC.2 below. Recall the square root matrix $\mathbf{\Sigma}^{\frac{1}{2}}$ and the pseudo-inverse square root matrix $\mathbf{\Sigma}^{-\frac{1}{2}}$.

Theorem EC.2 (Optimal Control Variates) *For notational simplicity, we reload the notation $\mathbf{M}$ and define $\mathbf{M}$ to be a symmetric matrix*

$$\begin{aligned} \mathbf{M} = & \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{\Pi}(1)^{-1} (\mu^2(1) \mathbf{1}_n \mathbf{1}_n^\top + \sigma^2(1) \mathbf{I}_n) \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \\ & + \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{\Pi}(1)^{-1} (\mu(1)\mu(0) \mathbf{1}_n \mathbf{1}_n^\top + \sigma(1, 0) \mathbf{I}_n) \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \\ & + \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{\Pi}(0)^{-1} (\mu(1)\mu(0) \mathbf{1}_n \mathbf{1}_n^\top + \sigma(1, 0) \mathbf{I}_n) \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \\ & + \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{\Pi}(0)^{-1} (\mu^2(0) \mathbf{1}_n \mathbf{1}_n^\top + \sigma^2(0) \mathbf{I}_n) \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}^{\frac{1}{2}}, \end{aligned}$$

where $\mathbf{I}_n$ is a $(n \times n)$ identity matrix and $\mathbf{1}_n$ is a n -dimensional vector with each element equal to 1. Let $\lambda_l(\mathbf{M})$ be the l -th largest eigenvalue of $\mathbf{M}$, and let $\mathbf{u}_l(\mathbf{M})$ be the eigenvector that corresponds to $\lambda_l(\mathbf{M})$. Under Assumptions 1*-(i) and 3*, the optimal bases are given by

$$\mathbf{A}^* = \mathbf{\Sigma}^{-\frac{1}{2}} [\mathbf{u}_1(\mathbf{M}), \mathbf{u}_2(\mathbf{M})] \mathbf{C}$$

where $\mathbf{C}$ is any invertible, symmetric and semidefinite matrix. And in expectation, the optimal variance reduction is given by

$$\mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\text{Var}(\hat{\tau}^{\text{HT}}) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right] = \frac{\lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M})}{n^2}.$$

The proof of Theorem EC.2 is given in Section EC.4. Similar to Theorem 2, Theorem EC.2 also combines a design-based perspective and a model-based perspective, where we assume a model for the unknown potential outcomes $\mathbf{Y}(1), \mathbf{Y}(0)$ to guide the choice of bases. But we rely on the experimental design as the only source of randomness when we estimate the average treatment effect. We point it out that Assumption 3* is only required to show optimality of the bases $\mathbf{A}^*$. The basis $\mathbf{A}^*$ is still well defined without Assumption 3*.

In Theorem EC.2, the optimal bases $\mathbf{A}^*$ depends on the unknown moments of the joint distribution $\mathcal{Y}_{\text{PO}}$. In practice, we may use domain knowledge to estimate the moments of the joint distribution $\mathcal{Y}_{\text{PO}}$, and then use them to find the optimal bases $\mathbf{A}^*$. Even an imperfect guess of the moments will yield a set of bases that can still reduce the variance over the baseline Horvitz-Thompson estimator.

Theorem EC.2 involves a scaling matrix $\mathbf{C}$ which makes the optimal bases $\mathbf{A}^*$ not unique. We could always choose an invertible, symmetric and semidefinite matrix $\mathbf{C}$ that satisfies some nice regularity conditions. See Proposition 2 below.

Proposition 2 (Orthonormal Bases) *Let $\mathbf{M}$ be as defined in Theorem EC.2. When $\mathbf{C} = n^{\frac{1}{2}} \mathbf{I}_2$ where $\mathbf{I}_2$ stands for a 2×2 identity matrix, the optimal bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$ in Theorem EC.2 are given by*

$$\mathbf{a}(1) = n^{\frac{1}{2}} \boldsymbol{\Sigma}^{-\frac{1}{2}} \mathbf{u}_1(\mathbf{M}), \quad \text{and} \quad \mathbf{a}(0) = -n^{\frac{1}{2}} \boldsymbol{\Sigma}^{-\frac{1}{2}} \mathbf{u}_2(\mathbf{M}),$$

and satisfies the following regularity conditions

$$\mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(0) = 0, \quad \text{and} \quad \mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(1) = \mathbf{a}(0)^\top \boldsymbol{\Sigma} \mathbf{a}(0) = n.$$

The proof of Proposition 2 is given in Section EC.4. Proposition 2 recommends one specific set of bases to choose. This set of bases satisfies two properties: (i) the two control variates $\hat{X}(1)$ and $\hat{X}(0)$ are uncorrelated, that is, $\text{Cov}(\hat{X}(1), \hat{X}(0)) = 0$; and (ii) the two control variates $\hat{X}(1)$ and $\hat{X}(0)$ have appropriate variances, that is, $\text{Var}(\hat{X}(1)) = \text{Var}(\hat{X}(0)) = n^{-1}$.

To conclude this section, we consider a remarkable special case of Theorem EC.2 when $\sigma^2(1) = \sigma^2(0) = \sigma(1, 0) = 0$.

Corollary 3 (Noiseless Potential Outcomes) *Under Assumption 1*-(i), and assuming that the potential outcomes $Y_i(1) = y(1)$ and $Y_i(0) = y(0)$ take the same unknown constants, then one set of optimal bases is given, for any $i \in [n]$, by*

$$a_i(1) = \frac{1}{\pi_i}, \quad \text{and} \quad a_i(0) = -\frac{1}{1 - \pi_i}.$$

The proof of Corollary 3 is given in Section EC.4. In this special case when all potential outcomes take the same two constants, the only fluctuation of the Horvitz-Thompson estimator comes from the random components $\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=1\}}{\pi_i}$ and $\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=0\}}{1-\pi_i}$. Corollary 1 suggests to correct it by using the exact same terms $\hat{X}(1) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=1\}}{\pi_i}$ and $\hat{X}(0) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=0\}}{1-\pi_i}$ as the control variates. This draws a connection between the control variate estimator and the Hajek estimator, which we will discuss in Section EC.2.5 below.

EC.2.5. Connections to Hajek Estimator

A Naive Control Variate Estimator. As an alternative to the variance minimizing coefficients provided in Lemma EC.1, we can consider a naive control variate estimator using coefficients $\gamma^{\text{Naive}}(1)$ and $\gamma^{\text{Naive}}(0)$ defined as follows.

$$\gamma^{\text{Naive}}(1) = \frac{\text{Cov}(\hat{\mu}^{\text{HT}}(1), \hat{X}(1))}{\text{Var}(\hat{X}(1))}, \quad \text{and} \quad \gamma^{\text{Naive}}(0) = -\frac{\text{Cov}(\hat{\mu}^{\text{HT}}(0), \hat{X}(0))}{\text{Var}(\hat{X}(0))}.$$

The naive coefficients consider the treatment and control components of the Horvitz-Thompson estimator separately. Specifically, $\gamma^{\text{Naive}}(1)$ is chosen as if $\hat{X}(1)$ were only used to reduce the variance of $\hat{\mu}^{\text{HT}}(1)$, and $\gamma^{\text{Naive}}(0)$ is chosen as if $\hat{X}(0)$ were only used to reduce the variance of $\hat{\mu}^{\text{HT}}(0)$. By doing so, it ignores the correlation between $\hat{\mu}^{\text{HT}}(0)$ and $\hat{X}(1)$, as well as between $\hat{\mu}^{\text{HT}}(1)$ and $\hat{X}(0)$. Therefore, the variance of the naive control variate estimator is greater or equal to the optimal control variate estimator. The magnitude of this variance gap depends on the choice of bases $\mathbf{a}(1)$ and $\mathbf{a}(0)$. We illustrate this variance gap in Examples 8 and 9 below.

Example 8 *Let $\mathbf{M}$ be as defined in Theorem EC.2. Under Assumption 1*-(i) and using the bases*

$$\mathbf{a}(1) = n^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \mathbf{u}_1(\mathbf{M}), \quad \text{and} \quad \mathbf{a}(0) = -n^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \mathbf{u}_2(\mathbf{M}),$$

we have

$$\begin{aligned} & \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^{\text{Naive}}(1), \gamma^{\text{Naive}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \\ &= \frac{1}{n^2} \left((\mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \Sigma^{\frac{1}{2}} \mathbf{u}_1(\mathbf{M}))^2 + (\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \Sigma^{\frac{1}{2}} \mathbf{u}_2(\mathbf{M}))^2 \right) \geq 0. \end{aligned}$$

Example 9 Denote $\Theta(1) = \frac{n \sum_{i=1}^n \frac{1-\pi_i}{\pi_i} Y_i(1)}{\sum_{i=1}^n \frac{1-\pi_i}{\pi_i}} - \sum_{i=1}^n Y_i(1)$ and $\Theta(0) = \frac{n \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} Y_i(0)}{\sum_{i=1}^n \frac{\pi_i}{1-\pi_i}} - \sum_{i=1}^n Y_i(0)$. Under Assumption 1^{*}-(i) and using the bases

$$a_i(1) = \frac{1}{\pi_i}, \quad \text{and} \quad a_i(0) = -\frac{1}{1-\pi_i},$$

for any $i \in [n]$, we have

$$\begin{aligned} & \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^{\text{Naive}}(1), \gamma^{\text{Naive}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \\ &= \frac{1}{n^2 \left(\sum_{i=1}^n \frac{1-\pi_i}{\pi_i} \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} - n^2 \right)} \sum_{i=1}^n \left(\sqrt{\frac{1-\pi_i}{\pi_i}} \Theta(1) - \sqrt{\frac{\pi_i}{1-\pi_i}} \Theta(0) \right)^2 \geq 0, \end{aligned}$$

where the inequality takes equality if and only if one of the following two conditions hold:

1. $\pi_i = \pi$ for any $i \in [n]$;
2. $Y_i(1) = Y(1)$ and $Y_i(0) = Y(0)$ for any $i \in [n]$.

The proofs of Examples 8 and 9 are given in Section EC.4. Examples 8 and 9 quantify the loss in variance from using the naive coefficients $\gamma^{\text{Naive}}(1)$ and $\gamma^{\text{Naive}}(0)$ instead of the jointly optimal coefficients $\gamma^*(1)$ and $\gamma^*(0)$. In other words, they quantify the loss in variance from ignoring the correlation between $\hat{\mu}^{\text{HT}}(0)$ and $\hat{X}(1)$, as well as between $\hat{\mu}^{\text{HT}}(1)$ and $\hat{X}(0)$. But as Example 9 suggests, the loss in variance is equal to zero under the special bases $a_i(1) = \frac{1}{\pi_i}$ and $a_i(0) = -\frac{1}{1-\pi_i}$ for any $i \in [n]$ and when $\pi_i = \pi$ for any $i \in [n]$.

Hajek Estimator. In the causal inference setup and under SUTVA, the Hajek estimator is defined as

$$\hat{\tau}_{\text{Hajek}} = \frac{\hat{\mu}^{\text{HT}}(1)}{\hat{1}(1)} - \frac{\hat{\mu}^{\text{HT}}(0)}{\hat{1}(0)}, \quad \text{where} \quad \hat{1}(1) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=1\}}{\pi_i}, \quad \hat{1}(0) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=0\}}{1-\pi_i}.$$

The Hajek estimator in the causal inference setup simply combines two Hajek estimators in survey sampling to estimate the average treatment effect. Following similar discussions as in Section 2.5, we make a first order approximation by using Taylor expansion around $\hat{1}(1) \approx 1$ and $\hat{1}(0) \approx 1$, which gives

$$\hat{\tau}_{\text{Hajek}} \approx \hat{\tau}^{\text{HT}} - \hat{\mu}^{\text{HT}}(1) \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=1\}}{\pi_i} - 1 \right) + \hat{\mu}^{\text{HT}}(0) \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=0\}}{1-\pi_i} - 1 \right).$$

On the other hand, using the bases $a_i(1) = \frac{1}{\pi_i}$ and $a_i(0) = -\frac{1}{1-\pi_i}, \forall i \in [n]$, the control variate estimator using coefficients $\hat{\gamma}^{\text{Naive}}(1)$ and $\hat{\gamma}^{\text{Naive}}(0)$ is given as

$$\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{Naive}}(1), \hat{\gamma}^{\text{Naive}}(0)) = \hat{\tau}^{\text{HT}} - \hat{\gamma}^{\text{Naive}}(1) \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=1\}}{\pi_i} - 1 \right) + \hat{\gamma}^{\text{Naive}}(0) \left(\frac{1}{n} \sum_{i=1}^n \frac{\mathbb{1}\{W_i=0\}}{1-\pi_i} - 1 \right),$$

where we use the Horvitz-Thompson style estimators $\hat{\gamma}^{\text{Naive}}(1)$ and $\hat{\gamma}^{\text{Naive}}(0)$ to estimate the coefficients $\gamma^{\text{Naive}}(1)$ and $\gamma^{\text{Naive}}(0)$ in the naive estimator. Under Bernoulli randomized experiments and when all the treatment probabilities are equal $\pi_i = \pi$, we have

$$\hat{\gamma}^{\text{Naive}}(1) = \frac{1}{n} \sum_{i=1}^n \frac{Y_i \mathbb{1}\{W_i = 1\}}{\pi} = \hat{\mu}^{\text{HT}}(1), \quad \text{and} \quad \hat{\gamma}^{\text{Naive}}(0) = \frac{1}{n} \sum_{i=1}^n \frac{Y_i \mathbb{1}\{W_i = 0\}}{1 - \pi} = \hat{\mu}^{\text{HT}}(0).$$

Therefore, the first-order approximation to the Hajek estimator is a special case of the control variate estimator using the bases $a_i(1) = \frac{1}{\pi_i}$ and $a_i(0) = -\frac{1}{1-\pi_i}$, $\forall i \in [n]$. Combining with Corollary 3 and Example 9, it shows that the first-order approximation to the Hajek estimator is asymptotically optimal when the outcomes have near-zero variance, and when data is collected under Bernoulli randomized experiments with all the treatment probabilities being equal.

EC.3. Useful Lemmas

Lemma EC.2 (Dependency Neighborhood CLT, Ross (2011) Theorem 3.5) *Let $\xi_1, \xi_2, \dots, \xi_n$ be random variables with $E[\xi_i] = 0$ and $E[\xi_i^4] < +\infty$, and define $\sigma_n = \sqrt{\text{Var}\left(\sum_{i=1}^n \xi_i\right)}$ and $S_n = \frac{\sum_{i=1}^n \xi_i}{\sigma_n}$. Let $\xi_1, \xi_2, \dots, \xi_n$ satisfy the dependency neighborhood assumption (i.e., Assumption 1-(ii) in the paper) with $|\mathcal{N}_i| \leq d_n$ for all $i \in [n]$. Then for Z a standard normal random variable,*

$$d_W(S_n, Z) \leq \frac{d_n^2}{\sigma_n^3} \sum_{i=1}^n E[|\xi_i|^3] + \frac{26^{\frac{1}{2}} d_n^{\frac{3}{2}}}{\pi^{\frac{1}{2}} \sigma_n^2} \left(\sum_{i=1}^n E[\xi_i^4] \right)^{\frac{1}{2}},$$

where $D_W(S_n, Z)$ stands for the Wasserstein distance between S_n and Z .

Note that, in Lemma EC.2 $\pi \approx 3.14$. This should not be confused with the marginal probabilities π_i or their lower bound $\underline{\pi}$.

Lemma EC.3 (Rayleigh Quotient, Horn and Johnson (2012) Theorem 4.2.2) *Let $\mathcal{V}$ be a unit sphere defined as $\mathcal{V} = \{\mathbf{v} \in \mathbb{R}^n \mid \|\mathbf{v}\|_2 = 1\}$. For any symmetric real matrix $\mathbf{E} \in \mathbb{R}^{n \times n}$, its largest eigenvalue is given by*

$$\lambda_1(\mathbf{E}) = \max_{\mathbf{v} \in \mathcal{V}} \mathbf{v}^\top \mathbf{E} \mathbf{v}.$$

Its principal eigenvector, that is, the eigenvector corresponding to the largest eigenvalue, is given by

$$\mathbf{v}_1(\mathbf{E}) = \arg \max_{\mathbf{v} \in \mathcal{V}} \mathbf{v}^\top \mathbf{E} \mathbf{v}.$$

Lemma EC.4 (Fan's Principle, Fan (1949) Theorem 1) *Let $p < n$ be a constant. Let $\mathbf{V} \in \mathbb{R}^{n \times p}$ be a matrix whose column vectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ are orthonormal, that is, $\mathbf{V} \in \mathcal{V} = \{\mathbf{V} | \mathbf{V}^\top \mathbf{V} = \mathbf{I}_p\}$, where $\mathbf{I}_p$ is a $(p \times p)$ identity matrix. For any symmetric real matrix $\mathbf{E} \in \mathbb{R}^{n \times n}$, we have*

$$\max_{\mathbf{V} \in \mathcal{V}} \text{Tr}(\mathbf{V}^\top \mathbf{E} \mathbf{V}) = \sum_{i=1}^p \lambda_i(\mathbf{E}),$$

where $\text{Tr}(\mathbf{V}^\top \mathbf{E} \mathbf{V}) = \sum_{i=1}^p \mathbf{v}_i^\top \mathbf{E} \mathbf{v}_i$ stands for the trace of matrix $\mathbf{V}^\top \mathbf{E} \mathbf{V}$, and $\lambda_1(\mathbf{E}), \lambda_2(\mathbf{E}), \dots, \lambda_p(\mathbf{E})$ stand for the largest p eigenvalues of $\mathbf{E}$. The optimal solution to the above problem is given by $\mathbf{v}_1 = \mathbf{v}_1(\mathbf{E}), \mathbf{v}_2 = \mathbf{v}_2(\mathbf{E}), \dots, \mathbf{v}_p = \mathbf{v}_p(\mathbf{E})$, the eigenvectors corresponding to the largest p eigenvalues.

EC.4. Missing Proofs

EC.4.1. Proof of Lemma 1

Proof of Lemma 1. We follow (2) to find the expression of γ^* . We start with $\text{Cov}(\hat{\mu}^{\text{HT}}, \hat{X})$.

$$\begin{aligned} \text{Cov}(\hat{\mu}^{\text{HT}}, \hat{X}) &= \mathbb{E}[\hat{\mu}^{\text{HT}} \cdot \hat{X}] - \mathbb{E}[\hat{\mu}^{\text{HT}}] \cdot \mathbb{E}[\hat{X}] \\ &= \frac{1}{n^2} \mathbb{E} \left[\sum_{i=1}^n \frac{Y_i \mathbb{1}\{W_i = 1\}}{\pi_i} \cdot \sum_{i=1}^n a_i \mathbb{1}\{W_i = 1\} \right] - \frac{1}{n^2} \sum_{i=1}^n Y_i \cdot \sum_{i=1}^n a_i \pi_i \\ &= \frac{1}{n^2} \sum_{i=1}^n Y_i a_i \mathbb{E} \left[\frac{\mathbb{1}\{W_i = 1\}}{\pi_i} \right] + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} Y_i a_j \mathbb{E} \left[\frac{\mathbb{1}\{W_i = 1\} \mathbb{1}\{W_j = 1\}}{\pi_i} \right] \\ &\quad - \frac{1}{n^2} \sum_{i=1}^n Y_i a_i \pi_i - \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} Y_i a_j \pi_j \\ &= \frac{1}{n^2} \sum_{i=1}^n Y_i a_i (1 - \pi_i) + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} Y_i a_j \left(\frac{\pi_{ij}}{\pi_i} - \pi_j \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \frac{Y_i}{\pi_i} (\pi_i - \pi_i^2) a_i + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} \frac{Y_i}{\pi_i} (\pi_{ij} - \pi_i \pi_j) a_j \\ &= \frac{1}{n^2} \mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a}. \end{aligned}$$

Similarly, we have

$$\begin{aligned} \text{Var}(\hat{X}) &= \mathbb{E}[\hat{X}^2] - \mathbb{E}[\hat{X}]^2 \\ &= \frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n a_i \mathbb{1}\{W_i = 1\} \right)^2 \right] - \frac{1}{n^2} \left(\sum_{i=1}^n a_i \pi_i \right)^2 \\ &= \frac{1}{n^2} \sum_{i=1}^n a_i^2 \mathbb{E}[\mathbb{1}\{W_i = 1\}] + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} a_i a_j \mathbb{E}[\mathbb{1}\{W_i = 1\} \mathbb{1}\{W_j = 1\}] \\ &\quad - \frac{1}{n^2} \sum_{i=1}^n a_i^2 \pi_i^2 - \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} a_i \pi_i a_j \pi_j \\ &= \frac{1}{n^2} \sum_{i=1}^n a_i (\pi_i - \pi_i^2) a_i + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} a_i (\pi_{ij} - \pi_i \pi_j) a_j \\ &= \frac{1}{n^2} \mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}. \end{aligned}$$

Combining both, the variance minimizing coefficient can be expressed by

$$\gamma^* = \frac{\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a}}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}}.$$

Next, we study the variance of the control variate estimator $\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))$. Note that

$$\begin{aligned} \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) &= \text{Var}(\hat{\mu}^{\text{HT}}) - 2\gamma^* \text{Cov}(\hat{\mu}^{\text{HT}}, \hat{X}) + (\gamma^*)^2 \text{Var}(\hat{X}) \\ &= \text{Var}(\hat{\mu}^{\text{HT}}) - \frac{\text{Cov}(\hat{\mu}^{\text{HT}}, \hat{X})^2}{\text{Var}(\hat{X})}, \end{aligned}$$

where the second equality follows from (2). Next, we have

$$\begin{aligned} \text{Var}(\hat{\mu}^{\text{HT}}) &= \mathbb{E}[(\hat{\mu}^{\text{HT}})^2] - \mathbb{E}[\hat{\mu}^{\text{HT}}]^2 \\ &= \frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n \frac{Y_i \mathbb{1}\{W_i=1\}}{\pi_i} \right)^2 \right] - \frac{1}{n^2} \left(\sum_{i=1}^n Y_i \right)^2 \\ &= \frac{1}{n^2} \sum_{i=1}^n Y_i^2 \mathbb{E} \left[\frac{\mathbb{1}\{W_i=1\}}{\pi_i^2} \right] + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} Y_i Y_j \mathbb{E} \left[\frac{\mathbb{1}\{W_i=1\} \mathbb{1}\{W_j=1\}}{\pi_i \pi_j} \right] \\ &\quad - \frac{1}{n^2} \sum_{i=1}^n Y_i^2 - \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} Y_i Y_j \\ &= \frac{1}{n^2} \sum_{i=1}^n Y_i^2 \left(\frac{1}{\pi_i} - 1 \right) + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} Y_i Y_j \left(\frac{\pi_{ij}}{\pi_i \pi_j} - 1 \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \frac{Y_i}{\pi_i} (\pi_i - \pi_i^2) \frac{Y_i}{\pi_i} + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i} \frac{Y_i}{\pi_i} (\pi_{ij} - \pi_i \pi_j) \frac{Y_j}{\pi_j} \\ &= \frac{1}{n^2} \mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{\Pi}^{-1} \mathbf{Y}. \end{aligned}$$

Plugging in all the expressions above, we have

$$\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) = \frac{1}{n^2} \left(\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{\Pi}^{-1} \mathbf{Y} - \frac{(\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a})^2}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \right).$$

This finishes the proof. $\square$

EC.4.2. Proof of Theorem 1

We present the following Lemma EC.5 which will be useful in the proof of Theorem 1.

Lemma EC.5 *Under Assumptions 1 and 2, and assuming the outcomes $|Y_i| \leq \bar{y}$ are all bounded, the optimal coefficient γ^* is bounded, that is,*

$$|\gamma^*| \leq \frac{\bar{y} \bar{d}^{\frac{1}{2}}}{\pi \mathcal{C}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}}.$$

Proof of Lemma EC.5. Note that,

$$\begin{aligned} |\gamma^*| &= \left| \frac{\text{Cov}(\hat{\mu}^{\text{HT}}, \hat{X})}{\text{Var}(\hat{X})} \right| \leq \sqrt{\frac{\text{Var}(\hat{\mu}^{\text{HT}})}{\text{Var}(\hat{X})}} = \sqrt{\frac{\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{\Pi}^{-1} \mathbf{Y}}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}}} \\ &\leq \sqrt{\frac{1}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \frac{\bar{y}^2 \bar{d} n^{1+\alpha}}{\underline{\pi}^2}} \leq \sqrt{\frac{\bar{y}^2}{\underline{\pi}^2 \underline{c}_\Sigma} n^\alpha} = \frac{\bar{y}}{\underline{\pi}} \frac{\bar{d}^{\frac{1}{2}}}{\underline{c}_\Sigma^{\frac{1}{2}}} n^{\frac{\alpha}{2}}, \end{aligned}$$

where the first inequality is due to Cauchy-Schwarz inequality; the second inequality is due to Assumption 1-(i), the assumption that the outcomes $|Y_i| \leq \bar{y}$ are all bounded, and Assumption 1-(ii); the last inequality is due to Assumption 2-(ii). $\square$

Now we prove Theorem 1 as follows.

Proof of Theorem 1. We prove the three claims (consistency, asymptotic variance, and asymptotic normality) in Theorem 1 one by one. To start, we introduce a definition $\mathbf{R} = \text{diag}(\mathbb{1}\{W_1 = 1\}, \mathbb{1}\{W_2 = 1\}, \dots, \mathbb{1}\{W_n = 1\})$.

Claim 1: Consistency. In the proof of this claim, we assume $\alpha < \frac{1}{3}$. We prove consistency through two steps. First, we show that $\lim_{n \rightarrow +\infty} (\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) - \hat{\mu}^{\text{CV}}(\gamma^*)) \xrightarrow{p} 0$. To show this, recall that by definition,

$$\hat{\gamma}^{\text{HT}} = \frac{\mathbf{Y}^\top \mathbf{R} \mathbf{\Pi}^{-2} \mathbf{\Sigma} \mathbf{a}}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}}.$$

For any $i \in [n]$, define $Z_i = \frac{Y_i}{\pi_i} (\mathbf{\Sigma} \mathbf{a})_i$, where $(\mathbf{\Sigma} \mathbf{a})_i$ stands for the i -th element of vector $\mathbf{\Sigma} \mathbf{a}$. Note that we assume for any $i \in [n]$, $|Y_i| \leq \bar{y}$ (the additional assumption made in Theorem 1), $|a_i| \leq \bar{a}$ (Assumption 2-(i)), and $\pi_i \geq \underline{\pi}$ (Assumption 1-(i)). So there exists a constant $\bar{z} = \frac{\bar{y} \bar{a} \bar{d}}{\underline{\pi}}$ such that $|Z_i| \leq \bar{z} n^\alpha$.

Using the definition of Z_i , the Horvitz-Thompson estimator $\hat{\gamma}^{\text{HT}}$ can be written as

$$\hat{\gamma}^{\text{HT}} = \frac{1}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \sum_{i=1}^n Z_i \mathbb{1}\{W_i = 1\}.$$

It is easy to see that $\hat{\gamma}^{\text{HT}}$ is unbiased,

$$\begin{aligned} \mathbb{E}_{\mathcal{W}}[\hat{\gamma}^{\text{HT}}] &= \frac{1}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \mathbb{E}_{\mathcal{W}}[\mathbf{Y}^\top \mathbf{R} \mathbf{\Pi}^{-2} \mathbf{\Sigma} \mathbf{a}] \\ &= \frac{1}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \mathbb{E}_{\mathcal{W}}\left[\sum_{i=1}^n Z_i \mathbb{1}\{W_i = 1\}\right] = \frac{1}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \sum_{i=1}^n Z_i \pi_i = \frac{1}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a} = \gamma^*. \end{aligned}$$

Next, we calculate the variance of $\hat{\gamma}^{\text{HT}}$ as follows,

$$\begin{aligned} \text{Var}(\hat{\gamma}^{\text{HT}}) &= \text{Var}\left(\frac{1}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \sum_{i=1}^n Z_i \mathbb{1}\{W_i = 1\}\right) = \frac{1}{(\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a})^2} \sum_{i=1}^n \sum_{j=1}^n Z_i Z_j (\pi_{ij} - \pi_i \pi_j) \\ &\leq \frac{\bar{z}^2 n^{2\alpha}}{(\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a})^2} \sum_{i=1}^n \sum_{j=1}^n |\pi_{ij} - \pi_i \pi_j| \leq \frac{\bar{z}^2 \bar{d} n^{1+3\alpha}}{(\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a})^2} \leq \frac{\bar{z}^2 \bar{d}}{\underline{c}_\Sigma^2} n^{3\alpha-1}, \end{aligned}$$

where the first inequality is upper bounding $|Z_i| \leq \bar{z}n^\alpha$; the second inequality is due to Assumption 1-(ii); and the third inequality is due to Assumption 2-(ii). Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr(|\hat{\gamma}^{\text{HT}} - \gamma^*| \geq \epsilon) \leq \frac{\text{Var}(\hat{\gamma}^{\text{HT}})}{\epsilon^2} \leq \frac{\bar{z}^2 \bar{d}}{\underline{c}_\Sigma^2} n^{3\alpha-1} \frac{1}{\epsilon^2}.$$

Similarly, we calculate the variance of $\hat{X}$ as follows,

$$\begin{aligned} \text{Var}(\hat{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n a_i \mathbb{1}\{W_i = 1\}\right) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i a_j (\pi_{ij} - \pi_i \pi_j) \\ &\leq \frac{\bar{a}^2}{n^2} \sum_{i=1}^n \sum_{j=1}^n |\pi_{ij} - \pi_i \pi_j| \leq \bar{a}^2 \bar{d} n^{\alpha-1}, \end{aligned}$$

where the first inequality is due to Assumption 1-(i); and the second inequality is due to Assumption 1-(ii). Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr(|\hat{X} - \mathbb{E}[\hat{X}]| \geq \epsilon) \leq \frac{\text{Var}(\hat{X})}{\epsilon^2} \leq \bar{a}^2 \bar{d} n^{\alpha-1} \frac{1}{\epsilon^2}.$$

Next, we focus on

$$\begin{aligned} \Delta_n &= \hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) - \hat{\mu}^{\text{CV}}(\gamma^*) \\ &= (\hat{\mu}^{\text{HT}} - \hat{\gamma}^{\text{HT}}(\hat{X} - \mathbb{E}[\hat{X}])) - (\hat{\mu}^{\text{HT}} - \gamma^*(\hat{X} - \mathbb{E}[\hat{X}])) = -(\hat{\gamma}^{\text{HT}} - \gamma^*)(\hat{X} - \mathbb{E}[\hat{X}]). \end{aligned}$$

We have that, for any $\epsilon > 0$,

$$\begin{aligned} \lim_{n \rightarrow +\infty} \Pr(|\Delta_n| \geq \epsilon) &\leq \lim_{n \rightarrow +\infty} \left(\Pr(|\hat{\gamma}^{\text{HT}} - \gamma^*| \geq \epsilon^{\frac{1}{2}}) + \Pr(|\hat{X} - \mathbb{E}[\hat{X}]| \geq \epsilon^{\frac{1}{2}}) \right) \\ &\leq \lim_{n \rightarrow +\infty} \left(\frac{\bar{z}^2 \bar{d}}{\underline{c}_\Sigma^2} n^{3\alpha-1} \frac{1}{\epsilon} + \bar{a}^2 \bar{d} n^{\alpha-1} \frac{1}{\epsilon} \right) \\ &= 0, \end{aligned}$$

where the first inequality holds because for $|\Delta_n| \geq \epsilon$ to hold, either $|\hat{\gamma}^{\text{HT}} - \gamma^*| \geq \epsilon^{\frac{1}{2}}$ or $|\hat{X} - \mathbb{E}[\hat{X}]| \geq \epsilon^{\frac{1}{2}}$ needs to hold; the last equality holds because $\alpha < \frac{1}{3}$. So we conclude that

$$\lim_{n \rightarrow +\infty} \left(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) - \hat{\mu}^{\text{CV}}(\gamma^*) \right) \xrightarrow{p} 0. \quad (\text{EC.5})$$

Next, we show that $\hat{\mu}^{\text{CV}}(\gamma^*)$ is a consistent estimator of μ . To show this, recall that

$$\Pr(|\hat{X} - \mathbb{E}[\hat{X}]| \geq \epsilon) \leq \frac{\text{Var}(\hat{X})}{\epsilon^2} \leq \bar{a}^2 \bar{d} n^{\alpha-1} \frac{1}{\epsilon^2}.$$

Note also that

$$\begin{aligned}\text{Var}(\hat{\mu}^{\text{HT}}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n \frac{Y_i}{\pi_i} \mathbb{1}\{W_i = 1\}\right) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i Y_j}{\pi_i \pi_j} (\pi_{ij} - \pi_i \pi_j) \\ &\leq \frac{\bar{y}^2}{\pi^2 n^2} \sum_{i=1}^n \sum_{j=1}^n |\pi_{ij} - \pi_i \pi_j| \leq \frac{\bar{y}^2 \bar{d}}{\pi^2} n^{\alpha-1},\end{aligned}$$

where the first inequality is due to Assumption 2-(i); and the second inequality is due to Assumption 1-(ii). Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr(|\hat{\mu}^{\text{HT}} - \mu| \geq \epsilon) \leq \frac{\text{Var}(\hat{\mu}^{\text{HT}})}{\epsilon^2} \leq \frac{\bar{y}^2 \bar{d}}{\pi^2} n^{\alpha-1} \frac{1}{\epsilon^2}.$$

Next, we focus on

$$\begin{aligned}\lim_{n \rightarrow +\infty} \Pr(|\hat{\mu}^{\text{CV}}(\gamma^*) - \mu| \geq \epsilon) &= \lim_{n \rightarrow +\infty} \Pr(|(\hat{\mu}^{\text{HT}} - \mu) - \gamma^*(\hat{X} - \mathbb{E}[\hat{X}])| \geq \epsilon) \\ &= \lim_{n \rightarrow +\infty} \left(\Pr(|\hat{\mu}^{\text{HT}} - \mu| \geq \frac{\epsilon}{2}) + \Pr(|\gamma^*| \cdot |\hat{X} - \mathbb{E}[\hat{X}]| \geq \frac{\epsilon}{2}) \right) \\ &\leq \lim_{n \rightarrow +\infty} \left(\frac{\bar{y}^2 \bar{d}}{\pi^2} n^{\alpha-1} \frac{4}{\epsilon^2} + \bar{a}^2 \bar{d} n^{\alpha-1} \frac{4|\gamma^*|^2}{\epsilon^2} \right) \\ &\leq \lim_{n \rightarrow +\infty} \left(\frac{\bar{y}^2 \bar{d}}{\pi^2} n^{\alpha-1} \frac{4}{\epsilon^2} + \frac{\bar{y}^2 \bar{a}^2 \bar{d}^3}{\pi^2 c_\Sigma} n^{2\alpha-1} \frac{4}{\epsilon^2} \right) \\ &= 0,\end{aligned}$$

where the first equality holds because for $|(\hat{\mu}^{\text{HT}} - \mu) - \gamma^*(\hat{X} - \mathbb{E}[\hat{X}])| \geq \epsilon$ to hold, either $|\hat{\mu}^{\text{HT}} - \mu| \geq \frac{\epsilon}{2}$ or $|\gamma^*| \cdot |\hat{X} - \mathbb{E}[\hat{X}]| \geq \frac{\epsilon}{2}$ needs to hold; the second inequality holds because of Lemma EC.5; the last equality holds because $\alpha < \frac{1}{3}$. So we conclude that

$$\lim_{n \rightarrow +\infty} \hat{\mu}^{\text{CV}}(\gamma^*) \xrightarrow{p} \mu. \quad (\text{EC.6})$$

Combining (EC.5) and (EC.6) we have

$$\lim_{n \rightarrow +\infty} \hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) \xrightarrow{p} \mu.$$

Claim 2: Asymptotic Variance. In the proof of this claim, we assume $\alpha < \frac{1}{4}$.

Recall that we define $\Delta_n = \hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) - \hat{\mu}^{\text{CV}}(\gamma^*)$, so we have

$$\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})) = \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) + \text{Var}(\Delta_n) + 2\text{Cov}(\hat{\mu}^{\text{CV}}(\gamma^*), \Delta_n). \quad (\text{EC.7})$$

It suffices to prove that both $\lim_{n \rightarrow +\infty} n \text{Var}(\Delta_n) = 0$ and $\lim_{n \rightarrow +\infty} n \text{Cov}(\hat{\mu}^{\text{CV}}(\gamma^*), \Delta_n) = 0$.

We first show that $\lim_{n \rightarrow +\infty} n \text{Var}(\Delta_n) = 0$. It suffices to show that $\lim_{n \rightarrow +\infty} n \mathbb{E}[\Delta_n^2] = 0$ because $0 \leq \text{Var}(\Delta_n) \leq \mathbb{E}[\Delta_n^2]$. To show this, note that

$$\mathbb{E}[\Delta_n^2] = \mathbb{E}[(\hat{\gamma}^{\text{HT}} - \gamma^*)^2 \cdot (\hat{X} - \mathbb{E}[\hat{X}])^2] \leq \mathbb{E}[(\hat{\gamma}^{\text{HT}} - \gamma^*)^4]^{\frac{1}{2}} \cdot \mathbb{E}[(\hat{X} - \mathbb{E}[\hat{X}])^4]^{\frac{1}{2}},$$

where the inequality holds due to Cauchy-Schwarz inequality. We next upper bound the two terms separately. Note that

$$\begin{aligned}
\mathbb{E}[(\hat{\gamma}^{\text{HT}} - \gamma^*)^4] &= \mathbb{E}\left[\left(\frac{1}{\mathbf{a}^\top \Sigma \mathbf{a}} \sum_{i=1}^n Z_i (\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\
&\leq \frac{\bar{z}^4 n^{4\alpha}}{\underline{c}_\Sigma^4 n^4} \mathbb{E}\left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\
&\leq \frac{\bar{z}^4 n^{4\alpha}}{\underline{c}_\Sigma^4 n^4} 125 \bar{d}^2 n^{2+2\alpha} \\
&= \frac{125 \bar{y}^4 \bar{a}^4 \bar{d}^6}{\pi^4 \underline{c}_\Sigma^4} n^{6\alpha-2},
\end{aligned} \tag{EC.8}$$

where the first inequality is using the upper bound that $|Z_i| \leq \bar{z}n^\alpha$ and using Assumption 2-(ii); the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1-(ii). Similarly,

$$\begin{aligned}
\mathbb{E}[(\hat{X} - \mathbb{E}[\hat{X}])^4] &= \mathbb{E}\left[\left(\frac{1}{n} \sum_{i=1}^n a_i (\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\
&\leq \frac{\bar{a}^4}{n^4} \mathbb{E}\left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\
&\leq 125 \bar{a}^4 \bar{d}^2 n^{2\alpha-2},
\end{aligned} \tag{EC.9}$$

where the first inequality is using the upper bound that $|a_i| \leq \bar{a}$; the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1-(ii).

Combining (EC.8) and (EC.9) we have

$$\lim_{n \rightarrow +\infty} n \mathbb{E}[\Delta_n^2] \leq \lim_{n \rightarrow +\infty} \frac{125 \bar{y}^2 \bar{a}^4 \bar{d}^4}{\pi^2 \underline{c}_\Sigma^2} n^{4\alpha-1} = 0, \tag{EC.10}$$

where the last equality holds because $\alpha < \frac{1}{4}$.

Next, we show that $\lim_{n \rightarrow +\infty} n \text{Cov}(\hat{\mu}^{\text{CV}}(\gamma^*), \Delta_n) = 0$. To show this, note that

$$\begin{aligned}
\text{Cov}(\hat{\mu}^{\text{CV}}(\gamma^*), \Delta_n) &= \mathbb{E}[\hat{\mu}^{\text{CV}}(\gamma^*) \Delta_n] - \mathbb{E}[\hat{\mu}^{\text{CV}}(\gamma^*)] \mathbb{E}[\Delta_n] = \mathbb{E}[(\hat{\mu}^{\text{CV}}(\gamma^*) - \mu) \Delta_n] \\
&= -\mathbb{E}\left[(\hat{\mu}^{\text{HT}} - \mu - \gamma^*(\hat{X} - \mathbb{E}[\hat{X}])) \cdot (\hat{\gamma}^{\text{HT}} - \gamma^*) \cdot (\hat{X} - \mathbb{E}[\hat{X}])\right].
\end{aligned}$$

So we have

$$\left| \text{Cov}(\hat{\mu}^{\text{CV}}(\gamma^*), \Delta_n) \right| \leq \mathbb{E}\left[(\hat{\mu}^{\text{HT}} - \mu - \gamma^*(\hat{X} - \mathbb{E}[\hat{X}]))^3\right]^{\frac{1}{3}} \cdot \mathbb{E}\left[(\hat{\gamma}^{\text{HT}} - \gamma^*)^3\right]^{\frac{1}{3}} \cdot \mathbb{E}\left[(\hat{X} - \mathbb{E}[\hat{X}])^3\right]^{\frac{1}{3}},$$

where the inequality holds due to Holder inequality. We next upper bound the three terms separately. Note that,

$$\mathbb{E}\left[(\hat{\mu}^{\text{HT}} - \mu - \gamma^*(\hat{X} - \mathbb{E}[\hat{X}]))^3\right] = \mathbb{E}\left[\left(\frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i}{\pi_i} - \gamma^* a_i\right) (\mathbb{1}\{W_i = 1\} - \pi_i)\right)^3\right]$$

$$\begin{aligned}
&\leq \frac{1}{n^3} \left(\frac{\bar{y}}{\underline{\pi}} + \frac{\bar{y} \bar{a} \bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{c}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \right)^3 \mathbb{E} \left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i) \right)^3 \right] \\
&\leq \frac{1}{n^3} \cdot \frac{8\bar{y}^3 \bar{a}^3 \bar{d}^{\frac{3}{2}}}{\underline{\pi}^3 \underline{c}_{\Sigma}^{\frac{3}{2}}} n^{\frac{3\alpha}{2}} \mathbb{E} \left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i) \right)^3 \right] \\
&\leq \frac{1}{n^3} \cdot \frac{8\bar{y}^3 \bar{a}^3 \bar{d}^{\frac{3}{2}}}{\underline{\pi}^3 \underline{c}_{\Sigma}^{\frac{3}{2}}} n^{\frac{3\alpha}{2}} 5\bar{d}^2 n^{1+2\alpha} \\
&= \frac{40\bar{y}^3 \bar{a}^3 \bar{d}^{\frac{7}{2}}}{\underline{\pi}^3 \underline{c}_{\Sigma}^{\frac{3}{2}}} n^{\frac{7}{2}\alpha-2}, \tag{EC.11}
\end{aligned}$$

where the first inequality is upper bounding $|\frac{Y_i}{\pi_i} - \gamma^* a_i|$ using Lemma EC.5; the second inequality holds when n is sufficiently large; the third inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1-(ii). Similarly,

$$\begin{aligned}
\mathbb{E} \left[(\hat{\gamma}^{\text{HT}} - \gamma^*)^3 \right] &= \mathbb{E} \left[\left(\frac{1}{\mathbf{a}^\top \Sigma \mathbf{a}} \sum_{i=1}^n Z_i (\mathbb{1}\{W_i = 1\} - \pi_i) \right)^3 \right] \\
&\leq \frac{\bar{z}^3 n^{3\alpha}}{\underline{c}_{\Sigma}^3 n^3} \mathbb{E} \left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i) \right)^3 \right] \\
&\leq \frac{\bar{z}^3 n^{3\alpha}}{\underline{c}_{\Sigma}^3 n^3} 5\bar{d}^2 n^{1+2\alpha} \\
&= \frac{5\bar{y}^3 \bar{a}^3 \bar{d}^5}{\underline{\pi}^3 \underline{c}_{\Sigma}^3} n^{5\alpha-2}, \tag{EC.12}
\end{aligned}$$

where the first inequality is using the upper bound that $|Z_i| \leq \bar{z} n^\alpha$ and using Assumption 2-(ii); the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1-(ii). Similarly,

$$\begin{aligned}
\mathbb{E} \left[(\hat{X} - \mathbb{E}[\hat{X}])^3 \right] &= \mathbb{E} \left[\left(\frac{1}{n} \sum_{i=1}^n a_i (\mathbb{1}\{W_i = 1\} - \pi_i) \right)^3 \right] \\
&\leq \frac{\bar{a}^3}{n^3} \mathbb{E} \left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i) \right)^3 \right] \\
&= 5\bar{a}^3 \bar{d}^2 n^{2\alpha-2}, \tag{EC.13}
\end{aligned}$$

where the first inequality is using the upper bound that $|a_i| \leq \bar{a}$; the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1-(ii).

Combining (EC.11), (EC.12), and (EC.13), we have

$$\lim_{n \rightarrow +\infty} n \text{Cov}(\hat{\mu}^{\text{CV}}(\gamma^*), \Delta_n) \leq \lim_{n \rightarrow +\infty} \frac{10\bar{y}^2 \bar{a}^3 \bar{d}^{\frac{7}{2}}}{\underline{\pi}^2 \underline{c}_{\Sigma}^{\frac{3}{2}}} n^{\frac{7}{2}\alpha-1} = 0,$$

where the last equality holds because $\alpha < \frac{1}{4}$.

Now that we have shown both $\lim_{n \rightarrow +\infty} n \text{Var}(\Delta_n) = 0$ and $\lim_{n \rightarrow +\infty} n \text{Cov}(\hat{\mu}^{\text{CV}}(\gamma^*), \Delta_n) = 0$, we conclude that

$$\lim_{n \rightarrow +\infty} n \text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})) = \lim_{n \rightarrow +\infty} n \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))$$

Claim 3: Asymptotic Normality. In the proof of this claim, we assume $\alpha < \frac{1}{7}$.

First, we show that

$$\lim_{n \rightarrow +\infty} \frac{\hat{\mu}^{\text{CV}}(\gamma^*) - \mu}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

To show this, we introduce some notations. Note that

$$\hat{\mu}^{\text{CV}}(\gamma^*) - \mu = \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i}{\pi_i} - \gamma^* a_i \right) (\mathbb{1}\{W_i = 1\} - \pi_i).$$

Now denote

$$\xi_i = \left(\frac{Y_i}{\pi_i} - \gamma^* a_i \right) (\mathbb{1}\{W_i = 1\} - \pi_i),$$

and denote $\bar{\xi} = \frac{2\bar{y}}{\pi} \frac{\bar{a}\bar{d}^{\frac{1}{2}}}{\underline{c}_{\Sigma}^{\frac{1}{2}}}$, so we have $|\xi_i| \leq \frac{\bar{y}}{\pi} + \frac{\bar{y}}{\pi} \frac{\bar{a}\bar{d}^{\frac{1}{2}}}{\underline{c}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \leq \bar{\xi} n^{\frac{\alpha}{2}}$ for sufficiently large n . Next, denote

$$\sigma_n = \sqrt{\text{Var}\left(\sum_{i=1}^n \xi_i\right)}, \quad \text{and} \quad S_n = \frac{\sum_{i=1}^n \xi_i}{\sigma_n}.$$

We will use Lemma EC.2 to show the central limit theorem. To do so, we bound each term in Lemma EC.2. First, recall that we additionally assume that for sufficiently large n , $n \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) \geq \underline{c}$. So we have

$$\sigma_n^2 = \text{Var}\left(\sum_{i=1}^n \xi_i\right) = n^2 \text{Var}\left(\frac{1}{n} \sum_{i=1}^n \xi_i\right) = n^2 \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) \geq \underline{c}n.$$

We also have

$$\sigma_n^3 \geq \underline{c}^{\frac{3}{2}} n^{\frac{3}{2}}.$$

Next, we have

$$\begin{aligned} \sum_{i=1}^n \mathbb{E}[|\xi_i|^3] &\leq \sum_{i=1}^n \bar{\xi}^3 n^{\frac{3}{2}\alpha} = \bar{\xi}^3 n^{1+\frac{3}{2}\alpha}, \\ \sum_{i=1}^n \mathbb{E}[\xi_i^4] &\leq \sum_{i=1}^n \bar{\xi}^4 n^{2\alpha} = \bar{\xi}^4 n^{1+2\alpha}. \end{aligned}$$

Putting all together into Lemma EC.2, for Z a standard normal random variable,

$$\begin{aligned} \lim_{n \rightarrow +\infty} d_W(S_n, Z) &\leq \lim_{n \rightarrow +\infty} \frac{\bar{d}^2 n^{2\alpha} \bar{\xi}^3 n^{1+\frac{3}{2}\alpha} + \frac{26^{\frac{1}{2}} \bar{d}^{\frac{3}{2}} n^{\frac{3}{2}\alpha}}{\pi^{\frac{1}{2}} \underline{c} n} \left(\bar{\xi}^4 n^{1+2\alpha} \right)^{\frac{1}{2}}}{\underline{c}^{\frac{3}{2}} n^{\frac{3}{2}}} \\ &= \lim_{n \rightarrow +\infty} \frac{\bar{d}^2 \bar{\xi}^3}{\underline{c}^{\frac{3}{2}}} n^{\frac{7}{2}\alpha - \frac{1}{2}} + \frac{26^{\frac{1}{2}} \bar{d}^{\frac{3}{2}} \bar{\xi}^2}{\pi^{\frac{1}{2}} \underline{c}} n^{\frac{5}{2}\alpha - \frac{1}{2}} \\ &= 0, \end{aligned}$$

where where $D_W(S_n, Z)$ stands for the Wasserstein distance between S_n and Z ; the first inequality holds using Lemma EC.2; the last equality holds because $\alpha < \frac{1}{7}$. So we have

$$\lim_{n \rightarrow +\infty} S_n = \lim_{n \rightarrow +\infty} \frac{\hat{\mu}^{\text{CV}}(\gamma^*) - \mu}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (\text{EC.14})$$

Next, we establish the connection between $\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})$ and $\hat{\mu}^{\text{CV}}(\gamma^*)$. Note that,

$$\begin{aligned} \frac{\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) - \mu}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))}} &= \frac{\hat{\mu}^{\text{CV}}(\gamma^*) - \mu}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))}} + \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))}} \\ &= \frac{\hat{\mu}^{\text{CV}}(\gamma^*) - \mu}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}} \frac{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))}} + \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}} \frac{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))}}. \quad (\text{EC.15}) \end{aligned}$$

From here, note that

$$\mathbb{E} \left[\frac{\Delta_n}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}} \right] \leq \mathbb{E} \left[\frac{\Delta_n^2}{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))} \right]^{\frac{1}{2}} \leq \sqrt{\frac{125 \bar{y}^2 \bar{a}^4 \bar{d}^4}{\pi^2 \underline{c}_{\Sigma}^2 \underline{c}}} n^{4\alpha-1},$$

where the first inequality is Cauchy-Schwarz inequality; and the second inequality is using (EC.10) and assuming $\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) \geq \underline{c} n^{-1}$. Using Markov inequality, for any $\epsilon > 0$,

$$\Pr \left(\left| \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}} \right| \geq \epsilon \right) \leq \sqrt{\frac{125 \bar{y}^2 \bar{a}^4 \bar{d}^4}{\pi^2 \underline{c}_{\Sigma}^2 \underline{c}}} \cdot n^{4\alpha-1} \cdot \frac{1}{\epsilon}.$$

So we have

$$\lim_{n \rightarrow +\infty} \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}} \xrightarrow{p} 0.$$

Note also that $\lim_{n \rightarrow +\infty} n \text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})) = \lim_{n \rightarrow +\infty} n \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))$ and for sufficiently large n we assume $n \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) \geq \underline{c}$. This ensures that $n \text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})) \geq \underline{c} > 0$ for sufficiently large n . So we have

$$\begin{aligned} \lim_{n \rightarrow +\infty} \left| \frac{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))}{\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))} - 1 \right| &= \lim_{n \rightarrow +\infty} \frac{|n \text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})) - n \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))|}{n \text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))} \\ &\leq \lim_{n \rightarrow +\infty} \frac{|n \text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}})) - n \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*))|}{\underline{c}} = 0, \end{aligned}$$

where the last equality is Claim 2 in Theorem 1.

Putting the above and (EC.14) into (EC.15), and using the Slutsky theorem we have

$$\lim_{n \rightarrow +\infty} \frac{\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}) - \mu}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\hat{\gamma}^{\text{HT}}))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

This finishes the proof. $\square$

EC.4.3. Proof of Theorem 2

Proof of Theorem 2. Recall that the objective function of (5) can be written as

$$\min_{\mathbf{a}} \mathbb{E}_{\mathbf{Y} \sim \mathcal{Y}^n} \left[\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{\Pi}^{-1} \mathbf{Y} - \frac{(\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a})^2}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \right].$$

Note that $\mathbb{E}_{\mathbf{Y} \sim \mathcal{Y}^n} [\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{\Pi}^{-1} \mathbf{Y}]$ does not depend on $\mathbf{a}$. So this minimization problem can be written as a maximization problem

$$\max_{\mathbf{a}} \mathbb{E}_{\mathbf{Y} \sim \mathcal{Y}^n} \left[\frac{(\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a})^2}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}} \right] = \max_{\mathbf{a}} \frac{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{\Pi}^{-1} \mathbb{E}_{\mathbf{Y} \sim \mathcal{Y}^n} [\mathbf{Y} \mathbf{Y}^\top] \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a}}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}}. \quad (\text{EC.16})$$

Note that $\mathbb{E}_{\mathbf{Y} \sim \mathcal{Y}^n} [\mathbf{Y} \mathbf{Y}^\top] = \sigma^2 \mathbf{I}_n + \mu^2 \mathbf{1}_n \mathbf{1}_n^\top$, where $\mathbf{I}_n$ is a $(n \times n)$ identity matrix and $\mathbf{1}_n \mathbf{1}_n^\top$ is a $(n \times n)$ matrix with all elements equal to 1. Using the above notation, (EC.16) can be written as

$$\max_{\mathbf{a}} \frac{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{\Pi}^{-1} (\sigma^2 \mathbf{I}_n + \mu^2 \mathbf{1}_n \mathbf{1}_n^\top) \mathbf{\Pi}^{-1} \mathbf{\Sigma} \mathbf{a}}{\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a}}. \quad (\text{EC.17})$$

Now we introduce a variable transformation. For any vector $\mathbf{a} \in \mathbb{R}^n$, let $\|\mathbf{a}\|_2$ be its L_2 norm, that is, $\|\mathbf{a}\|_2 = (\sum_{i=1}^n a_i^2)^{\frac{1}{2}}$. Next, because $\mathbf{\Sigma}$ is symmetric and positive semidefinite, we can define its square root matrix as $\mathbf{\Sigma}^{\frac{1}{2}}$ and its pseudo-inverse square root matrix as $\mathbf{\Sigma}^{-\frac{1}{2}}$. See Section EC.1 for their detailed definitions and properties. Using the above definitions, we define

$$\mathbf{v} = \frac{\mathbf{\Sigma}^{\frac{1}{2}} \mathbf{a}}{\|\mathbf{\Sigma}^{\frac{1}{2}} \mathbf{a}\|_2}$$

to be a unit length vector. Under this variable transformation, (EC.17) can be written as

$$\max_{\mathbf{v} \in \mathcal{V}} \mathbf{v}^\top \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{\Pi}^{-1} (\sigma^2 \mathbf{I}_n + \mu^2 \mathbf{1}_n \mathbf{1}_n^\top) \mathbf{\Pi}^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{v}, \quad (\text{EC.18})$$

where $\mathcal{V}$ is a unit sphere defined as $\mathcal{V} = \{\mathbf{v} \in \mathbb{R}^n \mid \|\mathbf{v}\|_2 = 1\}$.

Now that we know $\mathbf{\Sigma}^{\frac{1}{2}}$ is symmetric because $\mathbf{\Sigma}$ is symmetric and positive semidefinite, $\mathbf{\Pi}^{-1}$ is diagonal thus symmetric, and $(\sigma^2 \mathbf{I}_n + \mu^2 \mathbf{1}_n \mathbf{1}_n^\top)$ is also symmetric, we have

$$\left(\mathbf{\Sigma}^{\frac{1}{2}} \mathbf{\Pi}^{-1} (\sigma^2 \mathbf{I}_n + \mu^2 \mathbf{1}_n \mathbf{1}_n^\top) \mathbf{\Pi}^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \right)^\top = \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{\Pi}^{-1} (\sigma^2 \mathbf{I}_n + \mu^2 \mathbf{1}_n \mathbf{1}_n^\top) \mathbf{\Pi}^{-1} \mathbf{\Sigma}^{\frac{1}{2}}$$

so it is also symmetric. Now denote $\mathbf{M} = \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{\Pi}^{-1} (\sigma^2 \mathbf{I}_n + \mu^2 \mathbf{1}_n \mathbf{1}_n^\top) \mathbf{\Pi}^{-1} \mathbf{\Sigma}^{\frac{1}{2}}$. Using Lemma EC.3 the Rayleigh Quotient (Horn and Johnson (2012) Theorem 4.2.2), the optimal objective value of

(EC.18) is the largest eigenvalue of $\mathbf{M}$, denoted as $\lambda_1(\mathbf{M})$. So the optimal variance reduction is given by

$$\mathbb{E}_{\mathbf{Y} \sim \mathcal{Y}^n} \left[\text{Var}(\hat{\mu}^{\text{HT}}) - \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) \right] = \frac{\lambda_1(\mathbf{M})}{n^2}.$$

Additionally, using Lemma EC.3 the Rayleigh Quotient (Horn and Johnson (2012) Theorem 4.2.2), the optimal solution to (EC.18) is the principal eigenvector that corresponds to $\lambda_1(\mathbf{M})$, denoted as $\mathbf{u}_1(\mathbf{M})$. Substituting this back to the definition of $\mathbf{v}$ we have that the optimal basis is

$$\mathbf{a}^* = c \boldsymbol{\Sigma}^{-\frac{1}{2}} \mathbf{u}_1(\mathbf{M})$$

where $c \neq 0$ is any constant. □

EC.4.4. Proof of Proposition 1

We present the following Lemma EC.6 which will be useful in the proof of Proposition 1. To introduce Lemma EC.6, we denote the following notation. For any matrix $\mathbf{E} \in \mathbb{R}^{n \times n}$, denote $\text{range}(\mathbf{E})$ to be the column space of matrix $\mathbf{E}$, defined as

$$\text{range}(\mathbf{E}) = \{ \mathbf{E} \mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n \}.$$

Lemma EC.6 *Let $\mathbf{E} \in \mathbb{R}^{n \times n}$ be any symmetric matrix. Let $\lambda_1(\mathbf{M})$ be the largest eigenvalue of $\mathbf{M}$ and $\mathbf{u}_1(\mathbf{M})$ be the eigenvector that corresponds to $\lambda_1(\mathbf{M})$, where $\mathbf{M}$ is a symmetric matrix defined as $\mathbf{M} = \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{E} \boldsymbol{\Sigma}^{\frac{1}{2}}$. When $\lambda_1(\mathbf{M}) > 0$, we have*

$$\mathbf{u}_1(\mathbf{M}) \in \text{range}(\boldsymbol{\Sigma}).$$

Proof of Lemma EC.6. Note that $\mathbf{u}_1(\mathbf{M}) \in \text{range}(\mathbf{M})$ because $\mathbf{u}_1(\mathbf{M})$ is an eigenvector of the symmetric matrix $\mathbf{M}$. Next, we show that $\text{range}(\mathbf{M}) \subseteq \text{range}(\boldsymbol{\Sigma})$. To see this, note that for any vector $\mathbf{x}$,

$$\mathbf{M} \mathbf{x} = \boldsymbol{\Sigma}^{\frac{1}{2}} \left(\mathbf{E} \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{x} \right).$$

Because $(\mathbf{E} \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{x})$ may not span the entire space of $\mathbb{R}^n$, we have that

$$\text{range}(\mathbf{M}) \subseteq \text{range}(\boldsymbol{\Sigma}^{\frac{1}{2}}).$$

Next, because $\boldsymbol{\Sigma} = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^{-1}$ and $\boldsymbol{\Sigma}^{\frac{1}{2}} = \mathbf{U} \boldsymbol{\Lambda}^{\frac{1}{2}} \mathbf{U}^{-1}$ have the same directions along the eigenvectors corresponding to positive eigenvalues, we have

$$\text{range}(\boldsymbol{\Sigma}^{\frac{1}{2}}) = \text{range}(\boldsymbol{\Sigma}).$$

Putting all above together, we have

$$\mathbf{u}_1(\mathbf{M}) \in \text{range}(\mathbf{M}) \subseteq \text{range}(\boldsymbol{\Sigma}^{\frac{1}{2}}) = \text{range}(\boldsymbol{\Sigma}).$$

This finishes the proof. □

Now we prove Proposition 1 as follows.

Proof of Proposition 1. As a succinct notation, we drop $\mathbf{M}$ and write $\mathbf{u}_1(\mathbf{M}) = \mathbf{u}_1$. Denote the following diagonal matrix $\mathbb{1}^+ = \mathbf{diag}(\mathbb{1}\{\lambda_1 > 0\}, \mathbb{1}\{\lambda_2 > 0\}, \dots, \mathbb{1}\{\lambda_n > 0\})$, where λ_i stands for the i -th largest eigenvalue of matrix Σ . Denote $\mathbf{P}_\Sigma = \Sigma^{-\frac{1}{2}} \Sigma \Sigma^{-\frac{1}{2}}$.

Recall the eigen-decomposition $\Sigma = \mathbf{U} \Lambda \mathbf{U}^{-1}$. So we know that

$$\mathbf{P}_\Sigma = \Sigma^{-\frac{1}{2}} \Sigma \Sigma^{-\frac{1}{2}} = \mathbf{U} \mathbb{1}^+ \mathbf{U}^{-1}$$

is a projection matrix, that is, if a vector $\mathbf{x} \in \text{range}(\Sigma)$, we have $\mathbf{P}_\Sigma \mathbf{x} = \mathbf{x}$.

Using Lemma EC.6, we have

$$\mathbf{u}_1 \in \text{range}(\Sigma).$$

Consequently, $\mathbf{P}_\Sigma \mathbf{u}_1 = \mathbf{u}_1$. Next, we have

$$(\mathbf{a}^*)^\top \Sigma \mathbf{a}^* = n \mathbf{u}_1^\top \Sigma^{-\frac{1}{2}} \Sigma \Sigma^{-\frac{1}{2}} \mathbf{u}_1 = n \mathbf{u}_1^\top \mathbf{u}_1 = n,$$

where the last equality is because $\mathbf{u}_1$ is an eigenvector so it is unit length. $\square$

EC.4.5. Proof of Corollary 1

Proof of Corollary 1. The proof of Corollary 1 is very straightforward. Recall from Section 2.4 that if the outcomes $\mathbf{Y}$ were known to take values $Y_i = y_i$ for any $i \in [n]$, the problem

$$\min_{\mathbf{a}} \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*)) = \min_{\mathbf{a}} \frac{1}{n^2} \left(\mathbf{y}^\top \mathbf{\Pi}^{-1} \Sigma \mathbf{\Pi}^{-1} \mathbf{y} - \frac{(\mathbf{y}^\top \mathbf{\Pi}^{-1} \Sigma \mathbf{a})^2}{\mathbf{a}^\top \Sigma \mathbf{a}} \right)$$

can be easily solved by setting $\mathbf{a} = \mathbf{\Pi}^{-1} \mathbf{y}$. Now because we assume that the outcomes $Y_i = y$ take the same unknown constant, one optimal solution is given by $\mathbf{a} = \mathbf{\Pi}^{-1} \mathbf{1}_n \cdot y$. Because the optimal basis is scale invariant, so for any $c \neq 0$, the optimal basis takes the form of $\mathbf{a}^* = c \mathbf{\Pi}^{-1} \mathbf{1}_n$. $\square$

EC.4.6. Proof of Lemma 2

Proof of Lemma 2. Recall that we have defined the $2n$ -dimensional exposure condition indicators $\mathbf{D} = (\mathbb{1}\{\mathcal{E}_1(1)\}, \dots, \mathbb{1}\{\mathcal{E}_n(1)\}, \mathbb{1}\{\mathcal{E}_1(0)\}, \dots, \mathbb{1}\{\mathcal{E}_n(0)\})^\top$ and defined the $(2n \times 2n)$ covariance matrix $\Omega = \text{Var}_{\mathcal{W}}(\mathbf{D})$.

For notational convenience, we define the following matrices,

$$\Sigma(1,1) = \begin{bmatrix} \text{Var}(\mathbb{1}\{\mathcal{E}_1(1)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_2(1)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) \\ \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_2(1)\}) & \text{Var}(\mathbb{1}\{\mathcal{E}_2(1)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(1)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(1)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) & \dots & \text{Var}(\mathbb{1}\{\mathcal{E}_n(1)\}) \end{bmatrix},$$

$$\mathbf{\Sigma}(1,0) = \begin{bmatrix} \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_1(0)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_2(0)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) \\ \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_2(0)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(1)\}, \mathbb{1}\{\mathcal{E}_2(0)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(1)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\mathbb{1}\{\mathcal{E}_1(1)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(1)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_n(1)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) \end{bmatrix},$$

$$\mathbf{\Sigma}(0,1) = \begin{bmatrix} \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_1(1)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_2(1)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) \\ \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_2(1)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(0)\}, \mathbb{1}\{\mathcal{E}_2(1)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(0)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(0)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_n(0)\}, \mathbb{1}\{\mathcal{E}_n(1)\}) \end{bmatrix},$$

and

$$\mathbf{\Sigma}(0,0) = \begin{bmatrix} \text{Var}(\mathbb{1}\{\mathcal{E}_1(0)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_2(0)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) \\ \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_2(0)\}) & \text{Var}(\mathbb{1}\{\mathcal{E}_2(0)\}) & \dots & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(0)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\mathbb{1}\{\mathcal{E}_1(0)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) & \text{Cov}(\mathbb{1}\{\mathcal{E}_2(0)\}, \mathbb{1}\{\mathcal{E}_n(0)\}) & \dots & \text{Var}(\mathbb{1}\{\mathcal{E}_n(0)\}) \end{bmatrix}.$$

So we have

$$\mathbf{\Omega} = \begin{bmatrix} \mathbf{\Sigma}(1,1) & \mathbf{\Sigma}(1,0) \\ \mathbf{\Sigma}(0,1) & \mathbf{\Sigma}(0,0) \end{bmatrix}.$$

Next, we use the above notations to find the expressions of $\gamma^*(1)$ and $\gamma^*(0)$. We start with $\text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1))$.

$$\begin{aligned} & \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ &= \text{E}[\hat{\tau}^{\text{HT}} \hat{X}(1)] - \tau \text{E}[\hat{X}(1)] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{E} \left[\left(\frac{Y_i(1) \mathbb{1}\{\mathcal{E}_i(1)\}}{\text{Pr}(\mathcal{E}_i(1))} - \frac{Y_i(0) \mathbb{1}\{\mathcal{E}_i(0)\}}{\text{Pr}(\mathcal{E}_i(0))} \right) a_j(1) \mathbb{1}\{\mathcal{E}_j(1)\} \right] - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (Y_i(1) - Y_i(0)) a_j(1) \text{Pr}(\mathcal{E}_j(1)) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(1)}{\text{Pr}(\mathcal{E}_i(1))} \text{Cov}(\mathbb{1}\{\mathcal{E}_i(1)\}, \mathbb{1}\{\mathcal{E}_j(1)\}) a_j(1) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(0)}{\text{Pr}(\mathcal{E}_i(0))} \text{Cov}(\mathbb{1}\{\mathcal{E}_i(0)\}, \mathbb{1}\{\mathcal{E}_j(1)\}) a_j(1) \\ &= \frac{1}{n^2} \mathbf{Y}(\delta(1))^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}(1,1) \mathbf{a}(1) - \frac{1}{n^2} \mathbf{Y}(\delta(0))^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}(0,1) \mathbf{a}(1). \end{aligned}$$

Similarly, we derive the expression for $\text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0))$.

$$\begin{aligned} & \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \\ &= \text{E}[\hat{\tau}^{\text{HT}} \hat{X}(0)] - \tau \text{E}[\hat{X}(0)] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{E} \left[\left(\frac{Y_i(1) \mathbb{1}\{\mathcal{E}_i(1)\}}{\text{Pr}(\mathcal{E}_i(1))} - \frac{Y_i(0) \mathbb{1}\{\mathcal{E}_i(0)\}}{\text{Pr}(\mathcal{E}_i(0))} \right) a_j(0) \mathbb{1}\{\mathcal{E}_j(0)\} \right] - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (Y_i(1) - Y_i(0)) a_j(0) \text{Pr}(\mathcal{E}_j(0)) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(1)}{\text{Pr}(\mathcal{E}_i(1))} \text{Cov}(\mathbb{1}\{\mathcal{E}_i(1)\}, \mathbb{1}\{\mathcal{E}_j(0)\}) a_j(0) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(0)}{\text{Pr}(\mathcal{E}_i(0))} \text{Cov}(\mathbb{1}\{\mathcal{E}_i(0)\}, \mathbb{1}\{\mathcal{E}_j(0)\}) a_j(0) \\ &= \frac{1}{n^2} \mathbf{Y}(\delta(1))^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}(1,0) \mathbf{a}(0) - \frac{1}{n^2} \mathbf{Y}(\delta(0))^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}(0,0) \mathbf{a}(0). \end{aligned}$$

Next, we derive the expression for $\text{Cov}(\widehat{X}(1), \widehat{X}(0))$.

$$\begin{aligned}
\text{Cov}(\widehat{X}(1), \widehat{X}(0)) &= \mathbb{E}[\widehat{X}(1)\widehat{X}(0)] - \mathbb{E}[\widehat{X}(1)]\mathbb{E}[\widehat{X}(0)] \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \mathbb{E}[\mathbb{1}\{\mathcal{E}_i(1), \mathcal{E}_j(0)\}] a_j(0) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \Pr(\mathcal{E}_i(1)) \Pr(\mathcal{E}_j(0)) a_j(0) \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \text{Cov}(\mathbb{1}\{\mathcal{E}_i(1)\}, \mathbb{1}\{\mathcal{E}_j(0)\}) a_j(0) \\
&= \frac{1}{n^2} \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 0) \mathbf{a}(0) \\
&= \frac{1}{n^2} \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 1) \mathbf{a}(1).
\end{aligned}$$

Similarly, we derive the expression for $\text{Var}(\widehat{X}(1))$.

$$\begin{aligned}
\text{Var}(\widehat{X}(1)) &= \mathbb{E}[\widehat{X}(1)^2] - \mathbb{E}[\widehat{X}(1)]^2 \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \mathbb{E}[\mathbb{1}\{\mathcal{E}_i(1), \mathcal{E}_j(1)\}] a_j(1) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \Pr(\mathcal{E}_i(1)) \Pr(\mathcal{E}_j(1)) a_j(1) \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \text{Cov}(\mathbb{1}\{\mathcal{E}_i(1)\}, \mathbb{1}\{\mathcal{E}_j(1)\}) a_j(1) \\
&= \frac{1}{n^2} \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 1) \mathbf{a}(1).
\end{aligned}$$

Finally, we derive the expression for $\text{Var}(\widehat{X}(0))$.

$$\begin{aligned}
\text{Var}(\widehat{X}(0)) &= \mathbb{E}[\widehat{X}(0)^2] - \mathbb{E}[\widehat{X}(0)]^2 \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(0) \mathbb{E}[\mathbb{1}\{\mathcal{E}_i(0), \mathcal{E}_j(0)\}] a_j(0) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(0) \Pr(\mathcal{E}_i(0)) \Pr(\mathcal{E}_j(0)) a_j(0) \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(0) \text{Cov}(\mathbb{1}\{\mathcal{E}_i(0)\}, \mathbb{1}\{\mathcal{E}_j(0)\}) a_j(0) \\
&= \frac{1}{n^2} \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 0) \mathbf{a}(0).
\end{aligned}$$

Putting all the above into the expression of $(\gamma^*(1), \gamma^*(0))$, we have

$$\begin{bmatrix} \gamma^*(1) \\ \gamma^*(0) \end{bmatrix} = \begin{bmatrix} \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 1) \mathbf{a}(1) & \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 0) \mathbf{a}(0) \\ \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 1) \mathbf{a}(1) & \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 0) \mathbf{a}(0) \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 1) \boldsymbol{\Pi}(1)^{-1} \mathbf{Y}(\delta(1)) - \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 0) \boldsymbol{\Pi}(0)^{-1} \mathbf{Y}(\delta(0)) \\ \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 1) \boldsymbol{\Pi}(1)^{-1} \mathbf{Y}(\delta(1)) - \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 0) \boldsymbol{\Pi}(0)^{-1} \mathbf{Y}(\delta(0)) \end{bmatrix}.$$

Note that

$$\begin{bmatrix} \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 1) \mathbf{a}(1) & \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 0) \mathbf{a}(0) \\ \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 1) \mathbf{a}(1) & \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 0) \mathbf{a}(0) \end{bmatrix} = \mathbf{B}^\top \boldsymbol{\Omega} \mathbf{B},$$

and that

$$\begin{bmatrix} \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 1) \boldsymbol{\Pi}(1)^{-1} \mathbf{Y}(\delta(1)) - \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 0) \boldsymbol{\Pi}(0)^{-1} \mathbf{Y}(\delta(0)) \\ \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 1) \boldsymbol{\Pi}(1)^{-1} \mathbf{Y}(\delta(1)) - \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 0) \boldsymbol{\Pi}(0)^{-1} \mathbf{Y}(\delta(0)) \end{bmatrix} = \mathbf{B}^\top \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y}.$$

These succinct notations leads to

$$(\gamma^*(1), \gamma^*(0))^\top = (\mathbf{B}^\top \boldsymbol{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y},$$

which finishes the proof for the coefficients $(\gamma^*(1), \gamma^*(0))^\top$.

Finally, we focus on the expression for $\text{Var}(\hat{\tau}^{\text{HT}})$.

$$\begin{aligned} & \text{Var}(\hat{\tau}^{\text{HT}}) \\ &= \mathbb{E}[(\hat{\tau}^{\text{HT}})^2] - \tau^2 \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{E} \left[\left(\frac{Y_i(1) \mathbb{1}\{\mathcal{E}_i(1)\}}{\Pr(\mathcal{E}_i(1))} - \frac{Y_i(0) \mathbb{1}\{\mathcal{E}_i(0)\}}{\Pr(\mathcal{E}_i(0))} \right) \left(\frac{Y_j(1) \mathbb{1}\{\mathcal{E}_j(1)\}}{\Pr(\mathcal{E}_j(1))} - \frac{Y_j(0) \mathbb{1}\{\mathcal{E}_j(0)\}}{\Pr(\mathcal{E}_j(0))} \right) \right] \\ &\quad - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (Y_i(1) - Y_i(0)) (Y_j(1) - Y_j(0)) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(1)}{\Pr(\mathcal{E}_i(1))} \text{Cov}(\mathbb{1}\{\mathcal{E}_i(1)\}, \mathbb{1}\{\mathcal{E}_j(1)\}) \frac{Y_j(1)}{\Pr(\mathcal{E}_j(1))} \\ &\quad - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(1)}{\Pr(\mathcal{E}_i(1))} \text{Cov}(\mathbb{1}\{\mathcal{E}_i(1)\}, \mathbb{1}\{\mathcal{E}_j(0)\}) \frac{Y_j(0)}{\Pr(\mathcal{E}_j(0))} \\ &\quad - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(0)}{\Pr(\mathcal{E}_i(0))} \text{Cov}(\mathbb{1}\{\mathcal{E}_i(0)\}, \mathbb{1}\{\mathcal{E}_j(1)\}) \frac{Y_j(1)}{\Pr(\mathcal{E}_j(1))} \\ &\quad + \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(0)}{\Pr(\mathcal{E}_i(0))} \text{Cov}(\mathbb{1}\{\mathcal{E}_i(0)\}, \mathbb{1}\{\mathcal{E}_j(0)\}) \frac{Y_j(0)}{\Pr(\mathcal{E}_j(0))} \\ &= \frac{1}{n^2} \mathbf{Y}(\delta(1))^\top \boldsymbol{\Pi}(1)^{-1} \boldsymbol{\Sigma}(1, 1) \boldsymbol{\Pi}(1)^{-1} \mathbf{Y}(\delta(1)) - \frac{1}{n^2} \mathbf{Y}(\delta(1))^\top \boldsymbol{\Pi}(1)^{-1} \boldsymbol{\Sigma}(1, 0) \boldsymbol{\Pi}(0)^{-1} \mathbf{Y}(\delta(0)) \\ &\quad - \frac{1}{n^2} \mathbf{Y}(\delta(0))^\top \boldsymbol{\Pi}(0)^{-1} \boldsymbol{\Sigma}(0, 1) \boldsymbol{\Pi}(1)^{-1} \mathbf{Y}(\delta(1)) + \frac{1}{n^2} \mathbf{Y}(\delta(0))^\top \boldsymbol{\Pi}(0)^{-1} \boldsymbol{\Sigma}(0, 0) \boldsymbol{\Pi}(0)^{-1} \mathbf{Y}(\delta(0)). \end{aligned}$$

Putting all the above into expression (EC.28), we have

$$\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) = \frac{1}{n^2} \left(\mathbf{Y}^\top \boldsymbol{\Pi}^{-1} \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y} - \mathbf{Y}^\top \boldsymbol{\Pi}^{-1} \boldsymbol{\Omega} \mathbf{B} (\mathbf{B}^\top \boldsymbol{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y} \right),$$

which finishes the proof for the variance $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$. $\square$

EC.4.7. Proof of Theorem 3

We present the following Lemma EC.7 which will be useful in the proof of Theorem 3. We first introduce a vector notation

$$\boldsymbol{\gamma}^* = (\gamma^*(1), \gamma^*(0))^\top.$$

Lemma EC.7 *Under Assumptions 1[†], 2[‡], and 4, and assuming the potential outcomes $|Y_i(\delta(1))| \leq \bar{y}$ and $|Y_i(\delta(0))| \leq \bar{y}$ are all bounded, the optimal coefficients $\gamma^*(1)$ and $\gamma^*(0)$ are bounded, that is,*

$$|\gamma^*(1)| \leq \frac{2^{\frac{1}{2}} \bar{y} \bar{d}^{\frac{1}{2}}}{\pi \lambda_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}}, \quad |\gamma^*(0)| \leq \frac{2^{\frac{1}{2}} \bar{y} \bar{d}^{\frac{1}{2}}}{\pi \lambda_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}}.$$

Proof of Lemma EC.7. Recall that the variance minimizing coefficients are given by

$$\boldsymbol{\gamma}^* = (\mathbf{B}^\top \boldsymbol{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y}.$$

By Assumption 1[†]-(i) and the boundedness of potential outcomes, every element of $\boldsymbol{\Pi}^{-1} \mathbf{Y}$ is bounded in absolute value by $\frac{\bar{y}}{\pi}$. So we have

$$\mathbf{Y}^\top \boldsymbol{\Pi}^{-1} \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y} \leq \frac{\bar{y}^2}{\pi^2} \sum_{k=1}^{2n} \sum_{l=1}^{2n} |\Omega_{kl}| \leq \frac{2\bar{y}^2 \bar{d}}{\pi^2} n^{1+\alpha},$$

where the last inequality is due to Assumption 1[†]-(ii).

Next, note that

$$(\boldsymbol{\gamma}^*)^\top \mathbf{B}^\top \boldsymbol{\Omega} \mathbf{B} \boldsymbol{\gamma}^* = \mathbf{Y}^\top \boldsymbol{\Pi}^{-1} \boldsymbol{\Omega} \mathbf{B} (\mathbf{B}^\top \boldsymbol{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y} \leq \mathbf{Y}^\top \boldsymbol{\Pi}^{-1} \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y},$$

where the inequality follows because the middle term is the variance reduction from projecting $\boldsymbol{\Pi}^{-1} \mathbf{Y}$ onto the column space of $\mathbf{B}$, and hence cannot exceed the total variance $\mathbf{Y}^\top \boldsymbol{\Pi}^{-1} \boldsymbol{\Omega} \boldsymbol{\Pi}^{-1} \mathbf{Y}$.

By Assumption 2[†]-(ii),

$$(\boldsymbol{\gamma}^*)^\top \mathbf{B}^\top \boldsymbol{\Omega} \mathbf{B} \boldsymbol{\gamma}^* \geq \underline{\lambda}_\Omega n \left((\gamma^*(1))^2 + (\gamma^*(0))^2 \right).$$

Putting the above inequalities together, we have

$$(\gamma^*(1))^2 + (\gamma^*(0))^2 \leq \frac{2\bar{y}^2 \bar{d}}{\pi^2 \underline{\lambda}_\Omega} n^\alpha.$$

Taking square root we finish the proof. $\square$

Now we prove Theorem 3 as follows.

Proof of Theorem 3. We prove the three claims in Theorem 3 one by one.

Claim 1: Consistency. In the proof of this claim, we assume $\alpha < \frac{1}{3}$. We prove consistency through two steps. First, we show that

$$\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) \xrightarrow{P} 0.$$

Note that for each i ,

$$\hat{Y}_i(\delta(1)) - Y_i(\delta(1)) = (\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))) \frac{Y_i(\delta(1))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))},$$

and

$$\hat{Y}_i(\delta(0)) - Y_i(\delta(0)) = (\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))) \frac{Y_i(\delta(0))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))}.$$

Next, define for $l \in \{1, 2\}$,

$$Z_{i,l}(1) = \frac{Y_i(\delta(1))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))^2} (\mathbf{\Omega B}(\mathbf{B}^\top \mathbf{\Omega B})^{-1})_{il}$$

and

$$Z_{i,l}(0) = \frac{Y_i(\delta(0))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))^2} (\mathbf{\Omega B}(\mathbf{B}^\top \mathbf{\Omega B})^{-1})_{(n+i)l}.$$

where we use $(\mathbf{\Omega B}(\mathbf{B}^\top \mathbf{\Omega B})^{-1})_{il}$ to stand for the element in the i -th row and the l -st column and $(\mathbf{\Omega B}(\mathbf{B}^\top \mathbf{\Omega B})^{-1})_{(n+i)l}$ to stand for the element in the $n+i$ -th row and the l -st column.

Using the above definitions, we have

$$\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1) = \sum_{i=1}^n Z_{i,1}(1) (\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))) - \sum_{i=1}^n Z_{i,1}(0) (\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))),$$

and

$$\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0) = \sum_{i=1}^n Z_{i,2}(1) (\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))) - \sum_{i=1}^n Z_{i,2}(0) (\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))).$$

We next upper bound $|Z_{i,l}(1)|$ and $|Z_{i,l}(0)|$ for $l \in \{1, 2\}$. By Assumption 1[†]-(i) and boundedness of the potential outcomes,

$$\left| \frac{Y_i(\delta(1))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))^2} \right| \leq \frac{\bar{y}}{\underline{\pi}^2}, \quad \left| \frac{Y_i(\delta(0))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))^2} \right| \leq \frac{\bar{y}}{\underline{\pi}^2}.$$

By Assumptions 1[†]-(ii) and 2[†]-(i), each element $(\mathbf{\Omega B})_{kl}$ in the $(2n \times 2)$ matrix $\mathbf{\Omega B}$ is bounded by

$$|(\mathbf{\Omega B})_{kl}| \leq \bar{a} \bar{d} n^\alpha.$$

By Assumption 2[†]-(ii),

$$\|(\mathbf{B}^\top \mathbf{\Omega B})^{-1}\|_2 \leq \frac{1}{\underline{\lambda}_\Omega n},$$

and so every element of the inverse matrix is upper bounded by $\frac{1}{\underline{\lambda}_\Omega n}$. Now define $\bar{z} = \frac{2\bar{y}\bar{a}\bar{d}}{\underline{\pi}^2 \underline{\lambda}_\Omega} > 0$.

We have for all $i \in [n]$ and $l \in \{1, 2\}$,

$$|Z_{i,l}(1)| \leq \bar{z} n^{\alpha-1}, \quad |Z_{i,l}(0)| \leq \bar{z} n^{\alpha-1}.$$

Next, it is easy to see that

$$\mathbb{E}_{\mathcal{W}}[\hat{\gamma}^{\text{HT}}(1)] = \gamma^*(1), \quad \mathbb{E}_{\mathcal{W}}[\hat{\gamma}^{\text{HT}}(0)] = \gamma^*(0).$$

We calculate $\text{Var}_{\mathcal{W}}(\hat{\gamma}^{\text{HT}}(1))$ and $\text{Var}_{\mathcal{W}}(\hat{\gamma}^{\text{HT}}(0))$ as follows. Note that,

$$\text{Var}_{\mathcal{W}}(\hat{\gamma}^{\text{HT}}(1)) \leq \bar{z}^2 n^{2\alpha-2} \sum_{k=1}^{2n} \sum_{l=1}^{2n} |\Omega_{kl}| \leq 2\bar{z}^2 \bar{d} n^{3\alpha-1},$$

where the last inequality is due to Assumption 1[†]-(ii). Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr \left(|\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1)| \geq \epsilon \right) \leq 2\bar{z}^2 \bar{d} n^{3\alpha-1} \frac{1}{\epsilon^2}.$$

Similarly, we have

$$\Pr \left(|\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0)| \geq \epsilon \right) \leq 2\bar{z}^2 \bar{d} n^{3\alpha-1} \frac{1}{\epsilon^2}.$$

Next, we calculate the variance of the two control variates. Note that

$$\text{Var}_{\mathcal{W}}(\hat{X}(1)) = \frac{1}{n^2} \text{Var}_{\mathcal{W}} \left(\sum_{i=1}^n a_i(1) \mathbb{1}\{\mathcal{E}_i(1)\} \right) \leq \bar{a}^2 \bar{d} n^{\alpha-1}.$$

Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr \left(|\hat{X}(1) - \mathbb{E}_{\mathcal{W}}[\hat{X}(1)]| \geq \epsilon \right) \leq \bar{a}^2 \bar{d} n^{\alpha-1} \frac{1}{\epsilon^2}.$$

Similarly,

$$\Pr \left(|\hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)]| \geq \epsilon \right) \leq \bar{a}^2 \bar{d} n^{\alpha-1} \frac{1}{\epsilon^2}.$$

Finally, define

$$\begin{aligned} \Delta_n &= \hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) \\ &= -(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))(\hat{X}(1) - \mathbb{E}_{\mathcal{W}}[\hat{X}(1)]) - (\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))(\hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)]). \end{aligned}$$

For any $\epsilon > 0$, we have

$$\begin{aligned} \lim_{n \rightarrow +\infty} \Pr(|\Delta_n| \geq \epsilon) &\leq \lim_{n \rightarrow +\infty} \left(\Pr \left(|\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1)| \geq \left(\frac{\epsilon}{2} \right)^{\frac{1}{2}} \right) + \Pr \left(|\hat{X}(1) - \mathbb{E}[\hat{X}(1)]| \geq \left(\frac{\epsilon}{2} \right)^{\frac{1}{2}} \right) \right. \\ &\quad \left. + \Pr \left(|\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0)| \geq \left(\frac{\epsilon}{2} \right)^{\frac{1}{2}} \right) + \Pr \left(|\hat{X}(0) - \mathbb{E}[\hat{X}(0)]| \geq \left(\frac{\epsilon}{2} \right)^{\frac{1}{2}} \right) \right) \\ &\leq \lim_{n \rightarrow +\infty} \left(4\bar{z}^2 \bar{d} n^{3\alpha-1} \frac{2}{\epsilon} + 2\bar{a}^2 \bar{d} n^{\alpha-1} \frac{2}{\epsilon} \right) \\ &= 0, \end{aligned}$$

where the first inequality holds because, if $|\Delta_n| \geq \epsilon$ holds, then at least one of the four conditions must hold; the last and only equality holds because $\alpha < \frac{1}{3}$. So we have shown that

$$\Delta_n \xrightarrow{p} 0.$$

Second, we show that $\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))$ is a consistent estimator of $\tau(\delta(1), \delta(0))$. Note that

$$\begin{aligned} \hat{\tau}^{\text{HT}} - \tau(\delta(1), \delta(0)) &= \frac{1}{n} \sum_{i=1}^n \frac{Y_i(\delta(1))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))} (\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))) \\ &\quad - \frac{1}{n} \sum_{i=1}^n \frac{Y_i(\delta(0))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))} (\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))). \end{aligned}$$

So we have

$$\text{Var}_{\mathcal{W}}(\hat{\tau}^{\text{HT}}) \leq \frac{1}{n^2} \frac{\bar{y}^2}{\bar{\pi}^2} \sum_{k=1}^{2n} \sum_{l=1}^{2n} |\Omega_{kl}| \leq \frac{2\bar{y}^2 \bar{d}}{\bar{\pi}^2} n^{\alpha-1}.$$

Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr\left(|\hat{\tau}^{\text{HT}} - \tau(\delta(1), \delta(0))| \geq \epsilon\right) \leq \frac{2\bar{y}^2 \bar{d}}{\bar{\pi}^2} n^{\alpha-1} \frac{1}{\epsilon^2}.$$

Next, note that

$$\begin{aligned} \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0)) \\ = (\hat{\tau}^{\text{HT}} - \tau(\delta(1), \delta(0))) - \gamma^*(1)(\hat{X}(1) - \mathbb{E}_{\mathcal{W}}[\hat{X}(1)]) - \gamma^*(0)(\hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)]). \end{aligned}$$

So, for any $\epsilon > 0$,

$$\begin{aligned} & \lim_{n \rightarrow +\infty} \Pr\left(|\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))| \geq \epsilon\right) \\ & \leq \lim_{n \rightarrow +\infty} \left(\Pr\left(|\hat{\tau}^{\text{HT}} - \tau(\delta(1), \delta(0))| \geq \frac{\epsilon}{3}\right) + \Pr\left(|\gamma^*(1)| |\hat{X}(1) - \mathbb{E}_{\mathcal{W}}[\hat{X}(1)]| \geq \frac{\epsilon}{3}\right) \right. \\ & \quad \left. + \Pr\left(|\gamma^*(0)| |\hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)]| \geq \frac{\epsilon}{3}\right) \right) \\ & \leq \lim_{n \rightarrow +\infty} \left(\frac{2\bar{y}^2 \bar{d}}{\bar{\pi}^2} n^{\alpha-1} \frac{9}{\epsilon^2} + 2\bar{a}^2 \bar{d} n^{\alpha-1} \frac{9}{\epsilon^2} \left(\frac{2^{\frac{1}{2}} \bar{y} \bar{d}^{\frac{1}{2}}}{\pi \lambda_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \right)^2 \right) \\ & = 0, \end{aligned}$$

where the first inequality holds because if $|\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))| \geq \epsilon$ holds, then at least one of the three conditions must hold; the second inequality is due to Lemma EC.7; the last equality holds because $\alpha < \frac{1}{3}$. Therefore,

$$\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) \xrightarrow{P} \tau(\delta(1), \delta(0)).$$

Combining this with $\Delta_n \xrightarrow{P} 0$, we have

$$\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) \xrightarrow{P} \tau(\delta(1), \delta(0)).$$

Claim 2: Asymptotic Variance. In the proof of this claim, we assume $\alpha < \frac{1}{4}$.

Recall that we have defined

$$\Delta_n = \hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)).$$

Then

$$\begin{aligned} & \text{Var}_{\mathcal{W}}\left(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))\right) \\ & = \text{Var}_{\mathcal{W}}\left(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))\right) + \text{Var}_{\mathcal{W}}\left(\Delta_n\right) + 2\text{Cov}_{\mathcal{W}}\left(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n\right). \quad (\text{EC.19}) \end{aligned}$$

It suffices to prove that both

$$n\text{Var}_{\mathcal{W}}(\Delta_n) \rightarrow 0$$

and

$$n\text{Cov}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n) \rightarrow 0.$$

We first show that $n\text{Var}_{\mathcal{W}}(\Delta_n) \rightarrow 0$. Because $0 \leq \text{Var}_{\mathcal{W}}(\Delta_n) \leq \text{E}_{\mathcal{W}}[\Delta_n^2]$, it suffices to show that $n\text{E}_{\mathcal{W}}[\Delta_n^2] \rightarrow 0$. Recall that

$$\Delta_n = -(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))(\hat{X}(1) - \text{E}[\hat{X}(1)]) - (\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))(\hat{X}(0) - \text{E}[\hat{X}(0)]).$$

So we have,

$$\begin{aligned} \text{E}_{\mathcal{W}}[\Delta_n^2] &\leq 2\text{E}_{\mathcal{W}}[(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))^2(\hat{X}(1) - \text{E}[\hat{X}(1)])^2] + 2\text{E}_{\mathcal{W}}[(\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))^2(\hat{X}(0) - \text{E}[\hat{X}(0)])^2] \\ &\leq 2\text{E}_{\mathcal{W}}[(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))^4]^{\frac{1}{2}} \text{E}_{\mathcal{W}}[(\hat{X}(1) - \text{E}[\hat{X}(1)])^4]^{\frac{1}{2}} \\ &\quad + 2\text{E}_{\mathcal{W}}[(\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))^4]^{\frac{1}{2}} \text{E}_{\mathcal{W}}[(\hat{X}(0) - \text{E}[\hat{X}(0)])^4]^{\frac{1}{2}}, \end{aligned}$$

where the second inequality is due to Cauchy-Schwarz inequality.

We next upper bound the four terms separately. Note that

$$\begin{aligned} &\text{E}_{\mathcal{W}}[(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))^4] \\ &= \text{E}_{\mathcal{W}}\left[\left(\sum_{i=1}^n Z_{i,1}(1)(\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1))) - \sum_{i=1}^n Z_{i,1}(0)(\mathbb{1}\{\mathcal{E}_i(0)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(0)))\right)^4\right] \\ &\leq \bar{z}^4 n^{4\alpha-4} \text{E}_{\mathcal{W}}\left[\left(\sum_{i=1}^n (\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1))) - \sum_{i=1}^n (\mathbb{1}\{\mathcal{E}_i(0)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(0)))\right)^4\right] \\ &\leq \bar{z}^4 n^{4\alpha-4} 500 \bar{d}^2 n^{2+2\alpha} \\ &= 500 \bar{z}^4 \bar{d}^2 n^{6\alpha-2}, \end{aligned}$$

where the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1))| \leq 1$ and $|\mathbb{1}\{\mathcal{E}_i(0)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(0))| \leq 1$, and using Assumption 1[†]-(ii). Similarly, we have

$$\text{E}_{\mathcal{W}}[(\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))^4] \leq 500 \bar{z}^4 \bar{d}^2 n^{6\alpha-2}.$$

Additionally, we have

$$\begin{aligned} \text{E}_{\mathcal{W}}[(\hat{X}(1) - \text{E}_{\mathcal{W}}[\hat{X}(1)])^4] &= \text{E}_{\mathcal{W}}\left[\left(\frac{1}{n} \sum_{i=1}^n a_i(1)(\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1)))\right)^4\right] \\ &\leq \bar{a}^4 n^{-4} \text{E}_{\mathcal{W}}\left[\left(\sum_{i=1}^n (\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1)))\right)^4\right] \\ &\leq 125 \bar{a}^4 \bar{d}^2 n^{2\alpha-2}, \end{aligned}$$

where the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))| \leq 1$, and using Assumption 1[†]-(ii). Similarly, we have

$$\mathbb{E}_{\mathcal{W}}[(\hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)])^4] \leq 125\bar{a}^4\bar{d}^2 n^{2\alpha-2}.$$

Combining these four bounds, we have

$$\lim_{n \rightarrow +\infty} n\mathbb{E}_{\mathcal{W}}[\Delta_n^2] \leq \lim_{n \rightarrow +\infty} 1000\bar{z}^2\bar{a}^2\bar{d}^2 n^{4\alpha-2} = 0, \quad (\text{EC.20})$$

where the equality holds because $\alpha < \frac{1}{4}$.

Next, we show that $n\text{Cov}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n) \rightarrow 0$. Note that

$$\begin{aligned} & \left| \text{Cov}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n) \right| \\ &= \left| \mathbb{E}_{\mathcal{W}} \left[(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))) \Delta_n \right] \right| \\ &= \left| \mathbb{E}_{\mathcal{W}} \left[(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))) (\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1)) (\hat{X}(1) - \mathbb{E}_{\mathcal{W}}[\hat{X}(1)]) \right] \right| \\ & \quad + \left| \mathbb{E}_{\mathcal{W}} \left[(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))) (\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0)) (\hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)]) \right] \right| \\ &\leq \mathbb{E}_{\mathcal{W}} \left[|\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))|^3 \right]^{\frac{1}{3}} \mathbb{E}_{\mathcal{W}} \left[|\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1)|^3 \right]^{\frac{1}{3}} \mathbb{E}_{\mathcal{W}} \left[|\hat{X}(1) - \mathbb{E}_{\mathcal{W}}[\hat{X}(1)]|^3 \right]^{\frac{1}{3}} \\ & \quad + \mathbb{E}_{\mathcal{W}} \left[|\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))|^3 \right]^{\frac{1}{3}} \mathbb{E}_{\mathcal{W}} \left[|\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0)|^3 \right]^{\frac{1}{3}} \mathbb{E}_{\mathcal{W}} \left[|\hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)]|^3 \right]^{\frac{1}{3}} \end{aligned}$$

where the first equality is because $\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))$ is unbiased; the second equality is using the definition of Δ_n ; the last inequality is using Holder's inequality.

We next upper bound the three moments above. First,

$$\begin{aligned} & \left| \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0)) \right| \\ &\leq \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_i(\delta(1))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(1))} - \gamma^*(1)a_i(1) \right| \cdot |\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))| \\ & \quad + \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_i(\delta(0))}{\Pr_{\mathcal{W}}(\mathcal{E}_i(0))} - \gamma^*(0)a_i(0) \right| \cdot |\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))| \\ &\leq \frac{1}{n} \sum_{i=1}^n \left(\frac{\bar{y}}{\pi} + \frac{2^{\frac{1}{2}}\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\pi \lambda_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \right) \cdot |\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))| + \frac{1}{n} \sum_{i=1}^n \left(\frac{\bar{y}}{\pi} + \frac{2^{\frac{1}{2}}\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\pi \lambda_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \right) \cdot |\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))| \\ &\leq \frac{3\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\pi \lambda_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}-1} \left(\sum_{i=1}^n |\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))| + \sum_{i=1}^n |\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))| \right) \end{aligned}$$

where the first inequality is by definition and upper bounding each component by its absolute value; the second inequality is due to Lemma EC.7; the third inequality holds because $1 + 2^{\frac{1}{2}} \leq 3$ and when n is sufficiently large. So we have

$$\mathbb{E}_{\mathcal{W}} \left[|\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))|^3 \right]$$

$$\begin{aligned}
&\leq \frac{27\bar{a}^3 \bar{y}^3 \bar{d}^{\frac{3}{2}}}{\pi^3 \underline{\lambda}_\Omega^{\frac{3}{2}}} n^{\frac{3\alpha}{2}-3} \mathbb{E}_{\mathcal{W}} \left[\left(\sum_{i=1}^n |\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))| + \sum_{i=1}^n |\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))| \right)^3 \right] \\
&\leq \frac{27\bar{a}^3 \bar{y}^3 \bar{d}^{\frac{3}{2}}}{\pi^3 \underline{\lambda}_\Omega^{\frac{3}{2}}} n^{\frac{3\alpha}{2}-3} \cdot 10\bar{d}^2 n^{1+2\alpha} \\
&= \frac{270\bar{a}^3 \bar{y}^3 \bar{d}^{\frac{7}{2}}}{\pi^3 \underline{\lambda}_\Omega^{\frac{3}{2}}} n^{\frac{7\alpha}{2}-2},
\end{aligned}$$

where the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))| \leq 1$ and $|\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))| \leq 1$, and using Assumption 1[†]-(ii).

Second, we have

$$\begin{aligned}
&\mathbb{E}_{\mathcal{W}} \left[|\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1)|^3 \right] \\
&= \mathbb{E}_{\mathcal{W}} \left[\left| \sum_{i=1}^n Z_{i,1}(1) (\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))) - \sum_{i=1}^n Z_{i,1}(0) (\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))) \right|^3 \right] \\
&\leq \bar{z}^3 n^{3\alpha-3} 10\bar{d}^2 n^{1+2\alpha} \\
&= 10\bar{z}^3 \bar{d}^2 n^{5\alpha-2},
\end{aligned}$$

where the second inequality is using the upper bound $|Z_{i,1}(1)| \leq \bar{z}n^{\alpha-1}$ and $|Z_{i,1}(0)| \leq \bar{z}n^{\alpha-1}$, doing the combinatorial counting, upper bounding $|\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))| \leq 1$ and $|\mathbb{1}\{\mathcal{E}_i(0)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(0))| \leq 1$, and using Assumption 1[†]-(ii). Similarly,

$$\mathbb{E}_{\mathcal{W}} \left[|\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0)|^3 \right] \leq 10\bar{z}^3 \bar{d}^2 n^{5\alpha-2}.$$

Third, we have

$$\begin{aligned}
\mathbb{E}_{\mathcal{W}} \left[|\hat{X}(1) - \mathbb{E}_{\mathcal{W}}[\hat{X}(1)]|^3 \right] &= \mathbb{E}_{\mathcal{W}} \left[\left| \frac{1}{n} \sum_{i=1}^n a_i(1) (\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))) \right|^3 \right] \\
&\leq \bar{a}^3 n^{-3} 10\bar{d}^2 n^{1+2\alpha} \\
&= 10\bar{a}^3 \bar{d}^2 n^{2\alpha-2},
\end{aligned}$$

where the second inequality is upper bounding $|a_i(1)| \leq \bar{a}$ and doing the combinatorial counting, upper bounding $|\mathbb{1}\{\mathcal{E}_i(1)\} - \Pr_{\mathcal{W}}(\mathcal{E}_i(1))| \leq 1$, and using Assumption 1[†]-(ii). Similarly,

$$\mathbb{E}_{\mathcal{W}} \left[|\hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)]|^3 \right] \leq 10\bar{a}^3 \bar{d}^2 n^{2\alpha-2}.$$

Combining these bounds, we have

$$\lim_{n \rightarrow +\infty} n \left| \text{Cov}_{\mathcal{W}} \left(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n \right) \right| \leq \lim_{n \rightarrow +\infty} \frac{30\bar{a}^2 \bar{y} \bar{z} \bar{d}^{\frac{5}{2}}}{\pi \underline{\lambda}_\Omega^{\frac{1}{2}}} n^{\frac{7\alpha}{2}-1} = 0, \quad (\text{EC.21})$$

where the equality holds because $\alpha < \frac{1}{4}$.

Putting (EC.20) and (EC.21) in (EC.19), we have

$$\lim_{n \rightarrow +\infty} n \left(\text{Var}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) - \text{Var}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right) \rightarrow 0.$$

Claim 3: Asymptotic Normality. In the proof of this claim, we assume $\alpha < \frac{1}{7}$.

First, we show that

$$\frac{\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))}{\sqrt{\text{Var}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

To show this, we introduce some notations. Define

$$\begin{aligned} \xi_i = & \left(\frac{Y_i(\delta(1))}{\text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1))} - \gamma^*(1)a_i(1) \right) (\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1))) \\ & - \left(\frac{Y_i(\delta(1))}{\text{Pr}_{\mathcal{W}}(\mathcal{E}_i(0))} - \gamma^*(0)a_i(0) \right) (\mathbb{1}\{\mathcal{E}_i(0)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(0))). \end{aligned}$$

Using this notation, we have

$$\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0)) = \frac{1}{n} \sum_{i=1}^n \xi_i.$$

We can upper bound each ξ_i for any $i \in [n]$ by

$$\begin{aligned} |\xi_i| & \leq \left(\frac{\bar{y}}{\underline{\pi}} + \frac{2^{\frac{1}{2}} \bar{a} \bar{y} \bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \right) \cdot \left(|\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1))| + |\mathbb{1}\{\mathcal{E}_i(0)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(0))| \right) \\ & \leq \frac{3 \bar{a} \bar{y} \bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \cdot \left(|\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1))| + |\mathbb{1}\{\mathcal{E}_i(0)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(0))| \right) \leq \frac{6 \bar{a} \bar{y} \bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Omega}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \end{aligned}$$

the first inequality is due to Lemma EC.7; the second inequality holds when n is sufficiently large; the third inequality is because $|\mathbb{1}\{\mathcal{E}_i(1)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(1))| \leq 1$ and $|\mathbb{1}\{\mathcal{E}_i(0)\} - \text{Pr}_{\mathcal{W}}(\mathcal{E}_i(0))| \leq 1$. From

here, denote $\bar{\xi} = \frac{6 \bar{a} \bar{y} \bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Omega}^{\frac{1}{2}}}$. Next, denote

$$\sigma_n = \sqrt{\text{Var}_{\mathcal{W}}\left(\sum_{i=1}^n \xi_i\right)}, \quad S_n = \frac{\sum_{i=1}^n \xi_i}{\sigma_n}.$$

We will use Lemma EC.2 to show the central limit theorem. To do so, we bound each term in Lemma EC.2.

First, recall that we additionally assume that for sufficiently large n , $n \text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c}$. So we have

$$\sigma_n^2 = \text{Var}_{\mathcal{W}}\left(\sum_{i=1}^n \xi_i\right) = n^2 \text{Var}_{\mathcal{W}}\left(\frac{1}{n} \sum_{i=1}^n \xi_i\right) = n^2 \text{Var}_{\mathcal{W}}(\hat{\mu}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c}n.$$

We also have

$$\sigma_n^3 \geq \underline{c}^{\frac{3}{2}} n^{\frac{3}{2}}.$$

Next, we have

$$\begin{aligned} \sum_{i=1}^n \mathbb{E}[|\xi_i|^3] &\leq \sum_{i=1}^n \bar{\xi}^3 n^{\frac{3}{2}\alpha} = \bar{\xi}^3 n^{1+\frac{3}{2}\alpha}, \\ \sum_{i=1}^n \mathbb{E}[\xi_i^4] &\leq \sum_{i=1}^n \bar{\xi}^4 n^{2\alpha} = \bar{\xi}^4 n^{1+2\alpha}. \end{aligned}$$

Putting all together into Lemma EC.2, for Z a standard normal random variable,

$$\begin{aligned} \lim_{n \rightarrow +\infty} d_W(S_n, Z) &\leq \lim_{n \rightarrow +\infty} \frac{\bar{d}^2 n^{2\alpha}}{\underline{c}^{\frac{3}{2}} n^{\frac{3}{2}}} \bar{\xi}^3 n^{1+\frac{3}{2}\alpha} + \frac{26^{\frac{1}{2}} \bar{d}^{\frac{3}{2}} n^{\frac{3}{2}\alpha}}{\pi^{\frac{1}{2}} \underline{c} n} \left(\bar{\xi}^4 n^{1+2\alpha} \right)^{\frac{1}{2}} \\ &= \lim_{n \rightarrow +\infty} \frac{\bar{d}^2 \bar{\xi}^3}{\underline{c}^{\frac{3}{2}}} n^{7\alpha - \frac{1}{2}} + \frac{26^{\frac{1}{2}} \bar{d}^{\frac{3}{2}} \bar{\xi}^2}{\pi^{\frac{1}{2}} \underline{c}} n^{\frac{5}{2}\alpha - \frac{1}{2}} \\ &= 0, \end{aligned}$$

where where $D_W(S_n, Z)$ stands for the Wasserstein distance between S_n and Z ; the first inequality holds using Lemma EC.2; the last equality holds because $\alpha < \frac{1}{7}$. So we have

$$\lim_{n \rightarrow +\infty} S_n = \lim_{n \rightarrow +\infty} \frac{\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Next, we establish the connection between $\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ and $\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))$. Note that,

$$\begin{aligned} &\frac{\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \tau(\delta(1), \delta(0))}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}} \\ &= \frac{\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau(\delta(1), \delta(0))}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \frac{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}} \\ &\quad + \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \frac{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}}. \quad (\text{EC.22}) \end{aligned}$$

From here, note that

$$\mathbb{E} \left[\frac{\Delta_n}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \right] \leq \mathbb{E} \left[\frac{\Delta_n^2}{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \right]^{\frac{1}{2}} \leq \sqrt{\frac{500 \bar{z}^2 \bar{a}^2 \bar{d}^2}{\underline{c}}} n^{4\alpha-1},$$

where the first inequality is Cauchy-Schwarz inequality; and the second inequality is using (EC.20) and assuming $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c}n^{-1}$. Using Markov inequality, for any $\epsilon > 0$,

$$\Pr \left(\left| \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \right| \geq \epsilon \right) \leq \sqrt{\frac{500 \bar{z}^2 \bar{a}^4 \bar{d}^4}{\underline{c}}} \cdot n^{4\alpha-1} \cdot \frac{1}{\epsilon}.$$

So we have

$$\lim_{n \rightarrow +\infty} \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \xrightarrow{p} 0.$$

Note also that $\lim_{n \rightarrow +\infty} n \text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) = \lim_{n \rightarrow +\infty} n \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$ and for sufficiently large n we assume $n \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c}$. This ensures that $n \text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) \geq \underline{c} > 0$ for sufficiently large n . So we have

$$\begin{aligned} & \lim_{n \rightarrow +\infty} \left| \frac{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))} - 1 \right| \\ &= \lim_{n \rightarrow +\infty} \frac{|n \text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) - n \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))|}{n \text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))} \\ &\leq \lim_{n \rightarrow +\infty} \frac{|n \text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) - n \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))|}{\underline{c}} \\ &= 0, \end{aligned}$$

where the last equality is Claim 2 in Theorem 3.

Putting the above together into (EC.22) and using the Slutsky theorem we have

$$\lim_{n \rightarrow +\infty} \frac{\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \tau(\delta(1), \delta(0))}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

This finishes the proof. $\square$

EC.4.8. Proof of Theorem 4

Proof of Theorem 4. Define the following block diagonal constraint

$$\mathcal{B} = \left\{ \mathbf{B} \in \mathbb{R}^{2n \times 2} \mid \mathbf{B} = \begin{bmatrix} \mathbf{a}(1) & \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{a}(0) \end{bmatrix} \right\}.$$

Recall that the objective function of (6) can be written as

$$\min_{\mathbf{B} \in \mathcal{B}} \mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y} - \mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y} \right].$$

Note that $\mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} [\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y}]$ does not depend on $\mathbf{B}$. So this minimization problem can be written as a maximization problem

$$\max_{\mathbf{B} \in \mathcal{B}} \mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y} \right]. \quad (\text{EC.23})$$

Next, note that

$$\begin{aligned} & \mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y} \right] \\ &= \mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\text{Tr}(\mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y}) \right] \\ &= \mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\text{Tr}((\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbf{Y} \mathbf{Y}^\top \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{B}) \right] \\ &= \text{Tr}((\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega} \mathbf{\Pi}^{-1} \mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} [\mathbf{Y} \mathbf{Y}^\top] \mathbf{\Pi}^{-1} \mathbf{\Omega} \mathbf{B}), \end{aligned}$$

where the first equality is because a scalar is equal to its trace; the second equality is because of the cyclic property of a trace; the third equality is because of linearity of expectations. Now we define

$$\begin{aligned} \mathbf{M} &= \mathbf{\Omega}^{\frac{1}{2}} \mathbf{\Pi}^{-1} \mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} [\mathbf{Y} \mathbf{Y}^\top] \mathbf{\Pi}^{-1} \mathbf{\Omega}^{\frac{1}{2}} \\ &= \mathbf{\Omega}^{\frac{1}{2}} \mathbf{\Pi}^{-1} \begin{bmatrix} \mu^2(1) \mathbf{1} \mathbf{1}^\top + \sigma^2(1) \mathbf{I}_n & -\mu(1)\mu(0) \mathbf{1} \mathbf{1}^\top - \sigma(1,0) \mathbf{I}_n \\ -\mu(1)\mu(0) \mathbf{1} \mathbf{1}^\top - \sigma(1,0) \mathbf{I}_n & \mu^2(0) \mathbf{1} \mathbf{1}^\top + \sigma^2(0) \mathbf{I}_n \end{bmatrix} \mathbf{\Pi}^{-1} \mathbf{\Omega}^{\frac{1}{2}}, \end{aligned}$$

where $\mathbf{I}_n$ is a $(n \times n)$ identity matrix and $\mathbf{1}$ is a n -dimensional vector with each element equal to 1. So (EC.23) can be written as

$$\max_{\mathbf{B} \in \mathcal{B}} \text{Tr} \left((\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega}^{\frac{1}{2}} \mathbf{M} \mathbf{\Omega}^{\frac{1}{2}} \mathbf{B} \right). \quad (\text{EC.24})$$

Now we consider an unconstrained version of (EC.24).

$$\max_{\mathbf{B}} \text{Tr} \left((\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{\Omega}^{\frac{1}{2}} \mathbf{M} \mathbf{\Omega}^{\frac{1}{2}} \mathbf{B} \right). \quad (\text{EC.25})$$

Below, we drop $\mathbf{B} \in \mathbb{R}^{2n \times 2}$ from the maximization above to stand for unconstrained maximization.

We introduce a variable transformation Note that $\mathbf{\Omega}$ is a symmetric and positive semidefinite matrix. So $\mathbf{B}^\top \mathbf{\Omega} \mathbf{B}$ is also a symmetric and positive semidefinite matrix. So we can define $(\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-\frac{1}{2}}$ as the pseudo inverse square root matrix of $\mathbf{B}^\top \mathbf{\Omega} \mathbf{B}$. Now define

$$\mathbf{V} = \mathbf{\Omega}^{\frac{1}{2}} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-\frac{1}{2}}.$$

Using the definition of $\mathbf{V}$, (EC.25) can be written as

$$\max_{\mathbf{B}} \text{Tr} \left(\mathbf{V}^\top \mathbf{M} \mathbf{V} \right).$$

Now we consider the rank of $\mathbf{V}$. There are two cases. First, if $\text{rank}(\mathbf{V}) = 2$, then

$$\mathbf{V}^\top \mathbf{V} = (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-\frac{1}{2}} \mathbf{B}^\top \mathbf{\Omega}^{\frac{1}{2}} \mathbf{\Omega}^{\frac{1}{2}} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-\frac{1}{2}} = \mathbf{I}_2.$$

Using Lemma EC.4 the Fan's Principle (Fan (1949) Theorem 1), the optimal objective value of (EC.25) is the sum of the two largest eigenvalues of $\mathbf{M}$, denoted as $\lambda_1(\mathbf{M})$ and $\lambda_2(\mathbf{M})$. So the optimal variance reduction is given by

$$\mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\text{Var}(\hat{\tau}^{\text{HT}}) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right] = \frac{\lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M})}{n^2}.$$

Second, if $\text{rank}(\mathbf{V}) = 1$, then

$$\mathbf{V}^\top \mathbf{V} = (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-\frac{1}{2}} \mathbf{B}^\top \mathbf{\Omega}^{\frac{1}{2}} \mathbf{\Omega}^{\frac{1}{2}} \mathbf{B} (\mathbf{B}^\top \mathbf{\Omega} \mathbf{B})^{-\frac{1}{2}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

In this case, using Lemma EC.4 the Fan's Principle (Fan (1949) Theorem 1), the optimal objective value of (EC.36) is the largest eigenvalue of $\mathbf{M}$, denoted as $\lambda_1(\mathbf{M})$. So the optimal variance reduction is given by

$$\mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\text{Var}(\hat{\tau}^{\text{HT}}) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right] = \frac{\lambda_1(\mathbf{M})}{n^2} \leq \frac{\lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M})}{n^2}.$$

So the variance reduction in Case 2 is always smaller or equal to the variance reduction in Case 1.

Finally, we compare the constrained maximization problem (EC.24) and the unconstrained maximization problem (EC.25). Since the constrained maximization problem is always smaller than or equal to the unconstrained problem, we conclude that

$$\mathbb{E}_{\mathbf{Y}(\delta(1)), \mathbf{Y}(\delta(0)) \sim \mathcal{Y}_\delta^n} \left[\text{Var}(\hat{\tau}^{\text{HT}}) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right] \leq \frac{\lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M})}{n^2},$$

for the constrained maximization problem (EC.24). $\square$

EC.4.9. Proof of Example 6

We show that, under SUTVA, for any $2n \times 2$ bases matrix $\mathbf{B} \in \mathbb{R}^{2n \times 2}$, we can explicitly construct a block diagonal bases matrix $\mathbf{B}^\circ \in \mathcal{B}$ that attains exactly the same objective value.

Proof of Example 6 We start with any $2n \times 2$ bases matrix

$$\mathbf{B} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \\ \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} \in \mathbb{R}^{2n \times 2},$$

where $\mathbf{u}_1, \mathbf{u}_2, \mathbf{v}_1, \mathbf{v}_2$ are all n -dimensional vectors. From here, we construct the block-diagonal matrix

$$\mathbf{B}^\circ = \begin{bmatrix} \mathbf{u}_1 - \mathbf{v}_1 & \mathbf{0}_n \\ \mathbf{0}_n & \mathbf{v}_2 - \mathbf{u}_2 \end{bmatrix} \in \mathcal{B}.$$

Now we consider $\mathbf{B} - \mathbf{B}^\circ$. For the first column,

$$\begin{bmatrix} \mathbf{u}_1 \\ \mathbf{v}_1 \end{bmatrix} - \begin{bmatrix} \mathbf{u}_1 - \mathbf{v}_1 \\ \mathbf{0}_n \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_1 \end{bmatrix},$$

and for the second column,

$$\begin{bmatrix} \mathbf{u}_2 \\ \mathbf{v}_2 \end{bmatrix} - \begin{bmatrix} \mathbf{0}_n \\ \mathbf{v}_2 - \mathbf{u}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{u}_2 \\ \mathbf{u}_2 \end{bmatrix}.$$

Recall the special block structure (8) that

$$\mathbf{\Omega} = \begin{bmatrix} \mathbf{\Sigma} & -\mathbf{\Sigma} \\ -\mathbf{\Sigma} & \mathbf{\Sigma} \end{bmatrix}.$$

Both vectors are in the null space of $\mathbf{\Omega}$. Therefore,

$$\mathbf{\Omega}^{\frac{1}{2}} \mathbf{B} = \mathbf{\Omega}^{\frac{1}{2}} \mathbf{B}^\circ.$$

Then we have

$$\mathbf{B}^\top \Omega \mathbf{B} = (\mathbf{B}^\circ)^\top \Omega \mathbf{B}^\circ$$

and

$$\mathbf{B}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \mathbf{B} = (\mathbf{B}^\circ)^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \mathbf{B}^\circ.$$

So $\mathbf{B}$ and $\mathbf{B}^\circ$ attain the same objective value. Since every $2n \times 2$ bases matrix corresponds to a block diagonal bases matrix with the same objective value, the constrained and unconstrained optimization problems have the same optimal value under SUTVA. This shows that Theorem 4 in the network interference setting nests Theorem EC.2 in the SUTVA setting as a special case. $\square$

EC.4.10. Proof of Corollary 2

Proof of Corollary 2. The proof of Corollary 2 is very straightforward. Recall from Section 3.4 that if the potential outcomes $\mathbf{Y}(\delta(1))$ and $\mathbf{Y}(\delta(0))$ were known to take values $Y_i(\delta(1)) = y_i(\delta(1))$ and $Y_i(\delta(0)) = y_i(\delta(0))$ for any $i \in [n]$, the problem

$$\min_{\mathbf{B} \in \mathcal{B}} \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) = \min_{\mathbf{B} \in \mathcal{B}} \frac{1}{n^2} \left(\mathbf{y}^\top \Pi^{-1} \Omega \Pi^{-1} \mathbf{y} - \mathbf{y}^\top \Pi^{-1} \Omega \mathbf{B} (\mathbf{B}^\top \Omega \mathbf{B})^{-1} \mathbf{B}^\top \Omega \Pi^{-1} \mathbf{y} \right).$$

can be easily solved by choosing $\mathbf{B}$ such that $\Pi^{-1} \mathbf{y}$ is in the column space of $\mathbf{B}$, such as $\mathbf{a}(1) = \Pi(1)^{-1} \mathbf{y}(\delta(1))$ and $\mathbf{a}(0) = -\Pi(0)^{-1} \mathbf{y}(\delta(0))$. Now because we assume that the potential outcomes $Y_i(\delta(1)) = y_i(\delta(1))$ and $Y_i(\delta(0)) = y_i(\delta(0))$ take the same unknown constant, then one set of optimal solution is given by $\mathbf{a}(1) = \Pi(1)^{-1} \mathbf{1}_n \cdot y(\delta(1))$ and $\mathbf{a}(0) = -\Pi(0)^{-1} \mathbf{1}_n \cdot y(\delta(0))$. $\square$

EC.4.11. Proof of Theorem 5

Proof of Theorem 5. The proof proceeds in three steps. First, we solve the problem

$$\max_{\mathbf{b} \in \mathcal{B}} \frac{\mathbf{b}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \mathbf{b}}{\mathbf{b}^\top \Omega \mathbf{b}}.$$

This is a Rayleigh quotient problem and the optimal objective value is given as $\lambda_1(\mathbf{M})$, the largest eigenvalue of $\mathbf{M}$, and the optimal solution is explicitly given as

$$\tilde{\mathbf{b}} = c \Omega^{-\frac{1}{2}} \mathbf{u}_1(\mathbf{M}),$$

where $c \neq 0$ is any constant, and $\mathbf{u}_1(\mathbf{M})$ is the eigenvector corresponding to $\lambda_1(\mathbf{M})$. The computation of $\tilde{\mathbf{b}}$ takes polynomial time. Now denote $\tilde{\mathbf{b}} = (\tilde{\mathbf{b}}_1^\top, \tilde{\mathbf{b}}_2^\top)^\top$ and we construct

$$\tilde{\mathbf{B}} = \begin{bmatrix} \tilde{\mathbf{b}}_1 & \mathbf{0}_n \\ \mathbf{0}_n & \tilde{\mathbf{b}}_2 \end{bmatrix}.$$

Apparently, $\tilde{\mathbf{B}} \in \mathcal{B}$ is a feasible solution.

Second, we evaluate the feasible solution $\tilde{\mathbf{B}}$. Denote the following 2×2 symmetric and positive semidefinite matrix

$$\mathbf{S} = (\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-\frac{1}{2}} \tilde{\mathbf{B}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{B}} (\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-\frac{1}{2}}.$$

Using the cyclic property of a trace, we have

$$\text{Tr} \left((\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-1} \tilde{\mathbf{B}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{B}} \right) = \text{Tr}(\mathbf{S}).$$

Next, note that $\tilde{\mathbf{b}} = \tilde{\mathbf{B}} \mathbf{1}_2$, where $\mathbf{1}_2 = (1, 1)^\top$ is a two-dimensional vector with both elements equal to 1. We denote the 2-dimensional vector $\mathbf{x} = (\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-\frac{1}{2}} \mathbf{1}_2$. Using this variable transformation, we have

$$\frac{\tilde{\mathbf{b}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{b}}}{\tilde{\mathbf{b}}^\top \Omega \tilde{\mathbf{b}}} = \frac{\mathbf{x}^\top \mathbf{S} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}.$$

So we have

$$\frac{\tilde{\mathbf{b}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{b}}}{\tilde{\mathbf{b}}^\top \Omega \tilde{\mathbf{b}}} = \frac{\mathbf{x}^\top \mathbf{S} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \leq \lambda_1(\mathbf{S}) \leq \sum_{l=1}^n \lambda_l(\mathbf{S}) = \text{Tr}(\mathbf{S}) = \text{Tr} \left((\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-1} \tilde{\mathbf{B}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{B}} \right),$$

where the first inequality is due to Lemma EC.3; the second inequality is because $\mathbf{S}$ is a positive semidefinite matrix so all the eigenvalues are non-negative; the second equality is because the trace of a matrix is the sum of all its eigenvalues. The above inequality shows that

$$\text{Tr} \left((\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-1} \tilde{\mathbf{B}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{B}} \right) \geq \frac{\tilde{\mathbf{b}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{b}}}{\tilde{\mathbf{b}}^\top \Omega \tilde{\mathbf{b}}} = \lambda_1(\mathbf{M}). \quad (\text{EC.26})$$

Third, we compare the feasible solution $\tilde{\mathbf{B}}$ with the optimal solution. Note that

$$\begin{aligned} \text{Tr} \left((\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-1} \tilde{\mathbf{B}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{B}} \right) &\leq \max_{\mathbf{B} \in \mathcal{B}} \text{Tr} \left((\mathbf{B}^\top \Omega \mathbf{B})^{-1} \mathbf{B}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \mathbf{B} \right) \\ &\leq \max_{\mathbf{B} \in \mathbb{R}^{2n \times 2}} \text{Tr} \left((\mathbf{B}^\top \Omega \mathbf{B})^{-1} \mathbf{B}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \mathbf{B} \right) = \lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M}) \leq 2\lambda_1(\mathbf{M}), \end{aligned} \quad (\text{EC.27})$$

where the first inequality is because we compare the feasible solution with the optimal solution; the second inequality is because we relax the constrained maximization problem (7) to the unconstrained maximization problem; the equality is due to Lemma EC.4.

Combining (EC.26) and (EC.27) we have

$$\text{Tr} \left((\tilde{\mathbf{B}}^\top \Omega \tilde{\mathbf{B}})^{-1} \tilde{\mathbf{B}}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \tilde{\mathbf{B}} \right) \geq \frac{1}{2} \max_{\mathbf{B} \in \mathcal{B}} \text{Tr} \left((\mathbf{B}^\top \Omega \mathbf{B})^{-1} \mathbf{B}^\top \Omega^{\frac{1}{2}} \mathbf{M} \Omega^{\frac{1}{2}} \mathbf{B} \right),$$

which finishes the proof. $\square$

EC.4.12. Proof of Theorem 6

Proof of Theorem 6. The proof proceeds in two steps. We first prove monotonicity. For any k , because $\mathbf{a}^k(1) \in \arg \max_{\mathbf{a}(1)} \psi(\mathbf{a}(1), \mathbf{a}^{k-1}(0))$ whereas $\mathbf{a}^{k-1}(1)$ is a feasible solution,

$$\psi(\mathbf{a}^k(1), \mathbf{a}^{k-1}(0)) \geq \psi(\mathbf{a}^{k-1}(1), \mathbf{a}^{k-1}(0)).$$

Next, because $\mathbf{a}^k(0) \in \arg \max_{\mathbf{a}(0)} \psi(\mathbf{a}^k(1), \mathbf{a}(0))$ whereas $\mathbf{a}^{k-1}(0)$ is a feasible solution,

$$\psi(\mathbf{a}^k(1), \mathbf{a}^k(0)) \geq \psi(\mathbf{a}^k(1), \mathbf{a}^{k-1}(0)).$$

Combining both inequalities, we have

$$\psi^k = \psi(\mathbf{a}^k(1), \mathbf{a}^k(0)) \geq \psi(\mathbf{a}^{k-1}(1), \mathbf{a}^{k-1}(0)) = \psi^{k-1}.$$

Next, we see from Theorem 4 that even the optimal objective value of (7) is upper bounded by

$$\psi(\mathbf{a}^*(1), \mathbf{a}^*(0)) \leq \lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M}).$$

This means that, for any k ,

$$\psi(\mathbf{a}^k(1), \mathbf{a}^k(0)) \leq \lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M}).$$

A monotone increasing and upper bounded sequence $\{\psi^k\}_{k \geq 0}$ must converge. $\square$

EC.4.13. Derivation of Expression (EC.2)

We state the expressions we want to derive again.

The variance minimizing coefficients $\gamma^*(1)$ and $\gamma^*(0)$ for any generic control variates $\hat{X}(1)$ and $\hat{X}(0)$ are given by

$$\begin{bmatrix} \gamma^*(1) \\ \gamma^*(0) \end{bmatrix} = \begin{bmatrix} \text{Var}(\hat{X}(1)) & \text{Cov}(\hat{X}(1), \hat{X}(0)) \\ \text{Cov}(\hat{X}(1), \hat{X}(0)) & \text{Var}(\hat{X}(0)) \end{bmatrix}^{-1} \cdot \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix}.$$

And the smallest variance is given by

$$\begin{aligned} \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) &= \text{Var}(\hat{\tau}^{\text{HT}}) - \\ &\quad \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix}^\top \cdot \begin{bmatrix} \text{Var}(\hat{X}(1)) & \text{Cov}(\hat{X}(1), \hat{X}(0)) \\ \text{Cov}(\hat{X}(1), \hat{X}(0)) & \text{Var}(\hat{X}(0)) \end{bmatrix}^{-1} \cdot \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix}. \end{aligned} \quad (\text{EC.28})$$

To derive expression (EC.2), note that

$$\begin{aligned} \text{Var}(\hat{\tau}^{\text{CV}}(\gamma(1), \gamma(0))) &= \text{Var}(\hat{\tau}^{\text{HT}}) - 2 \begin{bmatrix} \gamma(1) \\ \gamma(0) \end{bmatrix}^\top \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix} \\ &\quad + \begin{bmatrix} \gamma(1) \\ \gamma(0) \end{bmatrix}^\top \cdot \begin{bmatrix} \text{Var}(\hat{X}(1)) & \text{Cov}(\hat{X}(1), \hat{X}(0)) \\ \text{Cov}(\hat{X}(1), \hat{X}(0)) & \text{Var}(\hat{X}(0)) \end{bmatrix} \cdot \begin{bmatrix} \gamma(1) \\ \gamma(0) \end{bmatrix}. \end{aligned} \quad (\text{EC.29})$$

Because it is a quadratic function with respect to $\gamma(1)$ and $\gamma(0)$, first order condition ensures optimality. The first order condition is given by

$$-2 \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix} + 2 \begin{bmatrix} \text{Var}(\hat{X}(1)) & \text{Cov}(\hat{X}(1), \hat{X}(0)) \\ \text{Cov}(\hat{X}(1), \hat{X}(0)) & \text{Var}(\hat{X}(0)) \end{bmatrix} \cdot \begin{bmatrix} \gamma^*(1) \\ \gamma^*(0) \end{bmatrix} = 0.$$

Solving this system of linear equations yields

$$\begin{bmatrix} \gamma^*(1) \\ \gamma^*(0) \end{bmatrix} = \begin{bmatrix} \text{Var}(\hat{X}(1)) & \text{Cov}(\hat{X}(1), \hat{X}(0)) \\ \text{Cov}(\hat{X}(1), \hat{X}(0)) & \text{Var}(\hat{X}(0)) \end{bmatrix}^{-1} \cdot \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix}.$$

Putting the above into (EC.29) we derive expression (EC.28), that is,

$$\begin{aligned} \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) &= \text{Var}(\hat{\tau}^{\text{HT}}) - \\ &\quad \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix}^\top \cdot \begin{bmatrix} \text{Var}(\hat{X}(1)) & \text{Cov}(\hat{X}(1), \hat{X}(0)) \\ \text{Cov}(\hat{X}(1), \hat{X}(0)) & \text{Var}(\hat{X}(0)) \end{bmatrix}^{-1} \cdot \begin{bmatrix} \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \end{bmatrix}. \end{aligned}$$

EC.4.14. Proof of Lemma EC.1

Proof of Lemma EC.1. Recall that we have defined Σ as

$$\Sigma = \begin{bmatrix} \pi_{11} - \pi_1\pi_1 & \pi_{12} - \pi_1\pi_2 & \dots & \pi_{1n} - \pi_1\pi_n \\ \pi_{12} - \pi_1\pi_2 & \pi_{22} - \pi_2\pi_2 & \dots & \pi_{2n} - \pi_2\pi_n \\ \vdots & \vdots & \ddots & \vdots \\ \pi_{1n} - \pi_1\pi_n & \pi_{2n} - \pi_2\pi_n & \dots & \pi_{nn} - \pi_n\pi_n \end{bmatrix}.$$

Similarly, we define the following matrices,

$$\Sigma(1,1) = \begin{bmatrix} \text{Var}(\mathbb{1}\{W_1=1\}) & \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_2=1\}) & \dots & \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_n=1\}) \\ \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_2=1\}) & \text{Var}(\mathbb{1}\{W_2=1\}) & \dots & \text{Cov}(\mathbb{1}\{W_2=1\}, \mathbb{1}\{W_n=1\}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_n=1\}) & \text{Cov}(\mathbb{1}\{W_2=1\}, \mathbb{1}\{W_n=1\}) & \dots & \text{Var}(\mathbb{1}\{W_n=1\}) \end{bmatrix},$$

$$\Sigma(1,0) = \begin{bmatrix} \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_1=0\}) & \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_2=0\}) & \dots & \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_n=0\}) \\ \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_2=0\}) & \text{Cov}(\mathbb{1}\{W_2=1\}, \mathbb{1}\{W_2=0\}) & \dots & \text{Cov}(\mathbb{1}\{W_2=1\}, \mathbb{1}\{W_n=0\}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\mathbb{1}\{W_1=1\}, \mathbb{1}\{W_n=0\}) & \text{Cov}(\mathbb{1}\{W_2=1\}, \mathbb{1}\{W_n=0\}) & \dots & \text{Cov}(\mathbb{1}\{W_n=1\}, \mathbb{1}\{W_n=0\}) \end{bmatrix},$$

and

$$\Sigma(0,0) = \begin{bmatrix} \text{Var}(\mathbb{1}\{W_1=0\}) & \text{Cov}(\mathbb{1}\{W_1=0\}, \mathbb{1}\{W_2=0\}) & \dots & \text{Cov}(\mathbb{1}\{W_1=0\}, \mathbb{1}\{W_n=0\}) \\ \text{Cov}(\mathbb{1}\{W_1=0\}, \mathbb{1}\{W_2=0\}) & \text{Var}(\mathbb{1}\{W_2=0\}) & \dots & \text{Cov}(\mathbb{1}\{W_2=0\}, \mathbb{1}\{W_n=0\}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\mathbb{1}\{W_1=0\}, \mathbb{1}\{W_n=0\}) & \text{Cov}(\mathbb{1}\{W_2=0\}, \mathbb{1}\{W_n=0\}) & \dots & \text{Var}(\mathbb{1}\{W_n=0\}) \end{bmatrix}.$$

Under the Stable Unit Treatment Value Assumption (SUTVA, Rubin 1980, Holland 1986), we show that

$$\Sigma(1,1) = \Sigma(0,0) = -\Sigma(1,0) = \Sigma.$$

To see this, note that because $\mathbb{1}\{W_i = 1\} = 1 - \mathbb{1}\{W_i = 0\}$, we have

$$\text{Var}(\mathbb{1}\{W_i = 0\}) = \text{Var}(1 - \mathbb{1}\{W_i = 1\}) = \text{Var}(\mathbb{1}\{W_i = 1\}),$$

and we also have

$$\text{Cov}(\mathbb{1}\{W_i = 1\}, \mathbb{1}\{W_i = 0\}) = \text{Cov}(\mathbb{1}\{W_i = 1\}, 1 - \mathbb{1}\{W_i = 1\}) = -\text{Var}(\mathbb{1}\{W_i = 1\}).$$

Additionally, we have

$$\text{Cov}(\mathbb{1}\{W_i = 0\}, \mathbb{1}\{W_j = 0\}) = \text{Cov}(1 - \mathbb{1}\{W_i = 1\}, 1 - \mathbb{1}\{W_j = 1\}) = \text{Cov}(\mathbb{1}\{W_i = 1\}, \mathbb{1}\{W_j = 1\}),$$

and we also have

$$\text{Cov}(\mathbb{1}\{W_i = 1\}, \mathbb{1}\{W_j = 0\}) = \text{Cov}(\mathbb{1}\{W_i = 1\}, 1 - \mathbb{1}\{W_j = 1\}) = -\text{Cov}(\mathbb{1}\{W_i = 1\}, \mathbb{1}\{W_j = 1\}).$$

Combining above, we have

$$\mathbf{\Sigma}(1, 1) = \mathbf{\Sigma}(0, 0) = -\mathbf{\Sigma}(1, 0) = \mathbf{\Sigma}.$$

Next, we use the above notations and follow (EC.2) to find the expressions of $\gamma^*(1)$ and $\gamma^*(0)$. We start with $\text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1))$.

$$\begin{aligned} & \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(1)) \\ &= \text{E}[\hat{\tau}^{\text{HT}} \hat{X}(1)] - \tau \text{E}[\hat{X}(1)] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{E} \left[\left(\frac{Y_i(1) \mathbb{1}\{W_i = 1\}}{\pi_i} - \frac{Y_i(0) \mathbb{1}\{W_i = 0\}}{1 - \pi_i} \right) a_j(1) \mathbb{1}\{W_j = 1\} \right] - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (Y_i(1) - Y_i(0)) a_j(1) \pi_j \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(1)}{\pi_i} \text{Cov}(\mathbb{1}\{W_i = 1\}, \mathbb{1}\{W_j = 1\}) a_j(1) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(0)}{1 - \pi_i} \text{Cov}(\mathbb{1}\{W_i = 0\}, \mathbb{1}\{W_j = 1\}) a_j(1) \\ &= \frac{1}{n^2} \mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}(1, 1) \mathbf{a}(1) - \frac{1}{n^2} \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}(1, 0) \mathbf{a}(1) \\ &= \frac{1}{n^2} \mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma} \mathbf{a}(1) + \frac{1}{n^2} \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma} \mathbf{a}(1). \end{aligned}$$

Similarly, we derive the expression for $\text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0))$.

$$\begin{aligned} & \text{Cov}(\hat{\tau}^{\text{HT}}, \hat{X}(0)) \\ &= \text{E}[\hat{\tau}^{\text{HT}} \hat{X}(0)] - \tau \text{E}[\hat{X}(0)] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{E} \left[\left(\frac{Y_i(1) \mathbb{1}\{W_i = 1\}}{\pi_i} - \frac{Y_i(0) \mathbb{1}\{W_i = 0\}}{1 - \pi_i} \right) a_j(0) \mathbb{1}\{W_j = 0\} \right] - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (Y_i(1) - Y_i(0)) a_j(0) (1 - \pi_j) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(1)}{\pi_i} \text{Cov}(\mathbb{1}\{W_i = 1\}, \mathbb{1}\{W_j = 0\}) a_j(0) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(0)}{1 - \pi_i} \text{Cov}(\mathbb{1}\{W_i = 0\}, \mathbb{1}\{W_j = 0\}) a_j(0) \\ &= \frac{1}{n^2} \mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}(1, 0) \mathbf{a}(0) - \frac{1}{n^2} \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}(0, 0) \mathbf{a}(0) \\ &= -\frac{1}{n^2} \mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma} \mathbf{a}(0) - \frac{1}{n^2} \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma} \mathbf{a}(0). \end{aligned}$$

Next, we derive the expression for $\text{Cov}(\widehat{X}(1), \widehat{X}(0))$.

$$\begin{aligned}
\text{Cov}(\widehat{X}(1), \widehat{X}(0)) &= \mathbb{E}[\widehat{X}(1)\widehat{X}(0)] - \mathbb{E}[\widehat{X}(1)]\mathbb{E}[\widehat{X}(0)] \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \mathbb{E}[\mathbb{1}\{W_i = 1, W_j = 0\}] a_j(0) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \pi_i (1 - \pi_j) a_j(0) \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \text{Cov}(\mathbb{1}\{W_i = 1\}, \mathbb{1}\{W_j = 0\}) a_j(0) \\
&= \frac{1}{n^2} \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 0) \mathbf{a}(0) \\
&= -\frac{1}{n^2} \mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(0).
\end{aligned}$$

Similarly, we derive the expression for $\text{Var}(\widehat{X}(1))$.

$$\begin{aligned}
\text{Var}(\widehat{X}(1)) &= \mathbb{E}[\widehat{X}(1)^2] - \mathbb{E}[\widehat{X}(1)]^2 \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \mathbb{E}[\mathbb{1}\{W_i = 1, W_j = 1\}] a_j(1) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \pi_i \pi_j a_j(1) \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1) \text{Cov}(\mathbb{1}\{W_i = 1\}, \mathbb{1}\{W_j = 1\}) a_j(1) \\
&= \frac{1}{n^2} \mathbf{a}(1)^\top \boldsymbol{\Sigma}(1, 1) \mathbf{a}(1) \\
&= \frac{1}{n^2} \mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(1).
\end{aligned}$$

Finally, we derive the expression for $\text{Var}(\widehat{X}(0))$.

$$\begin{aligned}
\text{Var}(\widehat{X}(0)) &= \mathbb{E}[\widehat{X}(0)^2] - \mathbb{E}[\widehat{X}(0)]^2 \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(0) \mathbb{E}[\mathbb{1}\{W_i = 0, W_j = 0\}] a_j(0) - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(0) (1 - \pi_i) (1 - \pi_j) a_j(0) \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(0) \text{Cov}(\mathbb{1}\{W_i = 0\}, \mathbb{1}\{W_j = 0\}) a_j(0) \\
&= \frac{1}{n^2} \mathbf{a}(0)^\top \boldsymbol{\Sigma}(0, 0) \mathbf{a}(0) \\
&= \frac{1}{n^2} \mathbf{a}(0)^\top \boldsymbol{\Sigma} \mathbf{a}(0).
\end{aligned}$$

Putting all the above into expression (EC.2), we have

$$\begin{bmatrix} \gamma^*(1) \\ \gamma^*(0) \end{bmatrix} = \begin{bmatrix} \mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(1) & -\mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(0) \\ -\mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(0) & \mathbf{a}(0)^\top \boldsymbol{\Sigma} \mathbf{a}(0) \end{bmatrix}^{-1} \begin{bmatrix} (\mathbf{Y}(1)^\top \boldsymbol{\Pi}(1)^{-1} + \mathbf{Y}(0)^\top \boldsymbol{\Pi}(0)^{-1}) \boldsymbol{\Sigma} \mathbf{a}(1) \\ -(\mathbf{Y}(1)^\top \boldsymbol{\Pi}(1)^{-1} + \mathbf{Y}(0)^\top \boldsymbol{\Pi}(0)^{-1}) \boldsymbol{\Sigma} \mathbf{a}(0) \end{bmatrix}.$$

Note that

$$\begin{bmatrix} \mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(1) & -\mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(0) \\ -\mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(0) & \mathbf{a}(0)^\top \boldsymbol{\Sigma} \mathbf{a}(0) \end{bmatrix} = \mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A},$$

and that

$$\begin{bmatrix} (\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} + \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1}) \mathbf{\Sigma} \mathbf{a}(1) \\ -(\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} + \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1}) \mathbf{\Sigma} \mathbf{a}(0) \end{bmatrix} = \mathbf{A}^\top \mathbf{\Sigma} (\mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0)).$$

These succinct notations leads to

$$(\gamma^*(1), \gamma^*(0))^\top = (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{\Sigma} (\mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0)).$$

Using the definition of baseline outcomes $\mathbf{G} = \mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0)$ we finish the proof for the coefficients $(\gamma^*(1), \gamma^*(0))^\top$.

Finally, we focus on the expression for $\text{Var}(\hat{\tau}^{\text{HT}})$.

$$\begin{aligned} & \text{Var}(\hat{\tau}^{\text{HT}}) \\ &= \mathbb{E}[(\hat{\tau}^{\text{HT}})^2] - \tau^2 \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{E} \left[\left(\frac{Y_i(1) \mathbb{1}\{W_i=1\}}{\pi_i} - \frac{Y_i(0) \mathbb{1}\{W_i=0\}}{1-\pi_i} \right) \left(\frac{Y_j(1) \mathbb{1}\{W_j=1\}}{\pi_j} - \frac{Y_j(0) \mathbb{1}\{W_j=0\}}{1-\pi_j} \right) \right] \\ & \quad - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (Y_i(1) - Y_i(0))(Y_j(1) - Y_j(0)) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(1)}{\pi_i} \text{Cov}(\mathbb{1}\{W_i=1\}, \mathbb{1}\{W_j=1\}) \frac{Y_j(1)}{\pi_j} - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(1)}{\pi_i} \text{Cov}(\mathbb{1}\{W_i=1\}, \mathbb{1}\{W_j=0\}) \frac{Y_j(0)}{1-\pi_j} \\ & \quad - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(0)}{1-\pi_i} \text{Cov}(\mathbb{1}\{W_i=0\}, \mathbb{1}\{W_j=1\}) \frac{Y_j(1)}{\pi_j} + \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \frac{Y_i(0)}{1-\pi_i} \text{Cov}(\mathbb{1}\{W_i=0\}, \mathbb{1}\{W_j=0\}) \frac{Y_j(0)}{1-\pi_j} \\ &= \frac{1}{n^2} \mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}(1,1) \mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) - \frac{1}{n^2} \mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}(1,0) \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0) \\ & \quad - \frac{1}{n^2} \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}(1,0) \mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \frac{1}{n^2} \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}(0,0) \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0) \\ &= \frac{1}{n^2} (\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} + \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1}) \mathbf{\Sigma} (\mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0)). \end{aligned}$$

Putting all the above into expression (EC.28), we have

$$\begin{aligned} \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) &= \frac{1}{n^2} \left((\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} + \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1}) \mathbf{\Sigma} (\mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0)) \right. \\ & \quad \left. - (\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} + \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1}) \mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{\Sigma} (\mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0)) \right). \end{aligned}$$

Using the definition of baseline outcomes $\mathbf{G} = \mathbf{\Pi}(1)^{-1} \mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1} \mathbf{Y}(0)$ we finish the proof for the variance $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$. $\square$

EC.4.15. Proof of Theorem EC.1

We present the following Lemma EC.8 which will be useful in the proof of Theorem EC.1. We first introduce a vector notation $\boldsymbol{\gamma}^* = (\gamma^*(1), \gamma^*(0))^\top$.

Lemma EC.8 *Under Assumptions 1* and 2*, and assuming the potential outcomes $|Y_i(1)| \leq \bar{y}$ and $|Y_i(0)| \leq \bar{y}$ are all bounded, the optimal coefficients $\gamma^*(1)$ and $\gamma^*(0)$ are bounded, that is,*

$$|\gamma^*(1)| \leq \frac{2\bar{y}\bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}}, \quad |\gamma^*(0)| \leq \frac{2\bar{y}\bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}}.$$

Proof of Lemma EC.8. Recall that the variance minimizing coefficients are given by

$$\mathbf{G} = \mathbf{\Pi}(1)^{-1}\mathbf{Y}(1) + \mathbf{\Pi}(0)^{-1}\mathbf{Y}(0).$$

By Assumption 1*-(i) and the boundedness of potential outcomes, for each $i \in [n]$, $|G_i| \leq \frac{2\bar{y}}{\underline{\pi}}$. So we have

$$\mathbf{G}^{\top} \mathbf{\Sigma} \mathbf{G} \leq \frac{4\bar{y}^2}{\underline{\pi}^2} \sum_{i=1}^n \sum_{j=1}^n |\pi_{ij} - \pi_i \pi_j| \leq \frac{4\bar{y}^2 \bar{d}}{\underline{\pi}^2} n^{1+\alpha},$$

where the last inequality is due to Assumption 1*-(ii).

Next, note that

$$(\gamma^*)^{\top} \mathbf{A}^{\top} \mathbf{\Sigma} \mathbf{A} \gamma^* = \mathbf{G}^{\top} \mathbf{\Sigma} \mathbf{A} (\mathbf{A}^{\top} \mathbf{\Sigma} \mathbf{A})^{-1} \mathbf{A}^{\top} \mathbf{\Sigma} \mathbf{G} \leq \mathbf{G}^{\top} \mathbf{\Sigma} \mathbf{G},$$

where the inequality follows because the middle term is the variance reduction from projecting $\mathbf{G}$ onto the column space of $\mathbf{A}$, and hence cannot exceed the total variance $\mathbf{G}^{\top} \mathbf{\Sigma} \mathbf{G}$.

By Assumption 2*-(ii),

$$(\gamma^*)^{\top} \mathbf{A}^{\top} \mathbf{\Sigma} \mathbf{A} \gamma^* \geq \underline{\lambda}_{\Sigma} n \left((\gamma^*(1))^2 + (\gamma^*(0))^2 \right).$$

Putting the above inequalities together, we have

$$(\gamma^*(1))^2 + (\gamma^*(0))^2 \leq \frac{4\bar{y}^2 \bar{d}}{\underline{\pi}^2 \underline{\lambda}_{\Sigma}} n^{\alpha}.$$

Taking square root we finish the proof. □

Now we prove Theorem EC.1 as follows.

Proof of Theorem EC.1. We prove the three claims (consistency, asymptotic variance, and asymptotic normality) in Theorem EC.1 one by one.

Claim 1: Consistency. In the proof of this claim, we assume $\alpha < \frac{1}{3}$. We prove consistency through two steps. First, we show that

$$\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) \xrightarrow{p} 0.$$

For each $i \in [n]$, we have

$$\widehat{G}_i - G_i = (\mathbb{1}\{W_i = 1\} - \pi_i) \left(\frac{Y_i(1)}{\pi_i^2} + \frac{Y_i(0)}{(1 - \pi_i)^2} \right).$$

For each $i \in [n]$, define

$$Z_i(1) = \left(\frac{Y_i(1)}{\pi_i^2} + \frac{Y_i(0)}{(1 - \pi_i)^2} \right) (\mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1})_{i1},$$

and

$$Z_i(0) = \left(\frac{Y_i(1)}{\pi_i^2} + \frac{Y_i(0)}{(1 - \pi_i)^2} \right) (\mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1})_{i2},$$

where we use $(\mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1})_{i1}$ to stand for the element in the i -th row and the 1-st column, and $(\mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1})_{i2}$ to stand for the element in the i -th row and the 2-nd column. Using the above definitions, we have

$$\widehat{\gamma}^{\text{HT}}(1) - \gamma^*(1) = \sum_{i=1}^n Z_i(1) (\mathbb{1}\{W_i = 1\} - \pi_i),$$

and

$$\widehat{\gamma}^{\text{HT}}(0) - \gamma^*(0) = \sum_{i=1}^n Z_i(0) (\mathbb{1}\{W_i = 1\} - \pi_i),$$

We next upper bound $|Z_i(1)|$ and $|Z_i(0)|$. By Assumption 1*-(i) and boundedness of the potential outcomes,

$$\left| \frac{Y_i(1)}{\pi_i^2} + \frac{Y_i(0)}{(1 - \pi_i)^2} \right| \leq \frac{2\bar{y}}{\underline{\pi}^2}.$$

By Assumptions 1*-(ii) and 2*-(i), each element $(\mathbf{\Sigma} \mathbf{A})_{il}$ in the $(n \times 2)$ matrix $\mathbf{\Sigma} \mathbf{A}$ is bounded by

$$|(\mathbf{\Sigma} \mathbf{A})_{il}| = \left| \sum_{j=1}^n (\pi_{ij} - \pi_i \pi_j) a_j(l) \right| \leq \bar{a} \bar{d} n^\alpha.$$

By Assumption 2*-(ii),

$$\|(\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1}\|_2 \leq \frac{1}{\underline{\lambda}_\Sigma n}.$$

Therefore, there exists a constant $\bar{z} = \frac{4\bar{y}\bar{a}\bar{d}}{\underline{\pi}^2 \underline{\lambda}_\Sigma} > 0$ such that

$$|Z_i(1)| \leq \bar{z} n^{\alpha-1}, \quad |Z_i(0)| \leq \bar{z} n^{\alpha-1}.$$

Next, it is easy to see that

$$\mathbb{E}_{\mathcal{W}}[\widehat{\gamma}^{\text{HT}}(1)] = \gamma^*(1), \quad \mathbb{E}_{\mathcal{W}}[\widehat{\gamma}^{\text{HT}}(0)] = \gamma^*(0).$$

Next, we calculate $\text{Var}(\hat{\gamma}^{\text{HT}}(1))$ as

$$\text{Var}(\hat{\gamma}^{\text{HT}}(1)) = \sum_{i=1}^n \sum_{j=1}^n Z_i(1)Z_j(1)(\pi_{ij} - \pi_i\pi_j) \leq \bar{z}^2 n^{2\alpha-2} \sum_{i=1}^n \sum_{j=1}^n |\pi_{ij} - \pi_i\pi_j| \leq \bar{z}^2 \bar{d} n^{3\alpha-1},$$

where the last inequality is due to Assumption 1*-(ii). Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr\left(|\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1)| \geq \epsilon\right) \leq \frac{\text{Var}(\hat{\gamma}^{\text{HT}}(1))}{\epsilon^2} \leq \bar{z}^2 \bar{d} n^{3\alpha-1} \frac{1}{\epsilon^2}.$$

Similarly, we have

$$\Pr\left(|\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0)| \geq \epsilon\right) \leq \frac{\text{Var}(\hat{\gamma}^{\text{HT}}(0))}{\epsilon^2} \leq \bar{z}^2 \bar{d} n^{3\alpha-1} \frac{1}{\epsilon^2},$$

Next, we calculate the variance of the two control variates. Note that

$$\text{Var}(\hat{X}(1)) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n a_i(1) \mathbb{1}\{W_i = 1\}\right) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_i(1)a_j(1)(\pi_{ij} - \pi_i\pi_j) \leq \bar{a}^2 \bar{d} n^{\alpha-1}.$$

Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr\left(|\hat{X}(1) - \mathbb{E}[\hat{X}(1)]| \geq \epsilon\right) \leq \bar{a}^2 \bar{d} n^{\alpha-1} \frac{1}{\epsilon^2},$$

Similarly, because $\mathbb{1}\{W_i = 0\} - (1 - \pi_i) = -(\mathbb{1}\{W_i = 1\} - \pi_i)$, we have

$$\Pr\left(|\hat{X}(0) - \mathbb{E}[\hat{X}(0)]| \geq \epsilon\right) \leq \bar{a}^2 \bar{d} n^{\alpha-1} \frac{1}{\epsilon^2},$$

Finally, define

$$\begin{aligned} \Delta_n &= \hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) \\ &= -(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))(\hat{X}(1) - \mathbb{E}[\hat{X}(1)]) - (\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))(\hat{X}(0) - \mathbb{E}[\hat{X}(0)]). \end{aligned}$$

For any $\epsilon > 0$, we have

$$\begin{aligned} \lim_{n \rightarrow +\infty} \Pr(|\Delta_n| \geq \epsilon) &\leq \lim_{n \rightarrow +\infty} \left(\Pr\left(|\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1)| \geq \left(\frac{\epsilon}{2}\right)^{\frac{1}{2}}\right) + \Pr\left(|\hat{X}(1) - \mathbb{E}[\hat{X}(1)]| \geq \left(\frac{\epsilon}{2}\right)^{\frac{1}{2}}\right) \right. \\ &\quad \left. + \Pr\left(|\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0)| \geq \left(\frac{\epsilon}{2}\right)^{\frac{1}{2}}\right) + \Pr\left(|\hat{X}(0) - \mathbb{E}[\hat{X}(0)]| \geq \left(\frac{\epsilon}{2}\right)^{\frac{1}{2}}\right) \right) \\ &\leq \lim_{n \rightarrow +\infty} \left(2\bar{z}^2 \bar{d} n^{3\alpha-1} \frac{2}{\epsilon} + 2\bar{a}^2 \bar{d} n^{\alpha-1} \frac{2}{\epsilon} \right) \\ &= 0, \end{aligned}$$

where the first inequality holds because, if $|\Delta_n| \geq \epsilon$ holds, then at least one of the four conditions must hold; the last and only equality holds because $\alpha < \frac{1}{3}$. So we have shown that

$$\Delta_n \xrightarrow{p} 0.$$

Second, we show that $\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))$ is a consistent estimator of τ . Note that

$$\hat{\tau}^{\text{HT}} - \tau = \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i(1)}{\pi_i} + \frac{Y_i(0)}{1 - \pi_i} \right) (\mathbb{1}\{W_i = 1\} - \pi_i).$$

So we have,

$$\text{Var}(\hat{\tau}^{\text{HT}}) \leq \frac{1}{n^2} \frac{4\bar{y}^2}{\underline{\pi}^2} \text{Var}\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i)\right) \leq \frac{4\bar{y}^2 \bar{d}}{\underline{\pi}^2} n^{\alpha-1}.$$

Using Chebyshev inequality, for any $\epsilon > 0$,

$$\Pr\left(|\hat{\tau}^{\text{HT}} - \tau| \geq \epsilon\right) \leq \frac{\text{Var}(\hat{\tau}^{\text{HT}})}{\epsilon^2} \leq \frac{4\bar{y}^2 \bar{d}}{\underline{\pi}^2} n^{\alpha-1} \frac{1}{\epsilon^2}.$$

Next, note that

$$\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau = (\hat{\tau}^{\text{HT}} - \tau) - \gamma^*(1)(\hat{X}(1) - \mathbb{E}[\hat{X}(1)]) - \gamma^*(0)(\hat{X}(0) - \mathbb{E}[\hat{X}(0)]).$$

So we have, for any $\epsilon > 0$,

$$\begin{aligned} & \lim_{n \rightarrow +\infty} \Pr\left(|\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau| \geq \epsilon\right) \\ & \leq \lim_{n \rightarrow +\infty} \left(\Pr\left(|\hat{\tau}^{\text{HT}} - \tau| \geq \frac{\epsilon}{3}\right) + \Pr\left(|\gamma^*(1)(\hat{X}(1) - \mathbb{E}[\hat{X}(1)])| \geq \frac{\epsilon}{3}\right) + \Pr\left(|\gamma^*(0)(\hat{X}(0) - \mathbb{E}[\hat{X}(0)])| \geq \frac{\epsilon}{3}\right) \right) \\ & \leq \lim_{n \rightarrow +\infty} \left(\frac{4\bar{y}^2 \bar{d}}{\underline{\pi}^2} n^{\alpha-1} \frac{9}{\epsilon^2} + 2\bar{a}^2 \bar{d} n^{\alpha-1} \frac{9}{\epsilon^2} \left(\frac{2\bar{y} \bar{d}^{\frac{1}{2}}}{\underline{\pi} \lambda_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \right)^2 \right) \\ & = 0, \end{aligned}$$

where the first inequality holds because, if $|\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau| \geq \epsilon$ holds, then at least one of the three conditions must hold; the second inequality holds due to Lemma EC.8; the last equality holds because $\alpha < \frac{1}{3}$.

Therefore,

$$\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) \xrightarrow{P} \tau.$$

Combining this with $\Delta_n \xrightarrow{P} 0$ we have

$$\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) \xrightarrow{P} \tau.$$

Claim 2: Asymptotic Variance. In the proof of this claim, we assume $\alpha < \frac{1}{4}$.

Recall that we have defined

$$\Delta_n = \hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)).$$

Then

$$\begin{aligned} \text{Var}_{\mathcal{W}}\left(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))\right) \\ = \text{Var}_{\mathcal{W}}\left(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))\right) + \text{Var}_{\mathcal{W}}(\Delta_n) + 2\text{Cov}_{\mathcal{W}}\left(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n\right). \end{aligned} \quad (\text{EC.30})$$

It suffices to prove that both

$$n\text{Var}_{\mathcal{W}}(\Delta_n) \rightarrow 0$$

and

$$n\text{Cov}_{\mathcal{W}}\left(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n\right) \rightarrow 0.$$

We first show that $n\text{Var}_{\mathcal{W}}(\Delta_n) \rightarrow 0$. Because $0 \leq \text{Var}_{\mathcal{W}}(\Delta_n) \leq \text{E}_{\mathcal{W}}[\Delta_n^2]$, it suffices to show that $n\text{E}_{\mathcal{W}}[\Delta_n^2] \rightarrow 0$. Recall that

$$\Delta_n = -(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))(\hat{X}(1) - \text{E}[\hat{X}(1)]) - (\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))(\hat{X}(0) - \text{E}[\hat{X}(0)]).$$

So we have,

$$\begin{aligned} \text{E}_{\mathcal{W}}[\Delta_n^2] &\leq 2\text{E}_{\mathcal{W}}\left[(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))^2(\hat{X}(1) - \text{E}[\hat{X}(1)])^2\right] + 2\text{E}_{\mathcal{W}}\left[(\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))^2(\hat{X}(0) - \text{E}[\hat{X}(0)])^2\right] \\ &\leq 2\text{E}_{\mathcal{W}}\left[(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))^4\right]^{\frac{1}{2}} \text{E}_{\mathcal{W}}\left[(\hat{X}(1) - \text{E}[\hat{X}(1)])^4\right]^{\frac{1}{2}} \\ &\quad + 2\text{E}_{\mathcal{W}}\left[(\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))^4\right]^{\frac{1}{2}} \text{E}_{\mathcal{W}}\left[(\hat{X}(0) - \text{E}[\hat{X}(0)])^4\right]^{\frac{1}{2}}, \end{aligned}$$

where the second inequality is due to Cauchy-Schwarz inequality.

We next upper bound the four terms separately. Note that

$$\begin{aligned} \text{E}_{\mathcal{W}}[(\hat{\gamma}^{\text{HT}}(1) - \gamma^*(1))^4] &= \text{E}_{\mathcal{W}}\left[\left(\sum_{i=1}^n Z_i(1)(\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\ &\leq \bar{z}^4 n^{4\alpha-4} \text{E}_{\mathcal{W}}\left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\ &\leq \bar{z}^4 n^{4\alpha-4} 125 \bar{d}^2 n^{2+2\alpha} \\ &= 125 \bar{z}^4 \bar{d}^2 n^{6\alpha-2}, \end{aligned}$$

where the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1^{*}-(ii). Similarly, we have

$$\begin{aligned} \text{E}_{\mathcal{W}}[(\hat{\gamma}^{\text{HT}}(0) - \gamma^*(0))^4] &= \text{E}_{\mathcal{W}}\left[\left(\sum_{i=1}^n Z_i(0)(\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\ &\leq \bar{z}^4 n^{4\alpha-4} \text{E}_{\mathcal{W}}\left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\ &\leq \bar{z}^4 n^{4\alpha-4} 125 \bar{d}^2 n^{2+2\alpha} \\ &= 125 \bar{z}^4 \bar{d}^2 n^{6\alpha-2}, \end{aligned}$$

where the first equality is because $\mathbb{1}\{W_i = 0\} - (1 - \pi_i) = -(\mathbb{1}\{W_i = 1\} - \pi_i)$; the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1*-(ii). Additionally, we have

$$\begin{aligned} \mathbb{E}_{\mathcal{W}}[(\widehat{X}(1) - \mathbb{E}_{\mathcal{W}}[\widehat{X}(1)])^4] &= \mathbb{E}_{\mathcal{W}}\left[\left(\frac{1}{n} \sum_{i=1}^n a_i(1)(\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\ &\leq \bar{a}^4 n^{-4} \mathbb{E}_{\mathcal{W}}\left[\left(\sum_{i=1}^n (\mathbb{1}\{W_i = 1\} - \pi_i)\right)^4\right] \\ &\leq 125 \bar{a}^4 \bar{d}^2 n^{2\alpha-2}, \end{aligned}$$

where the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1*-(ii). Similarly, we have

$$\mathbb{E}_{\mathcal{W}}[(\widehat{X}(0) - \mathbb{E}_{\mathcal{W}}[\widehat{X}(0)])^4] \leq 125 \bar{a}^4 \bar{d}^2 n^{2\alpha-2}.$$

Combining these four bounds, we have

$$\lim_{n \rightarrow +\infty} n \mathbb{E}_{\mathcal{W}}[\Delta_n^2] \leq \lim_{n \rightarrow +\infty} 500 \bar{z}^2 \bar{a}^2 \bar{d}^2 n^{4\alpha-2} = 0, \quad (\text{EC.31})$$

where the equality holds because $\alpha < \frac{1}{4}$.

Next, we show that $n \text{Cov}_{\mathcal{W}}(\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n) \rightarrow 0$. Note that

$$\begin{aligned} & \left| \text{Cov}_{\mathcal{W}}(\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n) \right| \\ &= \left| \mathbb{E}_{\mathcal{W}}\left[(\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau) \Delta_n\right] \right| \\ &= \left| \mathbb{E}_{\mathcal{W}}\left[(\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau)(\widehat{\gamma}^{\text{HT}}(1) - \gamma^*(1))(\widehat{X}(1) - \mathbb{E}_{\mathcal{W}}[\widehat{X}(1)])\right] \right| \\ & \quad + \left| \mathbb{E}_{\mathcal{W}}\left[(\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau)(\widehat{\gamma}^{\text{HT}}(0) - \gamma^*(0))(\widehat{X}(0) - \mathbb{E}_{\mathcal{W}}[\widehat{X}(0)])\right] \right| \\ &\leq \mathbb{E}_{\mathcal{W}}\left[|\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau|^3\right]^{\frac{1}{3}} \mathbb{E}_{\mathcal{W}}\left[|\widehat{\gamma}^{\text{HT}}(1) - \gamma^*(1)|^3\right]^{\frac{1}{3}} \mathbb{E}_{\mathcal{W}}\left[|\widehat{X}(1) - \mathbb{E}_{\mathcal{W}}[\widehat{X}(1)]|^3\right]^{\frac{1}{3}} \\ & \quad + \mathbb{E}_{\mathcal{W}}\left[|\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau|^3\right]^{\frac{1}{3}} \mathbb{E}_{\mathcal{W}}\left[|\widehat{\gamma}^{\text{HT}}(0) - \gamma^*(0)|^3\right]^{\frac{1}{3}} \mathbb{E}_{\mathcal{W}}\left[|\widehat{X}(0) - \mathbb{E}_{\mathcal{W}}[\widehat{X}(0)]|^3\right]^{\frac{1}{3}} \end{aligned}$$

where the first equality is because $\widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))$ is unbiased; the second equality is using the definition of Δ_n ; the last inequality is using Holder's inequality.

We next upper bound the three moments above. First,

$$\begin{aligned} \left| \widehat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau \right| &\leq \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_i(1)}{\pi_i} - \frac{Y_i(0)}{1 - \pi_i} - \gamma^*(1)a_i(1) + \gamma^*(0)a_i(0) \right| \cdot |\mathbb{1}\{W_i = 1\} - \pi_i| \\ &\leq \frac{1}{n} \sum_{i=1}^n \left(\frac{2\bar{y}}{\underline{\pi}} + \frac{4\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \right) \cdot |\mathbb{1}\{W_i = 1\} - \pi_i| \\ &\leq \frac{1}{n} \sum_{i=1}^n \frac{6\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \cdot |\mathbb{1}\{W_i = 1\} - \pi_i| \end{aligned}$$

where the first inequality is by definition and upper bounding each component by its absolute value; the second inequality is due to Lemma EC.8; the third inequality holds when n is sufficiently large. So we have

$$\begin{aligned} \mathbb{E}_{\mathcal{W}} \left[\left| \hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau \right|^3 \right] &\leq \frac{216\bar{a}^3 \bar{y}^3 \bar{d}^{\frac{3}{2}}}{\pi^3 \underline{\lambda}_{\Sigma}^{\frac{3}{2}}} n^{\frac{3\alpha}{2}-3} \mathbb{E}_{\mathcal{W}} \left[\left| \sum_{i=1}^n \mathbb{1}\{W_i = 1\} - \pi_i \right|^3 \right] \\ &\leq \frac{216\bar{a}^3 \bar{y}^3 \bar{d}^{\frac{3}{2}}}{\pi^3 \underline{\lambda}_{\Sigma}^{\frac{3}{2}}} n^{\frac{3\alpha}{2}-3} 5\bar{d}^2 n^{1+2\alpha} \\ &= \frac{1080\bar{a}^3 \bar{y}^3 \bar{d}^{\frac{7}{2}}}{\pi^3 \underline{\lambda}_{\Sigma}^{\frac{3}{2}}} n^{\frac{7\alpha}{2}-2}, \end{aligned}$$

where the second inequality is doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1*-(ii).

Second, we have

$$\begin{aligned} \mathbb{E}_{\mathcal{W}} \left[\left| \hat{\gamma}^{\text{HT}}(1) - \gamma^*(1) \right|^3 \right] &= \mathbb{E}_{\mathcal{W}} \left[\left| \sum_{i=1}^n Z_i(1) (\mathbb{1}\{W_i = 1\} - \pi_i) \right|^3 \right] \\ &\leq \bar{z}^3 n^{3\alpha-3} 5\bar{d}^2 n^{1+2\alpha} \\ &= 5\bar{z}^3 \bar{d}^2 n^{5\alpha-2}, \end{aligned}$$

where the second inequality is using the upper bound $|Z_i(1)| \leq \bar{z}n^{\alpha-1}$ and doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1*-(ii). Similarly,

$$\mathbb{E}_{\mathcal{W}} \left[\left| \hat{\gamma}^{\text{HT}}(0) - \gamma^*(0) \right|^3 \right] \leq 5\bar{z}^3 \bar{d}^2 n^{5\alpha-2}.$$

Third, we have

$$\begin{aligned} \mathbb{E}_{\mathcal{W}} \left[\left| \hat{X}(1) - \mathbb{E}_{\mathcal{W}}[\hat{X}(1)] \right|^3 \right] &= \mathbb{E}_{\mathcal{W}} \left[\left| \frac{1}{n} \sum_{i=1}^n a_i(1) (\mathbb{1}\{W_i = 1\} - \pi_i) \right|^3 \right] \\ &\leq \bar{a}^3 n^{-3} 5\bar{d}^2 n^{1+2\alpha} \\ &= 5\bar{a}^3 \bar{d}^2 n^{2\alpha-2}, \end{aligned}$$

where the second inequality is upper bounding $|a_i(1)| \leq \bar{a}$ and doing the combinatorial counting, upper bounding $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$, and using Assumption 1*-(ii). Similarly,

$$\mathbb{E}_{\mathcal{W}} \left[\left| \hat{X}(0) - \mathbb{E}_{\mathcal{W}}[\hat{X}(0)] \right|^3 \right] \leq 5\bar{a}^3 \bar{d}^2 n^{2\alpha-2}.$$

Combining these bounds, we have

$$\lim_{n \rightarrow +\infty} n \left| \text{Cov}_{\mathcal{W}} \left(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)), \Delta_n \right) \right| \leq \lim_{n \rightarrow +\infty} \frac{30\bar{a}^2 \bar{y} \bar{z} \bar{d}^{\frac{5}{2}}}{\pi \underline{\lambda}_{\Sigma}^{\frac{1}{2}}} n^{\frac{7\alpha}{2}-1} = 0, \quad (\text{EC.32})$$

where the equality holds because $\alpha < \frac{1}{4}$.

Putting (EC.31) and (EC.32) in (EC.30), we have

$$\lim_{n \rightarrow +\infty} n \left(\text{Var}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) - \text{Var}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right) \rightarrow 0.$$

Claim 3: Asymptotic Normality. In the proof of this claim, we assume $\alpha < \frac{1}{7}$.

First, we show that

$$\frac{\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau}{\sqrt{\text{Var}_{\mathcal{W}}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

To show this, we introduce some notations. Define

$$\xi_i = \left(\frac{Y_i(1)}{\pi_i} - \frac{Y_i(0)}{1 - \pi_i} - \gamma^*(1)a_i(1) + \gamma^*(0)a_i(0) \right) (\mathbb{1}\{W_i = 1\} - \pi_i).$$

Using this notation, we have

$$\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau = \frac{1}{n} \sum_{i=1}^n \xi_i.$$

We can upper bound each ξ_i for any $i \in [n]$ by

$$|\xi_i| \leq \left(\frac{2\bar{y}}{\underline{\pi}} + \frac{4\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \right) \cdot |\mathbb{1}\{W_i = 1\} - \pi_i| \leq \frac{4\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}} \cdot |\mathbb{1}\{W_i = 1\} - \pi_i| \leq \frac{4\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Sigma}^{\frac{1}{2}}} n^{\frac{\alpha}{2}}$$

the first inequality is due to Lemma EC.8; the second inequality holds when n is sufficiently large; the third inequality is because $|\mathbb{1}\{W_i = 1\} - \pi_i| \leq 1$. From here, denote the constant $\bar{\xi} = \frac{4\bar{a}\bar{y}\bar{d}^{\frac{1}{2}}}{\underline{\pi} \underline{\lambda}_{\Sigma}^{\frac{1}{2}}}$.

Next, denote

$$\sigma_n = \sqrt{\text{Var}_{\mathcal{W}}\left(\sum_{i=1}^n \xi_i\right)}, \quad S_n = \frac{\sum_{i=1}^n \xi_i}{\sigma_n}.$$

We will use Lemma EC.2 to show the central limit theorem. To do so, we bound each term in Lemma EC.2.

First, recall that we additionally assume that for sufficiently large n , $n\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c}$.

So we have

$$\sigma_n^2 = \text{Var}_{\mathcal{W}}\left(\sum_{i=1}^n \xi_i\right) = n^2 \text{Var}_{\mathcal{W}}\left(\frac{1}{n} \sum_{i=1}^n \xi_i\right) = n^2 \text{Var}_{\mathcal{W}}(\hat{\mu}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c}n.$$

We also have

$$\sigma_n^3 \geq \underline{c}^{\frac{3}{2}} n^{\frac{3}{2}}.$$

Next, we have

$$\begin{aligned}\sum_{i=1}^n \mathbb{E}[|\xi_i|^3] &\leq \sum_{i=1}^n \bar{\xi}^3 n^{\frac{3}{2}\alpha} = \bar{\xi}^3 n^{1+\frac{3}{2}\alpha}, \\ \sum_{i=1}^n \mathbb{E}[\xi_i^4] &\leq \sum_{i=1}^n \bar{\xi}^4 n^{2\alpha} = \bar{\xi}^4 n^{1+2\alpha}.\end{aligned}$$

Putting all together into Lemma EC.2, for Z a standard normal random variable,

$$\begin{aligned}\lim_{n \rightarrow +\infty} d_W(S_n, Z) &\leq \lim_{n \rightarrow +\infty} \frac{\bar{d}^2 n^{2\alpha} \bar{\xi}^3 n^{1+\frac{3}{2}\alpha} + \frac{26^{\frac{1}{2}} \bar{d}^{\frac{3}{2}} n^{\frac{3}{2}\alpha}}{\pi^{\frac{1}{2}} \underline{c} n} \left(\bar{\xi}^4 n^{1+2\alpha} \right)^{\frac{1}{2}}}{\underline{c}^{\frac{3}{2}} n^{\frac{3}{2}}} \\ &= \lim_{n \rightarrow +\infty} \frac{\bar{d}^2 \bar{\xi}^3}{\underline{c}^{\frac{3}{2}}} n^{\frac{7}{2}\alpha - \frac{1}{2}} + \frac{26^{\frac{1}{2}} \bar{d}^{\frac{3}{2}} \bar{\xi}^2}{\pi^{\frac{1}{2}} \underline{c}} n^{\frac{5}{2}\alpha - \frac{1}{2}} \\ &= 0,\end{aligned}$$

where where $D_W(S_n, Z)$ stands for the Wasserstein distance between S_n and Z ; the first inequality holds using Lemma EC.2; the last equality holds because $\alpha < \frac{1}{7}$. So we have

$$\lim_{n \rightarrow +\infty} S_n = \lim_{n \rightarrow +\infty} \frac{\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau}{\sqrt{\text{Var}(\hat{\mu}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Next, we establish the connection between $\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))$ and $\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))$. Note that,

$$\begin{aligned}\frac{\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \tau}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}} &= \frac{\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)) - \tau}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \frac{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}} \\ &\quad + \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \frac{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}}. \quad (\text{EC.33})\end{aligned}$$

From here, note that

$$\mathbb{E} \left[\frac{\Delta_n}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \right] \leq \mathbb{E} \left[\frac{\Delta_n^2}{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))} \right]^{\frac{1}{2}} \leq \sqrt{\frac{500 \bar{z}^2 \bar{a}^4 \bar{d}^2}{\underline{c}} n^{4\alpha-1}},$$

where the first inequality is Cauchy-Schwarz inequality; and the second inequality is using (EC.31) and assuming $\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c} n^{-1}$. Using Markov inequality, for any $\epsilon > 0$,

$$\Pr \left(\left| \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \right| \geq \epsilon \right) \leq \sqrt{\frac{500 \bar{z}^2 \bar{a}^4 \bar{d}^4}{\underline{c}} \cdot n^{4\alpha-1}} \cdot \frac{1}{\epsilon}.$$

So we have

$$\lim_{n \rightarrow +\infty} \frac{\Delta_n}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}} \xrightarrow{p} 0.$$

Note also that $\lim_{n \rightarrow +\infty} n\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) = \lim_{n \rightarrow +\infty} n\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))$ and for sufficiently large n we assume $n\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \geq \underline{c}$. This ensures that $n\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) \geq \underline{c} > 0$ for sufficiently large n . So we have

$$\begin{aligned} & \lim_{n \rightarrow +\infty} \left| \frac{\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))}{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))} - 1 \right| \\ &= \lim_{n \rightarrow +\infty} \frac{|n\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) - n\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))|}{n\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))} \\ &\leq \lim_{n \rightarrow +\infty} \frac{|n\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0))) - n\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0)))|}{\underline{c}} \\ &= 0, \end{aligned}$$

where the last equality is Claim 2 in Theorem EC.1.

Putting the above together into (EC.33) and using the Slutsky theorem we have

$$\lim_{n \rightarrow +\infty} \frac{\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)) - \tau}{\sqrt{\text{Var}(\hat{\tau}^{\text{CV}}(\hat{\gamma}^{\text{HT}}(1), \hat{\gamma}^{\text{HT}}(0)))}} \xrightarrow{d} \mathcal{N}(0, 1).$$

This finishes the proof. $\square$

EC.4.16. Proof of Theorem EC.2

Proof of Theorem EC.2. Recall that the objective function of (EC.4) can be written as

$$\min_{\mathbf{A}} \mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\mathbf{G}^\top \mathbf{\Sigma} \mathbf{G} - \mathbf{G}^\top \mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{\Sigma} \mathbf{G} \right].$$

Note that $\mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} [\mathbf{G}^\top \mathbf{\Sigma} \mathbf{G}]$ does not depend on $\mathbf{A}$. So this minimization problem can be written as a maximization problem

$$\max_{\mathbf{A}} \mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\mathbf{G}^\top \mathbf{\Sigma} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{\Sigma} \mathbf{G} \right]. \quad (\text{EC.34})$$

Now we introduce a variable transformation. Note that $\mathbf{\Sigma}$ is a symmetric and positive semidefinite matrix. So $\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A}$ is also a symmetric and positive semidefinite matrix. So we can define $(\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-1}$ as the pseudo inverse matrix of $\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A}$, and $(\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-\frac{1}{2}}$ as the pseudo inverse square root matrix of $\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A}$. Now define

$$\mathbf{V} = \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{A} (\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A})^{-\frac{1}{2}}.$$

Using the definition of $\mathbf{V}$, (EC.34) can be written as

$$\max_{\mathbf{A}} \mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\mathbf{G}^\top \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{V} \mathbf{V}^\top \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{G} \right]. \quad (\text{EC.35})$$

Next, note that

$$\begin{aligned}
\mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\mathbf{G}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{V} \mathbf{V}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{G} \right] &= \mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\text{Tr} \left(\mathbf{G}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{V} \mathbf{V}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{G} \right) \right] \\
&= \mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\text{Tr} \left(\mathbf{V}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{G} \mathbf{G}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{V} \right) \right] \\
&= \text{Tr} \left(\mathbf{V}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\mathbf{G} \mathbf{G}^\top \right] \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{V} \right),
\end{aligned}$$

where the first equality is because a scalar is equal to its trace; the second equality is because of the cyclic property of a trace; the third equality is because of linearity of expectations. Now we define

$$\begin{aligned}
\mathbf{M} = \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\mathbf{G} \mathbf{G}^\top \right] \boldsymbol{\Sigma}^{\frac{1}{2}} &= \boldsymbol{\Sigma}^{\frac{1}{2}} \boldsymbol{\Pi}(1)^{-1} (\mu^2(1) \mathbf{1}_n \mathbf{1}_n^\top + \sigma^2(1) \mathbf{I}_n) \boldsymbol{\Pi}(1)^{-1} \boldsymbol{\Sigma}^{\frac{1}{2}} \\
&\quad + \boldsymbol{\Sigma}^{\frac{1}{2}} \boldsymbol{\Pi}(1)^{-1} (\mu(1) \mu(0) \mathbf{1}_n \mathbf{1}_n^\top + \sigma(1, 0) \mathbf{I}_n) \boldsymbol{\Pi}(0)^{-1} \boldsymbol{\Sigma}^{\frac{1}{2}} \\
&\quad + \boldsymbol{\Sigma}^{\frac{1}{2}} \boldsymbol{\Pi}(0)^{-1} (\mu(1) \mu(0) \mathbf{1}_n \mathbf{1}_n^\top + \sigma(1, 0) \mathbf{I}_n) \boldsymbol{\Pi}(1)^{-1} \boldsymbol{\Sigma}^{\frac{1}{2}} \\
&\quad + \boldsymbol{\Sigma}^{\frac{1}{2}} \boldsymbol{\Pi}(0)^{-1} (\mu^2(0) \mathbf{1}_n \mathbf{1}_n^\top + \sigma^2(0) \mathbf{I}_n) \boldsymbol{\Pi}(0)^{-1} \boldsymbol{\Sigma}^{\frac{1}{2}}.
\end{aligned}$$

So (EC.35) can be written as

$$\max_{\mathbf{V}} \text{Tr} \left(\mathbf{V}^\top \mathbf{M} \mathbf{V} \right). \tag{EC.36}$$

Now we consider the rank of $\mathbf{V}$. There are two cases. First, if $\text{rank}(\mathbf{V}) = 2$, then

$$\mathbf{V}^\top \mathbf{V} = (\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A})^{-\frac{1}{2}} \mathbf{A}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{A} (\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A})^{-\frac{1}{2}} = \mathbf{I}_2.$$

Using Lemma EC.4 the Fan's Principle (Fan (1949) Theorem 1), the optimal objective value of (EC.36) is the sum of the two largest eigenvalues of $\mathbf{M}$, denoted as $\lambda_1(\mathbf{M})$ and $\lambda_2(\mathbf{M})$. So the optimal variance reduction is given by

$$\mathbb{E}_{\mathbf{Y}(1), \mathbf{Y}(0) \sim \mathcal{Y}_{\text{PO}}^n} \left[\text{Var}(\hat{\tau}^{\text{HT}}) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right] = \frac{\lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M})}{n^2}.$$

Additionally, using Lemma EC.4 the Fan's Principle (Fan (1949) Theorem 1), the optimal solution to (EC.36) is the two eigenvectors that correspond to $\lambda_1(\mathbf{M})$ and $\lambda_2(\mathbf{M})$, denoted as $\mathbf{u}_1(\mathbf{M})$ and $\mathbf{u}_2(\mathbf{M})$. And we have the optimal solution being $\mathbf{V} = [\mathbf{u}_1(\mathbf{M}), \mathbf{u}_2(\mathbf{M})]$. Substituting this back to the definition of $\mathbf{V}$ we have that the optimal bases are

$$\mathbf{A}^* = \boldsymbol{\Sigma}^{-\frac{1}{2}} [\mathbf{u}_1(\mathbf{M}), \mathbf{u}_2(\mathbf{M})] \mathbf{C}$$

where $\mathbf{C}$ is any invertible, symmetric and semidefinite matrix.

Second, if $\text{rank}(\mathbf{V}) = 1$, then

$$\mathbf{V}^\top \mathbf{V} = (\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A})^{-\frac{1}{2}} \mathbf{A}^\top \boldsymbol{\Sigma}^{\frac{1}{2}} \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{A} (\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A})^{-\frac{1}{2}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

In this case, using Lemma EC.4 the Fan's Principle (Fan (1949) Theorem 1), the optimal objective value of (EC.36) is the largest eigenvalue of $\mathbf{M}$, denoted as $\lambda_1(\mathbf{M})$. So the optimal variance reduction is given by

$$\mathbb{E}_{\mathbf{Y}^{(1)}, \mathbf{Y}^{(0)} \sim \mathcal{Y}_{\text{PO}}^n} \left[\text{Var}(\hat{\tau}^{\text{HT}}) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right] = \frac{\lambda_1(\mathbf{M})}{n^2} \leq \frac{\lambda_1(\mathbf{M}) + \lambda_2(\mathbf{M})}{n^2}.$$

So the variance reduction in Case 2 is always smaller or equal to the variance reduction in Case 1. This means that we should choose the bases $\mathbf{A}$ to have $\text{rank}(\mathbf{A}) = 2$ full column rank, and the optimal solution comes from Case 1. $\square$

EC.4.17. Proof of Proposition 2

We present the following Lemma EC.9 which will be useful in the proof of Proposition 2. To introduce Lemma EC.9, we denote the following notation. For any matrix $\mathbf{E} \in \mathbb{R}^{n \times n}$, denote $\text{range}(\mathbf{E})$ to be the column space of matrix $\mathbf{E}$, defined as

$$\text{range}(\mathbf{E}) = \{ \mathbf{E}\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n \}.$$

Lemma EC.9 *Let $\mathbf{E} \in \mathbb{R}^{n \times n}$ be any symmetric matrix. For any $l \leq n$, let $\lambda_l(\mathbf{M})$ be the l -th largest eigenvalue of $\mathbf{M}$ and $\mathbf{u}_l(\mathbf{M})$ be the eigenvector that corresponds to $\lambda_l(\mathbf{M})$, where $\mathbf{M}$ is a symmetric matrix defined as $\mathbf{M} = \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{E} \mathbf{\Sigma}^{\frac{1}{2}}$. When $\lambda_l(\mathbf{M}) > 0$, we have*

$$\mathbf{u}_l(\mathbf{M}) \in \text{range}(\mathbf{\Sigma}).$$

Proof of Lemma EC.9. Note that $\mathbf{u}_l(\mathbf{M}) \in \text{range}(\mathbf{M})$ because $\mathbf{u}_l(\mathbf{M})$ is an eigenvector of the symmetric matrix $\mathbf{M}$. Next, we show that $\text{range}(\mathbf{M}) \subseteq \text{range}(\mathbf{\Sigma})$. To see this, note that for any vector $\mathbf{x}$,

$$\mathbf{M}\mathbf{x} = \mathbf{\Sigma}^{\frac{1}{2}} \left(\mathbf{E} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x} \right).$$

Because $(\mathbf{E} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x})$ may not span the entire space of $\mathbb{R}^n$, we have that

$$\text{range}(\mathbf{M}) \subseteq \text{range}(\mathbf{\Sigma}^{\frac{1}{2}}).$$

Next, because $\mathbf{\Sigma} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^{-1}$ and $\mathbf{\Sigma}^{\frac{1}{2}} = \mathbf{U} \mathbf{\Lambda}^{\frac{1}{2}} \mathbf{U}^{-1}$ have the same directions along the eigenvectors corresponding to positive eigenvalues, we have

$$\text{range}(\mathbf{\Sigma}^{\frac{1}{2}}) = \text{range}(\mathbf{\Sigma}).$$

Putting all above together, we have

$$\mathbf{u}_l(\mathbf{M}) \in \text{range}(\mathbf{M}) \subseteq \text{range}(\mathbf{\Sigma}^{\frac{1}{2}}) = \text{range}(\mathbf{\Sigma}).$$

This finishes the proof. $\square$

Now we prove Proposition 2 as follows.

Proof of Proposition 2. If we choose $\mathbf{C} = n^{\frac{1}{2}}\mathbf{I}_2$ where $\mathbf{I}_2$ stands for a 2×2 identity matrix, Theorem EC.2 suggests that

$$\mathbf{a}(1) = n^{\frac{1}{2}}\boldsymbol{\Sigma}^{-\frac{1}{2}}\mathbf{u}_1(\mathbf{M}), \quad \text{and} \quad \mathbf{a}(0) = -n^{\frac{1}{2}}\boldsymbol{\Sigma}^{-\frac{1}{2}}\mathbf{u}_2(\mathbf{M}).$$

Using the above expressions,

$$\mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(0) = (\boldsymbol{\Sigma}^{-\frac{1}{2}}\mathbf{u}_1(\mathbf{M}))^\top \boldsymbol{\Sigma} (-\boldsymbol{\Sigma}^{-\frac{1}{2}}\mathbf{u}_2(\mathbf{M})) = -\mathbf{u}_1(\mathbf{M})^\top \boldsymbol{\Sigma}^{-\frac{1}{2}}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-\frac{1}{2}}\mathbf{u}_2(\mathbf{M}).$$

Denote the following diagonal matrix $\mathbb{1}^+ = \mathbf{diag}(\mathbb{1}\{\lambda_1 > 0\}, \mathbb{1}\{\lambda_2 > 0\}, \dots, \mathbb{1}\{\lambda_n > 0\})$, where λ_i stands for the i -th largest eigenvalue of matrix $\boldsymbol{\Sigma}$. Denote $\mathbf{P}_\Sigma = \boldsymbol{\Sigma}^{-\frac{1}{2}}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-\frac{1}{2}}$. Recall the eigen-decomposition $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{-1}$. So we know that

$$\mathbf{P}_\Sigma = \boldsymbol{\Sigma}^{-\frac{1}{2}}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-\frac{1}{2}} = \mathbf{U}\mathbb{1}^+\mathbf{U}^{-1}$$

is a projection matrix, that is, if a vector $\mathbf{x} \in \text{range}(\boldsymbol{\Sigma})$, we have $\mathbf{P}_\Sigma \mathbf{x} = \mathbf{x}$.

Using Lemma EC.9, we have

$$\mathbf{u}_2(\mathbf{M}) \in \text{range}(\boldsymbol{\Sigma}).$$

Consequently, $\mathbf{P}_\Sigma \mathbf{u}_2(\mathbf{M}) = \mathbf{u}_2(\mathbf{M})$. This leads to

$$\mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(0) = -\mathbf{u}_1(\mathbf{M})^\top \mathbf{u}_2(\mathbf{M}) = 0,$$

where the last equality is because $\mathbf{u}_1(\mathbf{M})$ and $\mathbf{u}_2(\mathbf{M})$ are orthogonal.

Moreover, recall that using Lemma EC.9 we have

$$\mathbf{u}_1(\mathbf{M}), \mathbf{u}_2(\mathbf{M}) \in \text{range}(\boldsymbol{\Sigma}).$$

Consequently, $\mathbf{P}_\Sigma \mathbf{u}_1(\mathbf{M}) = \mathbf{u}_1(\mathbf{M})$ and $\mathbf{P}_\Sigma \mathbf{u}_2(\mathbf{M}) = \mathbf{u}_2(\mathbf{M})$. Next, we have

$$\mathbf{a}(1)^\top \boldsymbol{\Sigma} \mathbf{a}(1) = n\mathbf{u}_1(\mathbf{M})^\top \boldsymbol{\Sigma}^{-\frac{1}{2}}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-\frac{1}{2}}\mathbf{u}_1(\mathbf{M}) = n\mathbf{u}_1(\mathbf{M})^\top \mathbf{u}_1(\mathbf{M}) = n,$$

and

$$\mathbf{a}(0)^\top \boldsymbol{\Sigma} \mathbf{a}(0) = n\mathbf{u}_2(\mathbf{M})^\top \boldsymbol{\Sigma}^{-\frac{1}{2}}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-\frac{1}{2}}\mathbf{u}_2(\mathbf{M}) = n\mathbf{u}_2(\mathbf{M})^\top \mathbf{u}_2(\mathbf{M}) = n,$$

where the last equality in the above expressions holds because $\mathbf{u}_1(\mathbf{M})$ and $\mathbf{u}_2(\mathbf{M})$ are eigenvectors so they are unit length. $\square$

EC.4.18. Proof of Corollary 3

Proof of Corollary 3. The proof of Corollary 3 is very straightforward. Recall from Section EC.2.4 that if the potential outcomes $\mathbf{Y}(1)$ and $\mathbf{Y}(0)$ were known to take values $Y_i(1) = y_i(1)$ and $Y_i(0) = y_i(0)$ for any $i \in [n]$, the problem

$$\min_{\mathbf{A}} \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) = \min_{\mathbf{A}} \frac{1}{n^2} \left(\mathbf{g}^\top \Sigma \mathbf{g} - \mathbf{g}^\top \Sigma \mathbf{A} (\mathbf{A}^\top \Sigma \mathbf{A})^{-1} \mathbf{A}^\top \Sigma \mathbf{g} \right)$$

can be easily solved by choosing $\mathbf{A}$ such that $\mathbf{g}$ is in the column space of $\mathbf{A}$, such as $\mathbf{a}(1) = \mathbf{\Pi}(1)^{-1} \mathbf{y}(1)$ and $\mathbf{a}(0) = -\mathbf{\Pi}(0)^{-1} \mathbf{y}(0)$. Now because we assume that the potential outcomes $Y_i(1) = y_i(1)$ and $Y_i(0) = y_i(0)$ take the same unknown constant, then one set of optimal solution is given by $\mathbf{a}(1) = \mathbf{\Pi}(1)^{-1} \mathbf{1}_n \cdot y(1)$ and $\mathbf{a}(0) = -\mathbf{\Pi}(0)^{-1} \mathbf{1}_n \cdot y(0)$. $\square$

EC.4.19. Proof of Example 8

Proof of Example 8. Note that, for any $\gamma = [\gamma(1), \gamma(0)]^\top$, we have

$$\text{Var}(\hat{\tau}^{\text{CV}}(\gamma(1), \gamma(0))) = \text{Var}(\hat{\tau}^{\text{HT}}) - \frac{2}{n^2} \gamma^\top \mathbf{A}^\top \Sigma \mathbf{G} + \frac{1}{n^2} \gamma^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma.$$

So we have

$$\begin{aligned} & n^2 \left(\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^{\text{Naive}}(1), \gamma^{\text{Naive}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \right) \\ &= 2(\gamma^*)^\top \mathbf{A}^\top \Sigma \mathbf{G} - (\gamma^*)^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^* - 2(\gamma^{\text{Naive}})^\top \mathbf{A}^\top \Sigma \mathbf{G} + (\gamma^{\text{Naive}})^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^{\text{Naive}} \\ &= 2(\gamma^*)^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^* - (\gamma^*)^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^* - 2(\gamma^{\text{Naive}})^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^* + (\gamma^{\text{Naive}})^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^{\text{Naive}} \\ &= (\gamma^*)^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^* - 2(\gamma^{\text{Naive}})^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^* + (\gamma^{\text{Naive}})^\top \mathbf{A}^\top \Sigma \mathbf{A} \gamma^{\text{Naive}} \\ &= (\gamma^{\text{Naive}} - \gamma^*)^\top \mathbf{A}^\top \Sigma \mathbf{A} (\gamma^{\text{Naive}} - \gamma^*), \end{aligned} \tag{EC.37}$$

where the second equality is because $\gamma^* = (\mathbf{A}^\top \Sigma \mathbf{A})^{-1} \mathbf{A}^\top \Sigma \mathbf{G}$ so $\mathbf{A}^\top \Sigma \mathbf{G} = \mathbf{A}^\top \Sigma \mathbf{A} \gamma^*$.

Following Proposition 2, if we simply choose

$$\mathbf{a}(1) = n^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \mathbf{u}_1(\mathbf{M}), \quad \text{and} \quad \mathbf{a}(0) = -n^{\frac{1}{2}} \Sigma^{-\frac{1}{2}} \mathbf{u}_2(\mathbf{M}),$$

then $\mathbf{a}(1)^\top \Sigma \mathbf{a}(0) = 0$ and $\mathbf{a}(1)^\top \Sigma \mathbf{a}(1) = \mathbf{a}(0)^\top \Sigma \mathbf{a}(0) = n$ and we have that the variance minimizing coefficients in Lemma EC.1 can be simplified by

$$\gamma^*(1) = n^{-\frac{1}{2}} \mathbf{G}^\top \Sigma^{\frac{1}{2}} \mathbf{u}_1(\mathbf{M}), \quad \text{and} \quad \gamma^*(0) = n^{-\frac{1}{2}} \mathbf{G}^\top \Sigma^{\frac{1}{2}} \mathbf{u}_2(\mathbf{M}).$$

Next, note that

$$\gamma^{\text{Naive}}(1) = \frac{\text{Cov}(\hat{\mu}^{\text{HT}}(1), \hat{X}(1))}{\text{Var}(\hat{X}(1))} = \frac{\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \Sigma \mathbf{a}(1)}{\mathbf{a}(1)^\top \Sigma \mathbf{a}(1)} = n^{-\frac{1}{2}} \mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \Sigma^{\frac{1}{2}} \mathbf{u}_1(\mathbf{M})$$

and

$$\gamma^{\text{Naive}}(0) = -\frac{\text{Cov}(\hat{\mu}^{\text{HT}}(0), \hat{X}(0))}{\text{Var}(\hat{X}(0))} = -\frac{\mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma} \mathbf{a}(0)}{\mathbf{a}(0)^\top \mathbf{\Sigma} \mathbf{a}(0)} = n^{-\frac{1}{2}} \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{u}_2(\mathbf{M}).$$

So we have

$$\gamma^{\text{Naive}}(1) - \gamma^*(1) = -n^{-\frac{1}{2}} \mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{u}_1(\mathbf{M})$$

and

$$\gamma^{\text{Naive}}(0) - \gamma^*(0) = -n^{-\frac{1}{2}} \mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{u}_2(\mathbf{M}).$$

Putting the above into (EC.37) and recalling $\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A} = n \mathbf{I}_2$ where $\mathbf{I}_2$ is a 2×2 identity matrix, we have

$$\begin{aligned} & \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^{\text{Naive}}(1), \gamma^{\text{Naive}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \\ &= \frac{1}{n^2} \left((\mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{u}_1(\mathbf{M}))^2 + (\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{u}_2(\mathbf{M}))^2 \right) \geq 0, \end{aligned}$$

where the inequality takes equality if and only if when both $\mathbf{Y}(0)^\top \mathbf{\Pi}(0)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{u}_1(\mathbf{M}) = 0$ and $\mathbf{Y}(1)^\top \mathbf{\Pi}(1)^{-1} \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{u}_2(\mathbf{M}) = 0$. $\square$

EC.4.20. Proof of Example 9

Proof of Example 9. Using the bases $a_i(1) = \frac{1}{\pi_i}$ and $a_i(0) = -\frac{1}{1-\pi_i}$ for any $i \in [n]$, we have

$$\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A} = \begin{bmatrix} \sum_{i=1}^n \frac{1-\pi_i}{\pi_i} & n \\ n & \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} \end{bmatrix},$$

and

$$\mathbf{A}^\top \mathbf{\Sigma} \mathbf{G} = \begin{bmatrix} \sum_{i=1}^n (1-\pi_i) \left(\frac{Y_i(1)}{\pi_i} + \frac{Y_i(0)}{1-\pi_i} \right) \\ \sum_{i=1}^n \pi_i \left(\frac{Y_i(1)}{\pi_i} + \frac{Y_i(0)}{1-\pi_i} \right) \end{bmatrix}.$$

We further derive

$$\gamma^{\text{Naive}}(1) = \frac{\sum_{i=1}^n \frac{1-\pi_i}{\pi_i} Y_i(1)}{\sum_{i=1}^n \frac{1-\pi_i}{\pi_i}}, \quad \text{and} \quad \gamma^{\text{Naive}}(0) = \frac{\sum_{i=1}^n \frac{\pi_i}{1-\pi_i} Y_i(0)}{\sum_{i=1}^n \frac{\pi_i}{1-\pi_i}}.$$

Finally, we can show that

$$\det(\mathbf{A}^\top \mathbf{\Sigma} \mathbf{A}) = \left(\sum_{i=1}^n \frac{1-\pi_i}{\pi_i} \right) \left(\sum_{i=1}^n \frac{\pi_i}{1-\pi_i} \right) - n^2 \geq \left(\sum_{i=1}^n \sqrt{\frac{1-\pi_i}{\pi_i}} \cdot \sqrt{\frac{\pi_i}{1-\pi_i}} \right)^2 - n^2 = 0,$$

where the inequality is due to Cauchy-Schwarz inequality, and the inequality takes equality if and only if $\pi_i = \pi$ for any $i \in [n]$.

Note that,

$$\begin{aligned}
& \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^{\text{Naive}}(1), \gamma^{\text{Naive}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \\
&= \frac{1}{n^2} (\gamma^{\text{Naive}} - \gamma^*)^\top \mathbf{A}^\top \Sigma \mathbf{A} (\gamma^{\text{Naive}} - \gamma^*) \\
&= \frac{1}{n^2} (\mathbf{A}^\top \Sigma \mathbf{A} \gamma^{\text{Naive}} - \mathbf{A}^\top \Sigma \mathbf{G})^\top (\mathbf{A}^\top \Sigma \mathbf{A})^{-1} (\mathbf{A}^\top \Sigma \mathbf{A} \gamma^{\text{Naive}} - \mathbf{A}^\top \Sigma \mathbf{G}), \tag{EC.38}
\end{aligned}$$

where the first equality is due to (EC.37) in the proof of Example 8; the second equality is because $\gamma^* = (\mathbf{A}^\top \Sigma \mathbf{A})^{-1} \mathbf{A}^\top \Sigma \mathbf{G}$. From here, we focus on

$$\begin{aligned}
\mathbf{A}^\top \Sigma \mathbf{A} \gamma^{\text{Naive}} - \mathbf{A}^\top \Sigma \mathbf{G} &= \begin{bmatrix} \sum_{i=1}^n \frac{1-\pi_i}{\pi_i} & n \\ n & \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} \end{bmatrix} \cdot \begin{bmatrix} \gamma^{\text{Naive}}(1) \\ \gamma^{\text{Naive}}(0) \end{bmatrix} - \begin{bmatrix} \sum_{i=1}^n (1-\pi_i) \left(\frac{Y_i(1)}{\pi_i} + \frac{Y_i(0)}{1-\pi_i} \right) \\ \sum_{i=1}^n \pi_i \left(\frac{Y_i(1)}{\pi_i} + \frac{Y_i(0)}{1-\pi_i} \right) \end{bmatrix} \\
&= \begin{bmatrix} n\gamma^{\text{Naive}}(0) - \sum_{i=1}^n Y_i(0) \\ n\gamma^{\text{Naive}}(1) - \sum_{i=1}^n Y_i(1) \end{bmatrix}.
\end{aligned}$$

Now we distinguish two cases.

Case 1: $\det(\mathbf{A}^\top \Sigma \mathbf{A}) > 0$,

$$(\mathbf{A}^\top \Sigma \mathbf{A})^{-1} = \frac{1}{\det(\mathbf{A}^\top \Sigma \mathbf{A})} \begin{bmatrix} \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} & -n \\ -n & \sum_{i=1}^n \frac{1-\pi_i}{\pi_i} \end{bmatrix}.$$

Putting the above into (EC.38) we have

$$\begin{aligned}
& \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^{\text{Naive}}(1), \gamma^{\text{Naive}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) \\
&= \frac{1}{n^2 \left(\sum_{i=1}^n \frac{1-\pi_i}{\pi_i} \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} - n^2 \right)} \left(\sum_{i=1}^n \frac{1-\pi_i}{\pi_i} \Theta(1)^2 - 2n\Theta(1)\Theta(0) + \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} \Theta(0)^2 \right),
\end{aligned}$$

where we denote

$$\begin{aligned}
\Theta(1) &= \frac{n \sum_{i=1}^n \frac{1-\pi_i}{\pi_i} Y_i(1) - \left(\sum_{i=1}^n \frac{1-\pi_i}{\pi_i} \right) \left(\sum_{i=1}^n Y_i(1) \right)}{\sum_{i=1}^n \frac{1-\pi_i}{\pi_i}}; \\
\Theta(0) &= \frac{n \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} Y_i(0) - \left(\sum_{i=1}^n \frac{\pi_i}{1-\pi_i} \right) \left(\sum_{i=1}^n Y_i(0) \right)}{\sum_{i=1}^n \frac{\pi_i}{1-\pi_i}}.
\end{aligned}$$

Note that,

$$\begin{aligned}
& \sum_{i=1}^n \frac{1-\pi_i}{\pi_i} \Theta(1)^2 - 2n\Theta(1)\Theta(0) + \sum_{i=1}^n \frac{\pi_i}{1-\pi_i} \Theta(0)^2 \\
&= \sum_{i=1}^n \left(\frac{1-\pi_i}{\pi_i} \Theta(1)^2 - 2\Theta(1)\Theta(0) + \frac{\pi_i}{1-\pi_i} \Theta(0)^2 \right) \\
&= \sum_{i=1}^n \left(\sqrt{\frac{1-\pi_i}{\pi_i}} \Theta(1) - \sqrt{\frac{\pi_i}{1-\pi_i}} \Theta(0) \right)^2 \\
&\geq 0,
\end{aligned}$$

where inequality takes equality if and only if one of the following two conditions hold:

1. $\pi_i = \pi$ for any $i \in [n]$;
2. $Y_i(1) = Y(1)$ and $Y_i(0) = Y(0)$ for any $i \in [n]$.

But in this case, $\det(\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A}) > 0$ so the first condition does not hold. So the inequality takes equality if and only if the second condition holds.

Case 2: $\det(\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A}) = 0$. In this case, $\pi_i = \pi$ for any $i \in [n]$, and we have

$$\gamma^{\text{Naive}}(1) = \frac{1}{n} \sum_{i=1}^n Y_i(1), \quad \text{and} \quad \gamma^{\text{Naive}}(0) = \frac{1}{n} \sum_{i=1}^n Y_i(0).$$

So we know that

$$\mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\gamma}^{\text{Naive}} - \mathbf{A}^\top \boldsymbol{\Sigma} \mathbf{G} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Putting the above into (EC.38) we have

$$\text{Var}(\hat{\tau}^{\text{CV}}(\gamma^{\text{Naive}}(1), \gamma^{\text{Naive}}(0))) - \text{Var}(\hat{\tau}^{\text{CV}}(\gamma^*(1), \gamma^*(0))) = 0.$$

Combining both cases we finish the proof. □